Part 10
The Age of Apes

Human Origins



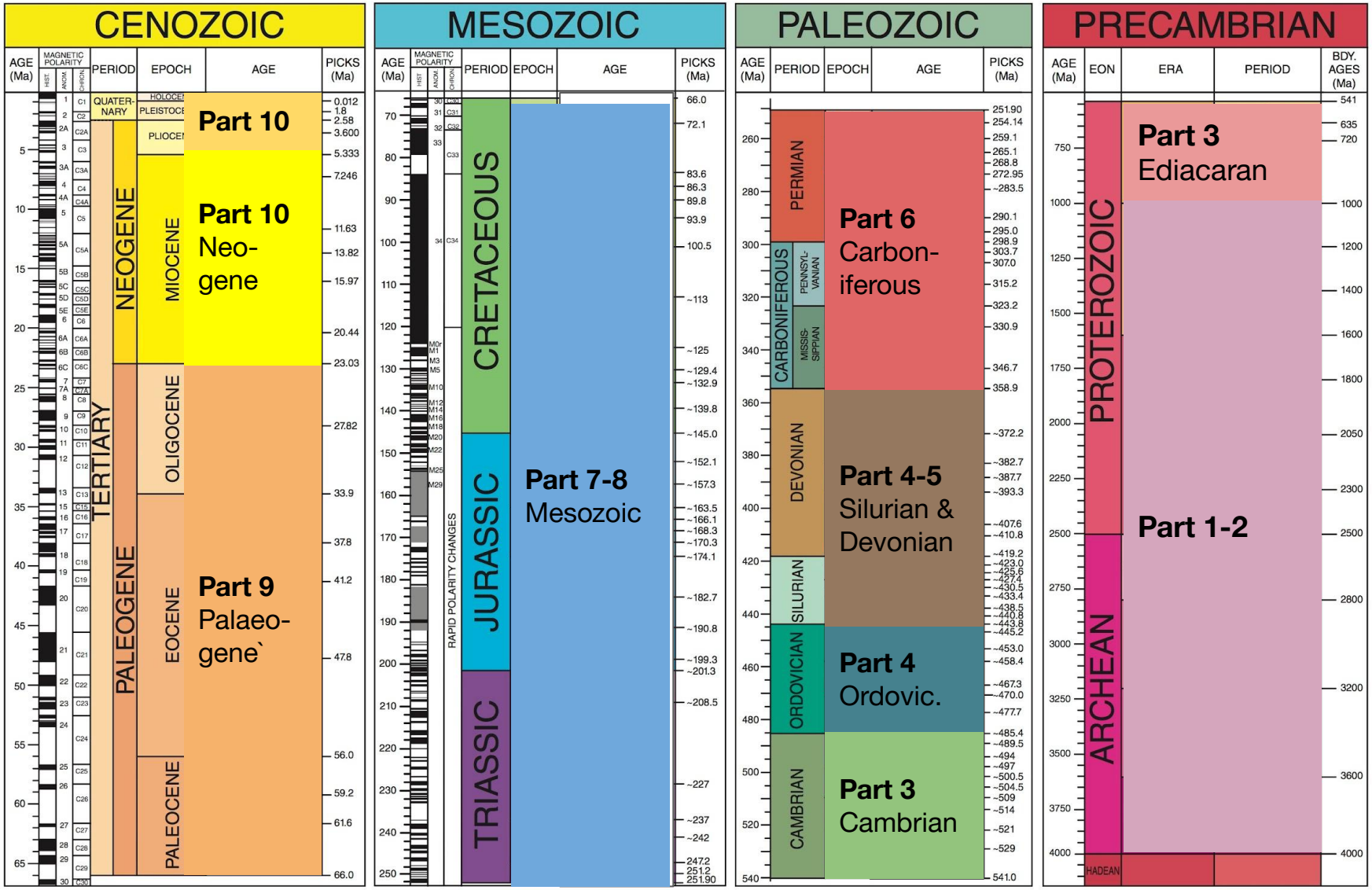
EART11100: How to Grow a Planet
Robert Sansom
University of Manchester

Palaeogene Summary

- Diversification of vertebrates after the end Cretaceous extinction
- Warm conditions in the Eocene, particularly the thermal maximum
- Spread of grasses



GSA GEOLOGIC TIME SCALE v. 5.0



Walker, J.D., Geissman, J.W., Bowring, S.A., and Babcock, L.E., compilers, 2018, Geologic Time Scale v. 5.0: Geological Society of America, https://doi.org/10.1130/2018.CTS005R3C.

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*The Pleistocene is divided into four ages, but only two are shown here. What is shown as Calabrian is actually three ages—Calabrian from 1.80 to 0.781 Ma, Middle from 0.781 to 0.126 Ma, and Late from 0.126 to 0.0117 Ma.

The Cenozoic, Mesozoic, and Paleozoic are the Eras of the Phanerozoic Eon. Names of units and age boundaries usually follow the Gradstein et al. (2012), Cohen et al. (2012) , and Cohen et al. (2013, updated) compilations. Numerical age estimates and picks of boundaries usually follow the Cohen et al. (2013, updated) compilation. The numbered epochs and ages of the Cambrian are provisional. A "~" before a numerical age estimate typically indicates an associated error of ±0.4 to over 1.6 Ma.

REFERENCES CITED

Cohen, K.M., Finney, S., and Gibbard, P.L., 2012, International Chronostratigraphic Chart: International Commission on Stratigraphy, www.stratigraphy.org (accessed May 2012). (Chart reproduced for the 34th International Geological Congress, Brisbane, Australia, 5–10 August 2012.)

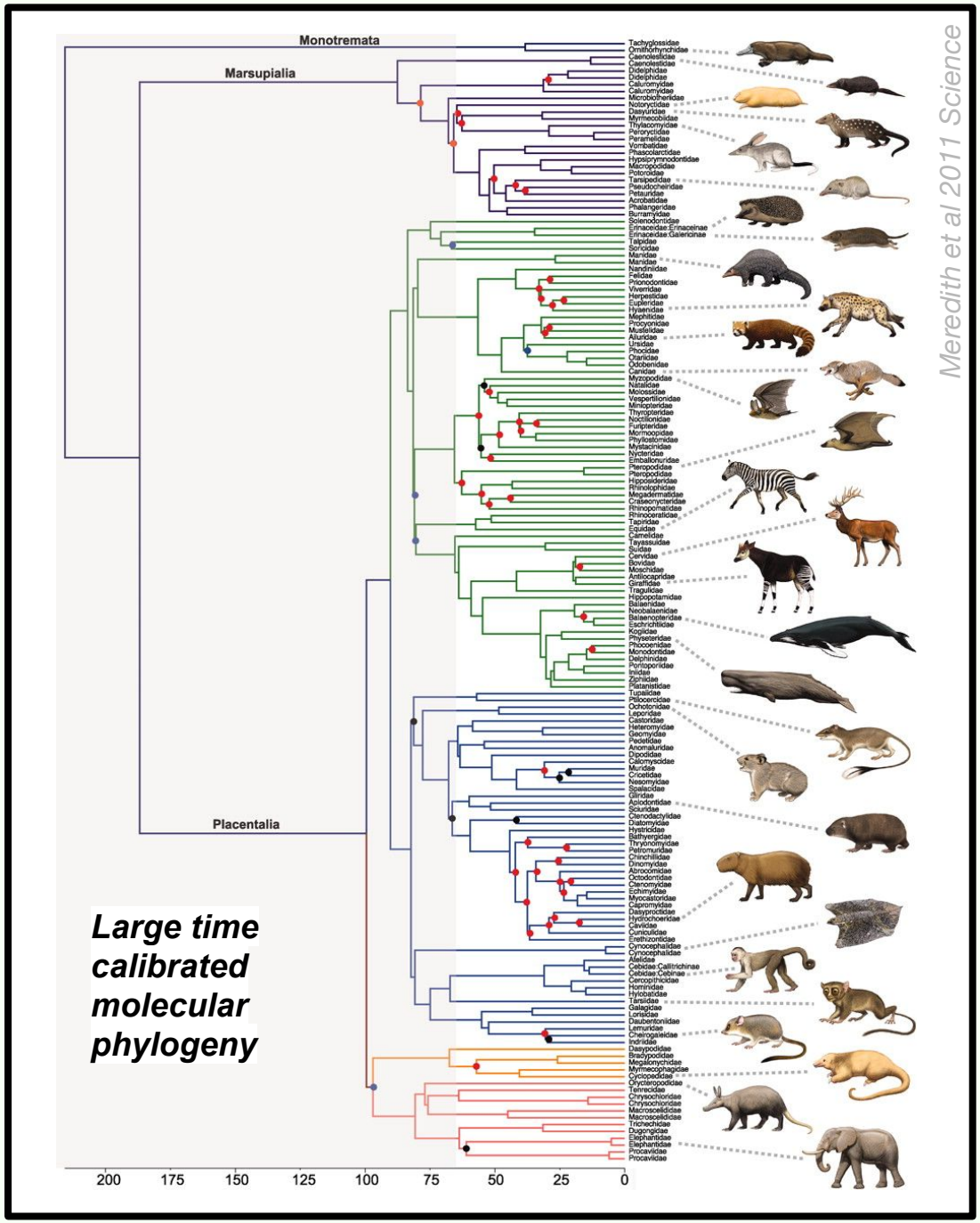
Cohen, K.M., Finney, S.C., Gibbard, P.L., and Fan, J.-X., 2013, The ICS International Chronostratigraphic Chart: Episodes v. 36, no. 3, p. 199–204 (updated 2017, v. 2, http://www.stratigraphy.org/index.php/ics-chart-timescale; accessed May 2018).

Gradstein, F.M, Ogg, J.G., Schmitz, M.D., et al., 2012, The Geologic Time Scale 2012: Boston, USA, Elsevier, https://doi.org/10.1016/B978-0-444-59425-9.00004-4.

Previous versions of the time scale and previously published papers about the time scale and its evolution are posted to http://www.geosociety.org/timescale.

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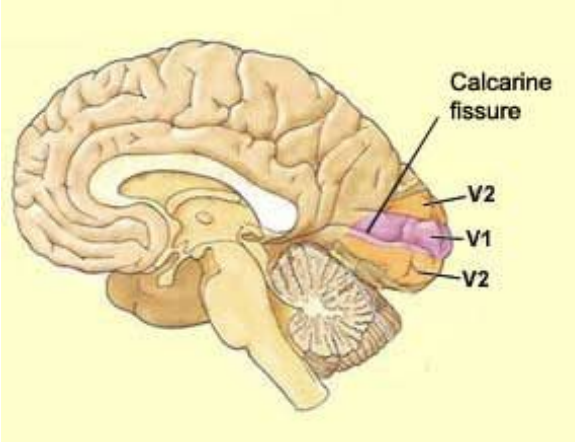
Mammal Evolution



Primates



Primates are a diverse group of mammals with relatively large brains, specialized for vision, and tactile grasping feet and hands





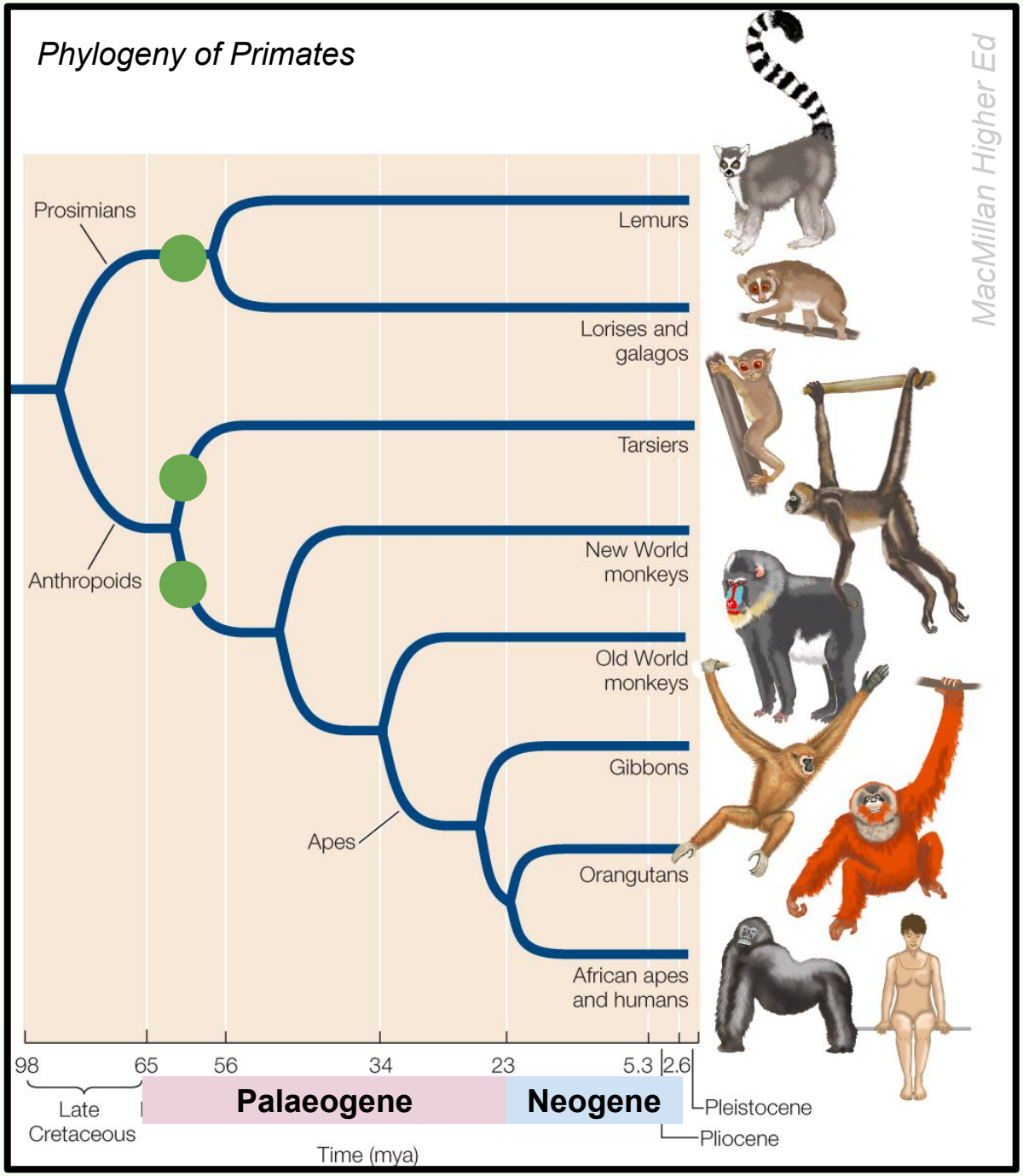
First diversified in Palaeogene, including the lemur-like *Darwinus*

Primates

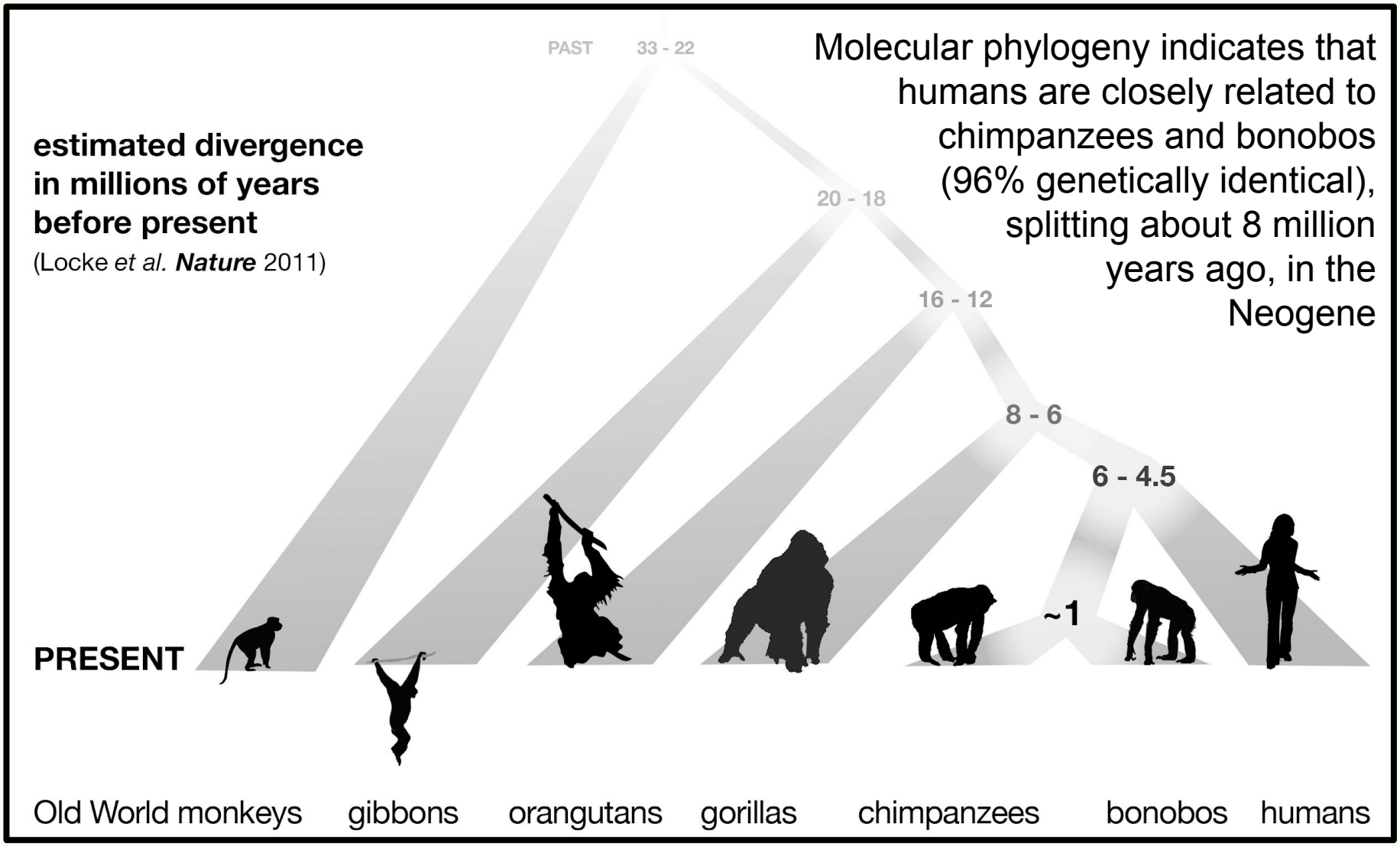
Primates have a long evolutionary history, divergences taking place in Palaeogene

Group comprises of Lemuriformes, Tarsiiformes, and Simiiformes

Primary specializations of primates relate to vision and thumbs



Hominidae


estimated divergence
in millions of years
before present
(Locke et al. Nature 2011)
Molecular phylogeny indicates that humans are closely related to chimpanzees and bonobos (96% genetically identical), splitting about 8 million years ago, in the Neogene
PAST
33 - 22
20 - 18
16 - 12
8 - 6
6 - 4.5
~1
PRESENT
Old World monkeys
gibbons
orangutans
gorillas
chimpanzees
bonobos
humans

Humans and Chimpanzees

The genus ***Pan*** has two living species, chimpanzee and bonobo





Both form highly social groups
Chimpanzees are larger and more aggressive, male dominated
Bonobos are smaller, more peaceful, and female dominated

There is one living species of Homo, ***Homo sapiens***



Compared to other apes:

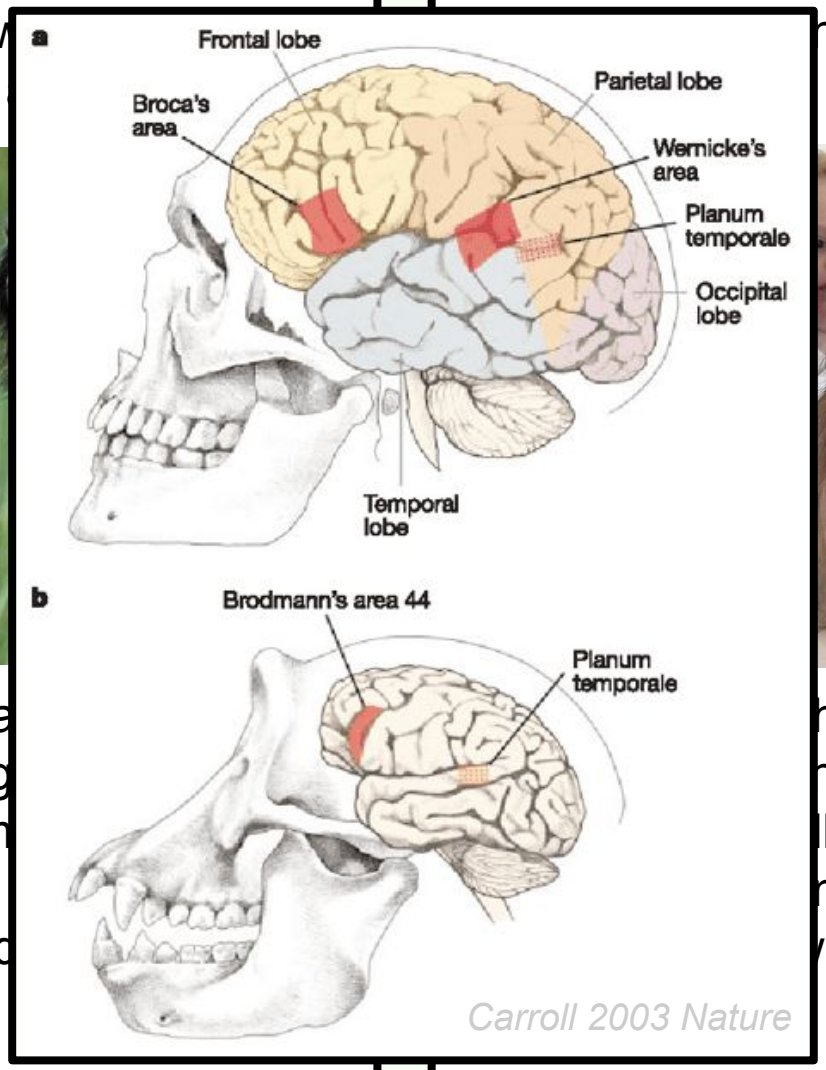
- Larger brains and skulls
- Bipedal walking gait
- Protruding nose and chin
- Smaller jaws and teeth

Humans and Chimpanzees

The genus ***Pan*** has tw
species, chimpanzee







Carroll 2003 Nature

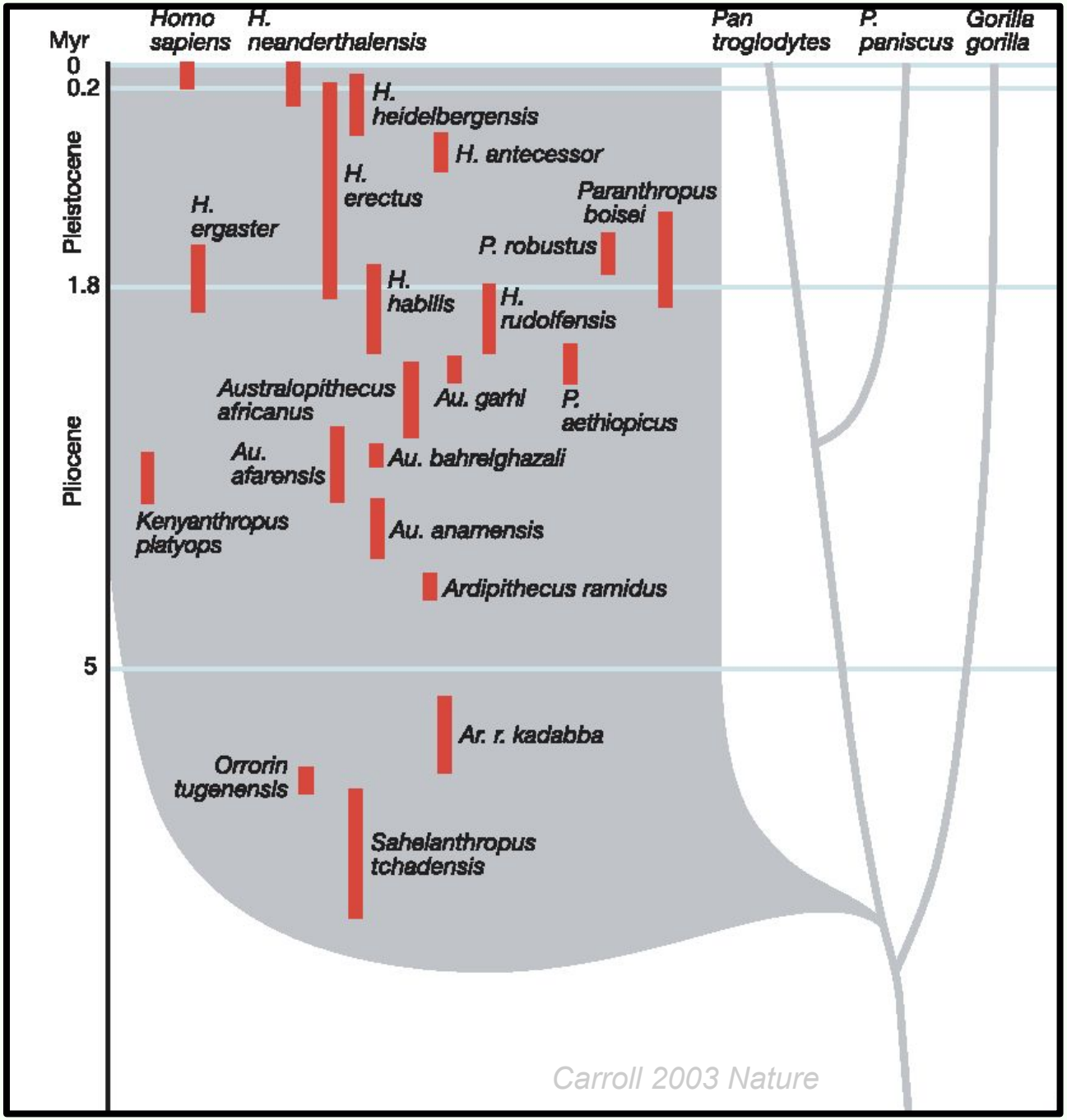
ng species of Homo,



Both form highly socia
Chimpanzees are larg
aggressive, male dom
Bonobos are smaller,
and female dominated

her apes:
ns and skulls
lking gait
nose and chin
s and teeth

Timescale & Fossil Evidence

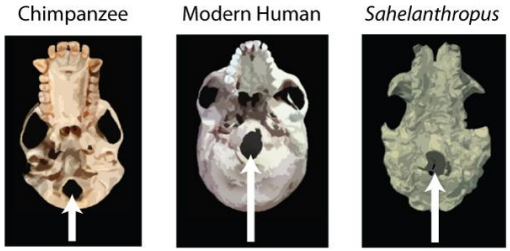


Early Apes



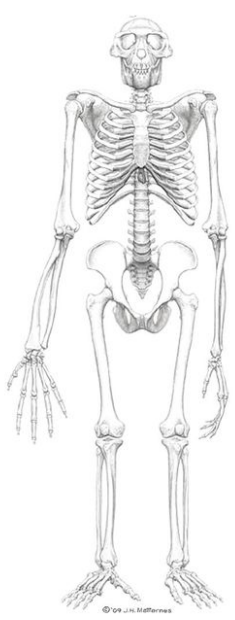
Sahelanthropus, 7 mya from Chad
A small brain and short canine teeth

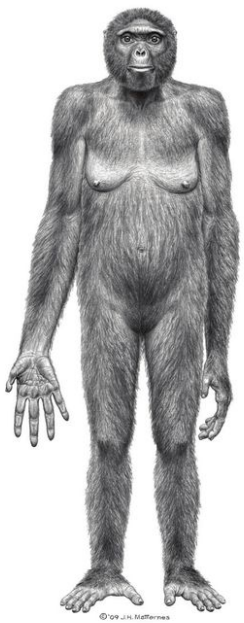
Potentially walked more upright



Su 2013 Nature Ed.







Ardipithecus, ~5 mya from Ethiopia
More reduced canines, walking-like hips but opposable toes, and flatter feet
Mosaic features of humans and chimps

B Asfaw, Boisserie 2010 An. d'Ethiopie

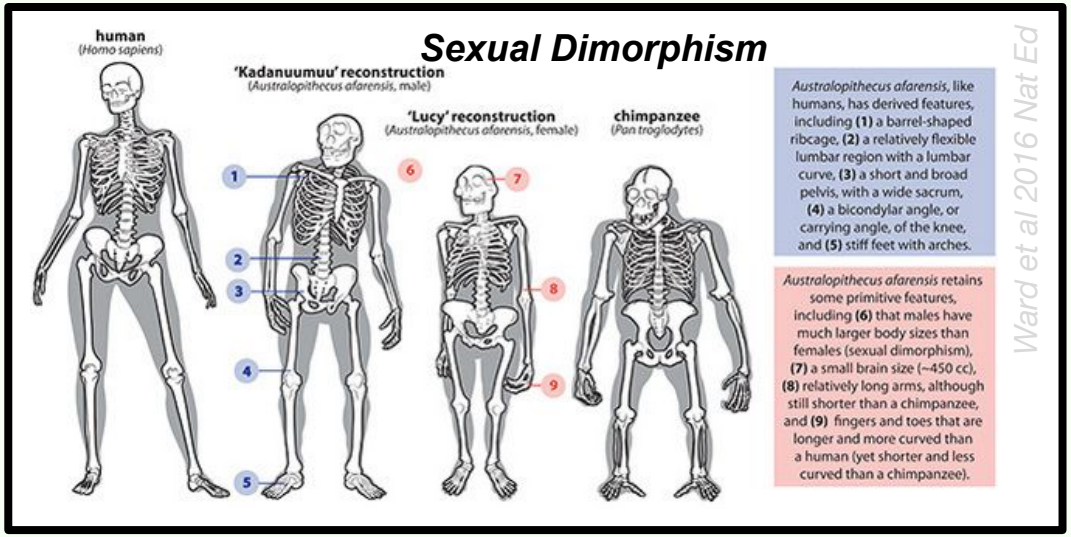
Australopithecines



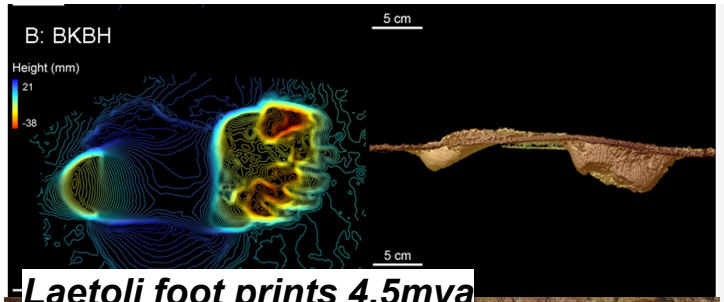


ESRF

Brain volume comparable to chimpanzee



Ward et al 2016 Nat Ed



Laetoli foot prints 4.5mya



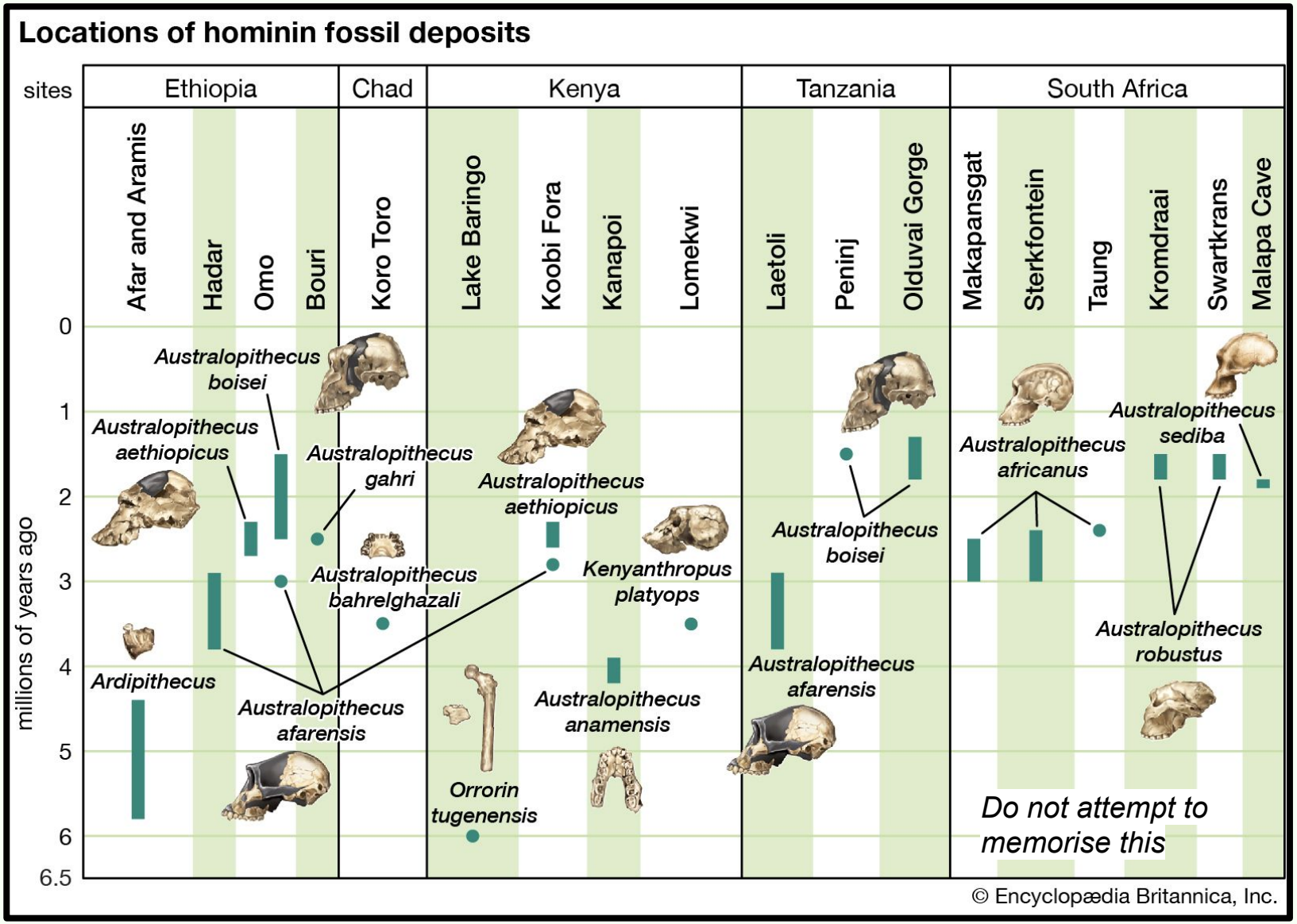


Raichlan et al 2010 PlosOne

Leakey Foundation

Foot prints and models demonstrate **bipedalism** in *Australopithecus*

Australopithecines



Australopithecus

Australopithecus

meaning "Southern Ape"
4 million to 2 million years ago
There are multiple species that make up this genus

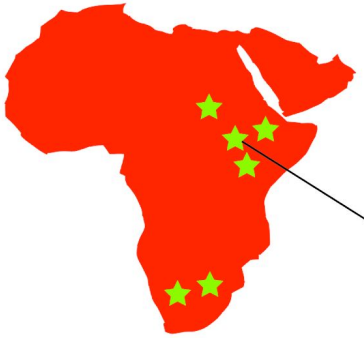


The most complete specimen of *Australopithecus* is nicknamed Lucy. The lower limbs and hip bones demonstrated *Australopithecus afarensis* was a "habitual biped" or walked like we do.

Brain volume
~400 cubic cm

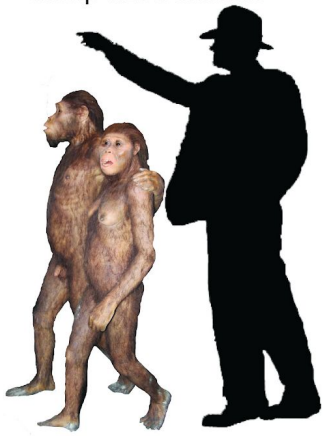


Lucy's skull shows she had a chimp-sized brain supported by legs like ours.



Australopithecus and its close relatives lived in Africa as did all our ancient ancestors. Stars show locations where fossils have been excavated. Lucy was found in Ethiopia.

Lucy and her relatives were relatively small, chimp sized animals

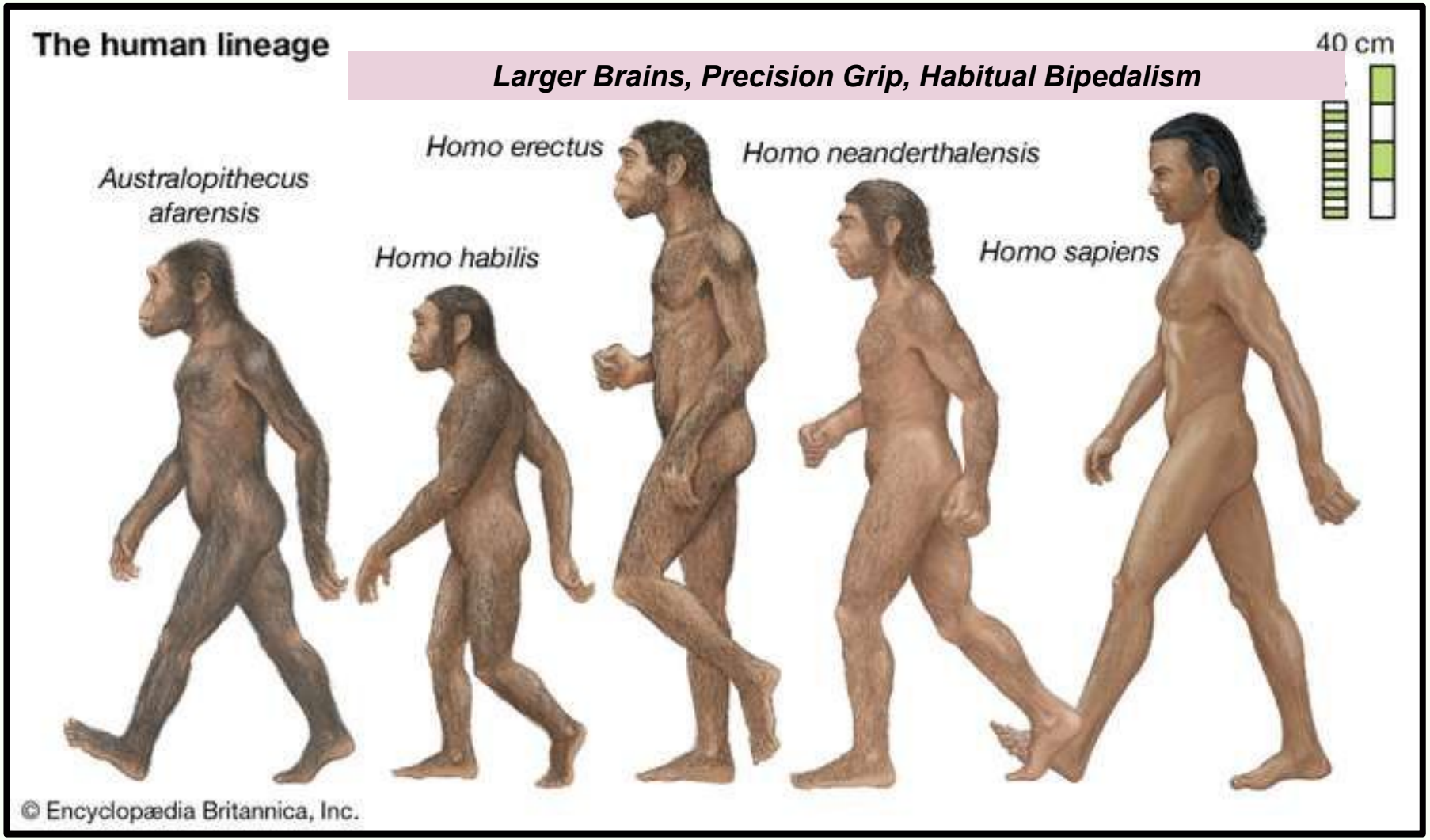


These footprints from Tanzania show Australopithecus-sized animals with feet built for walking on two-legs crossed the volcanic ash field 3.6 million years ago.



PastTime.org

Genus *Homo*


The human lineage
Larger Brains, Precision Grip, Habitual Bipedalism
40 cm
Australopithecus afarensis
Homo habilis
Homo erectus
Homo neanderthalensis
Homo sapiens
© Encyclopædia Britannica, Inc.

Homo erectus

Homo erectus

meaning "Upright man"

1.8 million to 140,000 years ago

Best-known specimen of *Homo erectus* is this specimen from Kenya nicknamed "Turkana Boy" (Also called Homo ergaster).

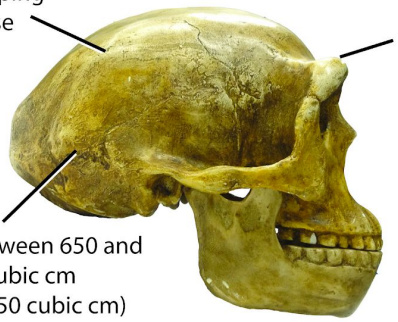


Weird forward-facing shoulder joint

Basically modern legs and hips

Low, sloping braincase

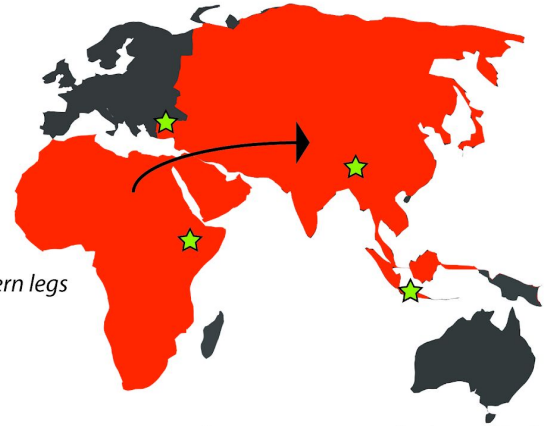
Pronounced brow ridges



Brain size between 650 and 1250 cubic cm (ours is ~1350 cubic cm)

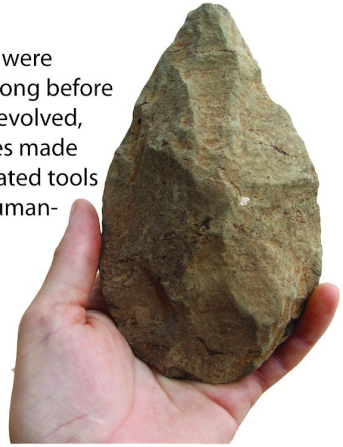


Unlike *Australopithecus, H. erectus* was the size of modern humans but built a little differently around the soulder



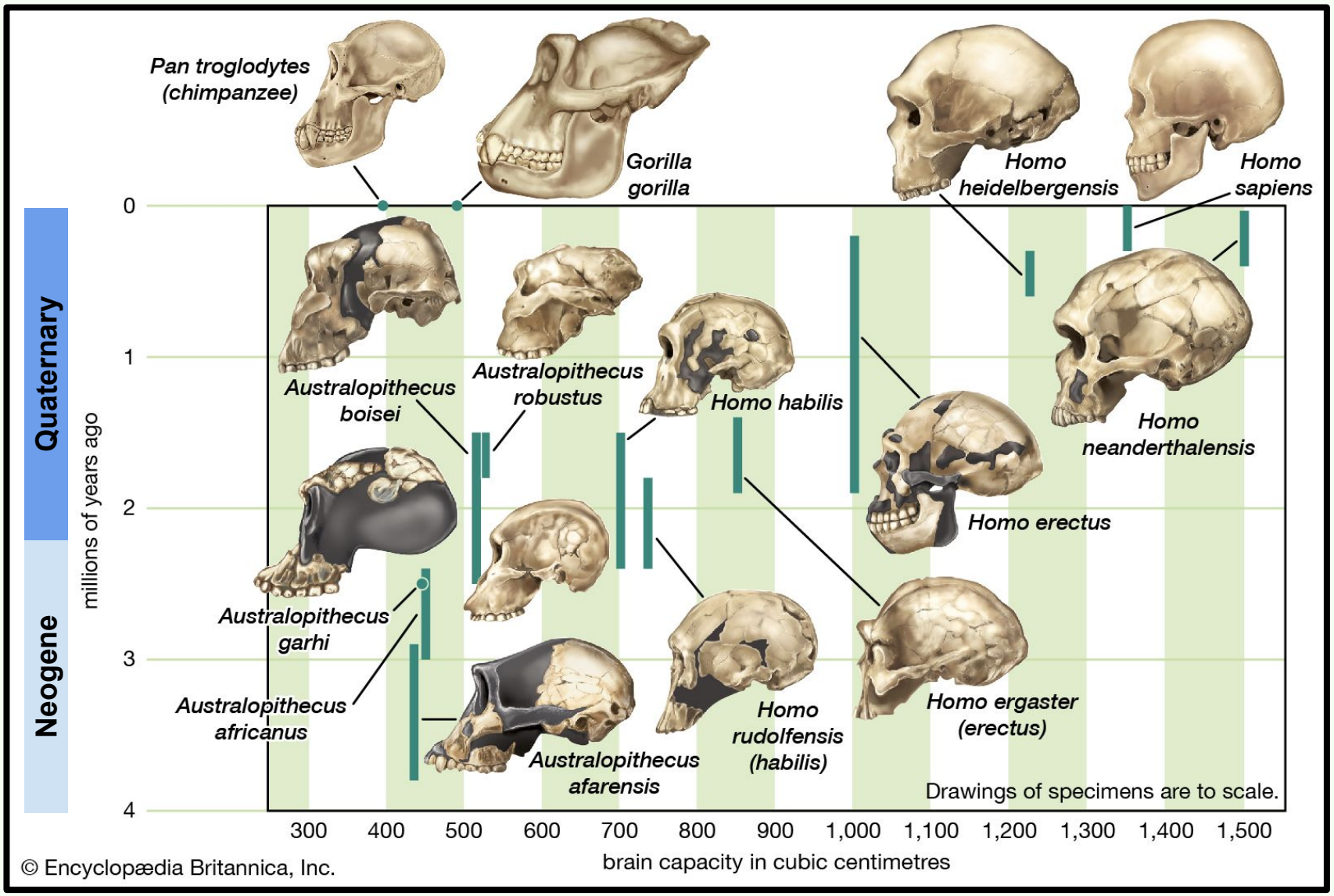
Homo erectus was our first ancestor to make it very far from Africa. The species evolved in Africa then expanded to Asia.

Our ancestors were making tools long before *Homo erectus* evolved, but this species made more complicated tools than earlier human-relatives.



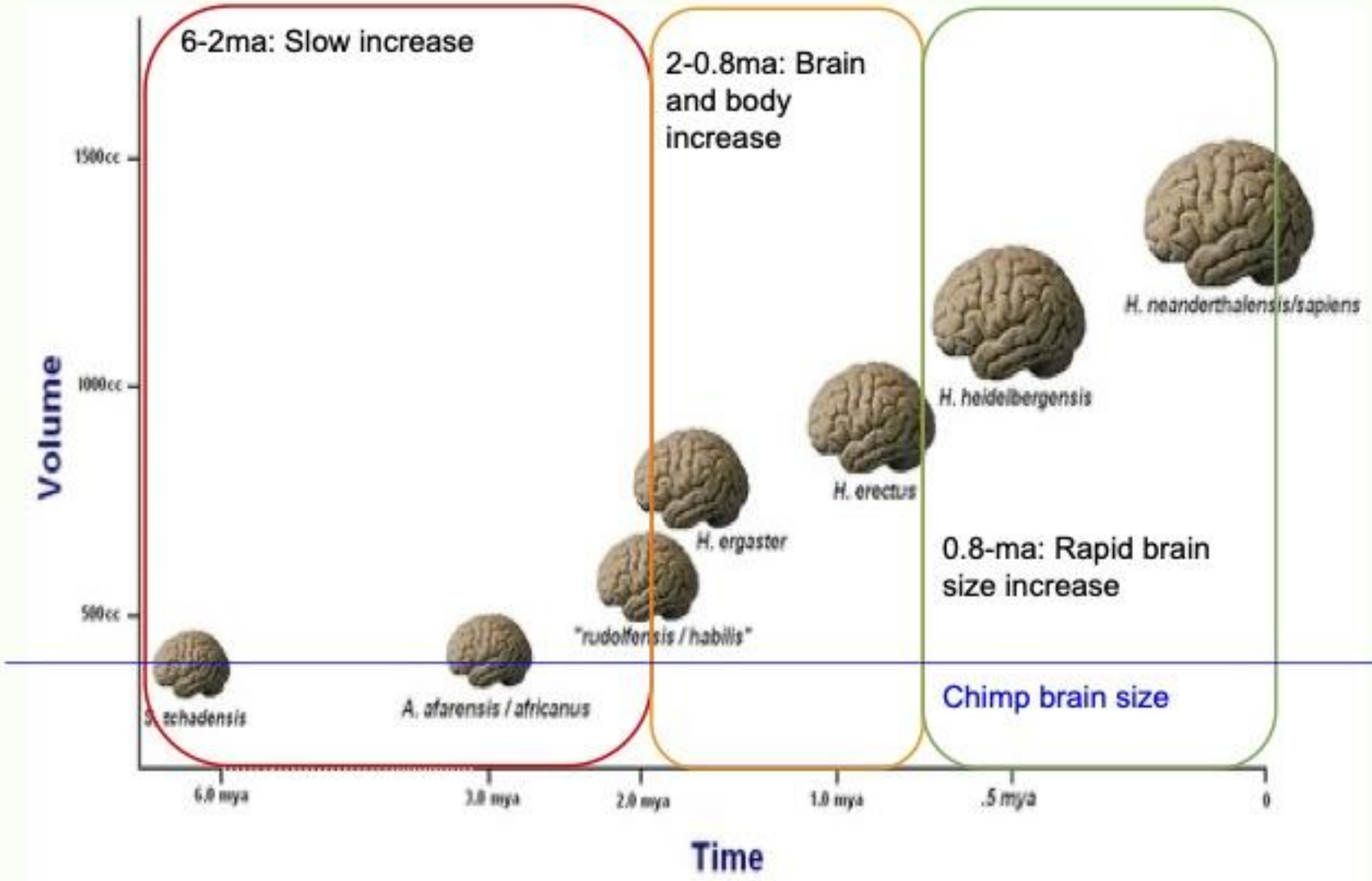
PastTime.org

Homo


Pan troglodytes (chimpanzee)
Gorilla gorilla
Homo heidelbergensis
Homo sapiens
Australopithecus boisei
Australopithecus robustus
Homo habilis
Homo neanderthalensis
Homo erectus
Australopithecus garhi
Australopithecus africanus
Australopithecus afarensis
Homo rudolfensis (habilis)
Homo ergaster (erectus)
Quaternary
Neogene
millions of years ago
0
1
2
3
4
300
400
500
600
700
800
900
1,000
1,100
1,200
1,300
1,400
1,500
brain capacity in cubic centimetres
Drawings of specimens are to scale.
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Hominin brain evolution

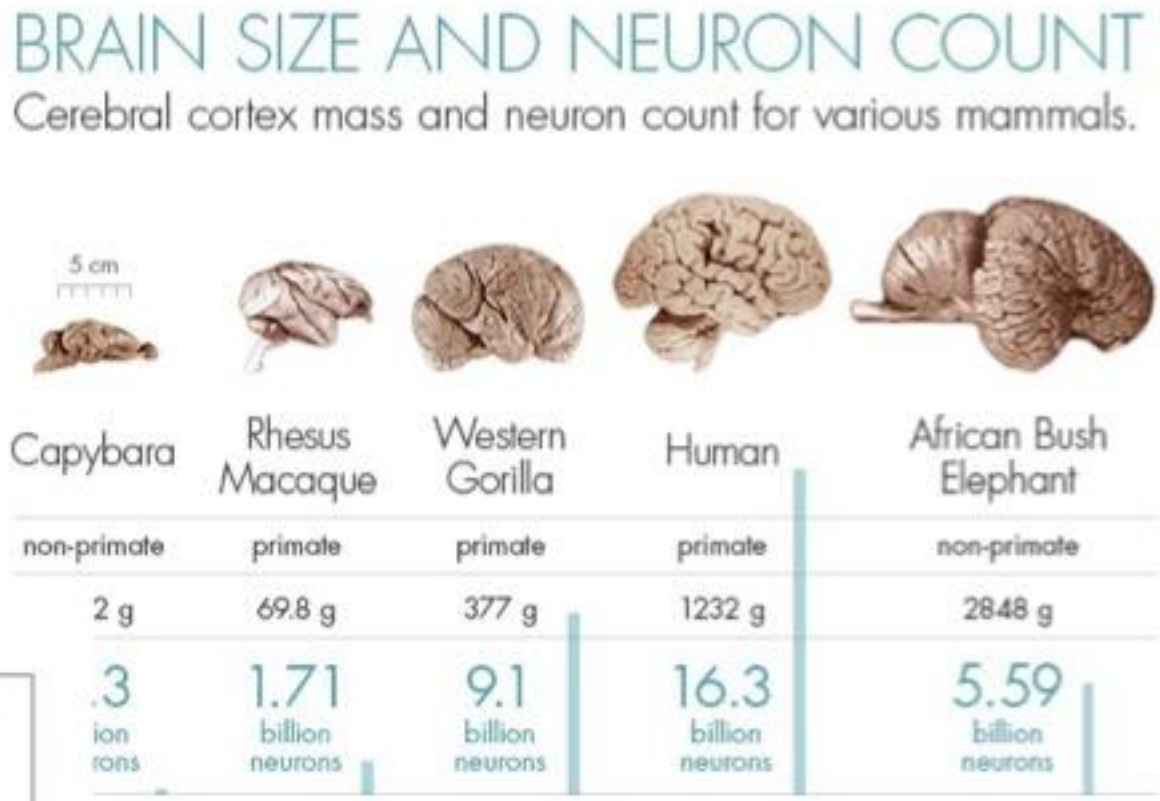
Brain size in hominins increased slowly at first, but underwent a '**brain boom**' in the last 800kyr

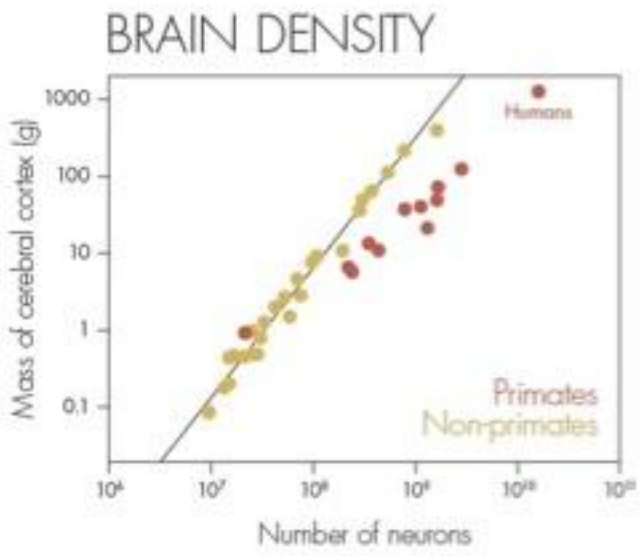


Size isn't everything

Despite having a smaller brain than elephants or dolphins, humans have **far more neurons**

… particularly in the **cortex** which is important for cognitive function

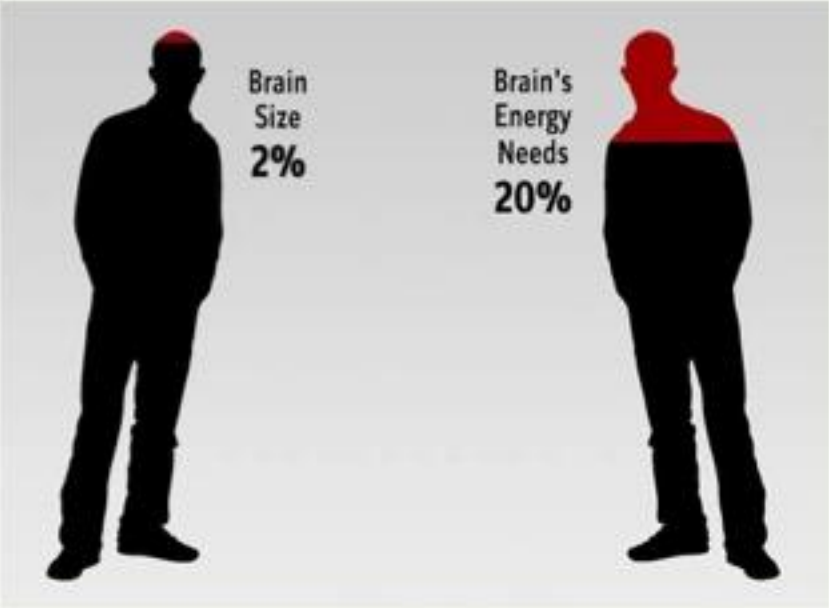


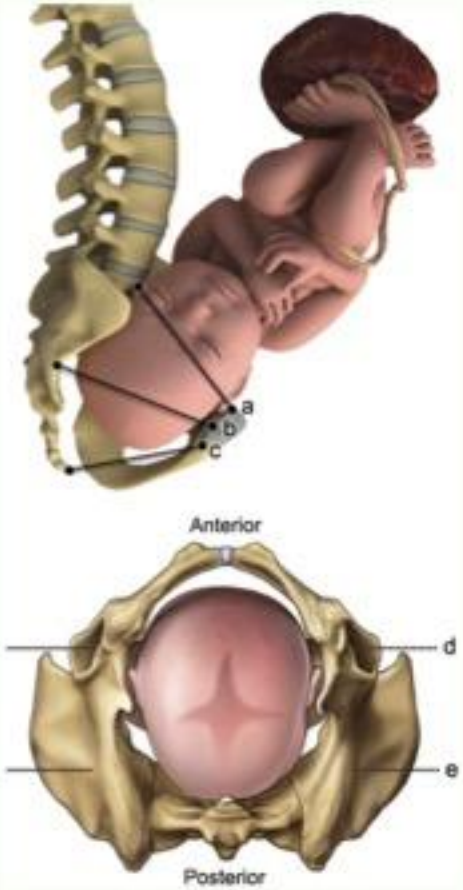


Primates achieve this with a much denser cortex

Herculano-Houzel et. al. (2014)

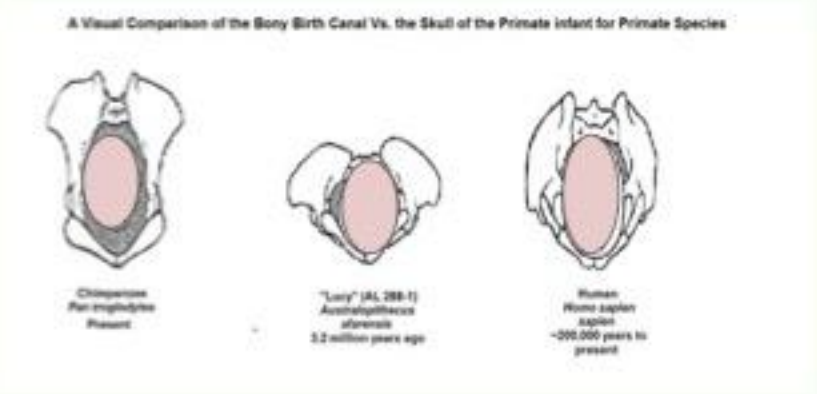
Problems with big brains





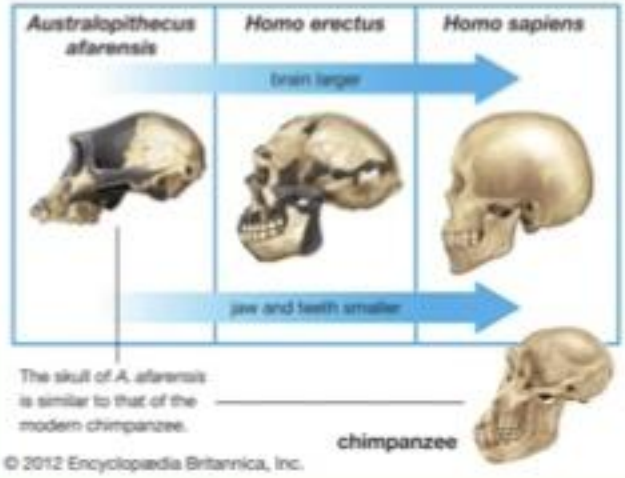
While big brains have many advantages, they are very **energetically expensive** to maintain.

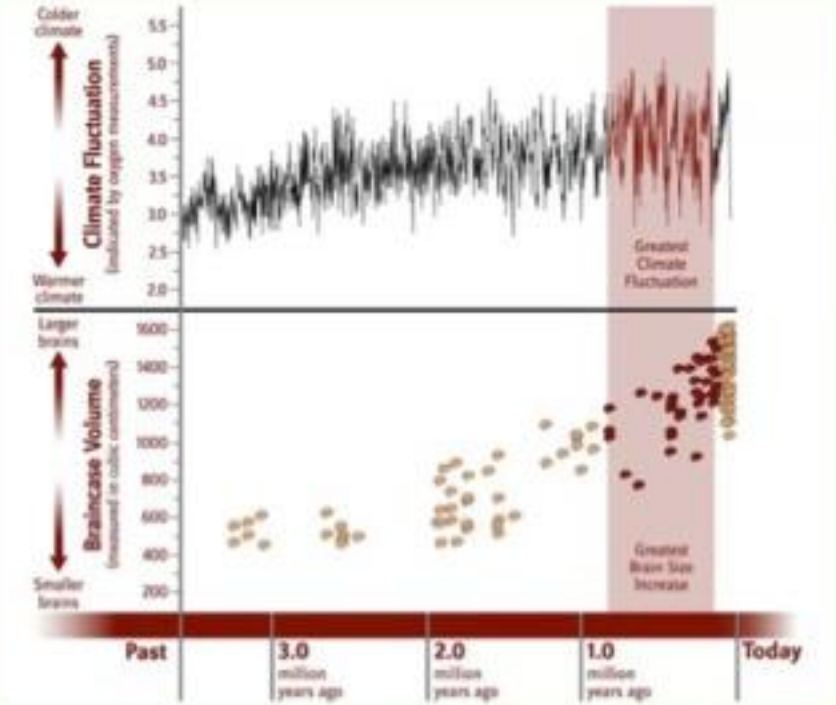
The **obstetrical dilemma** is a functional constraint on brain size based on hip width, which are narrower in bipedal species.

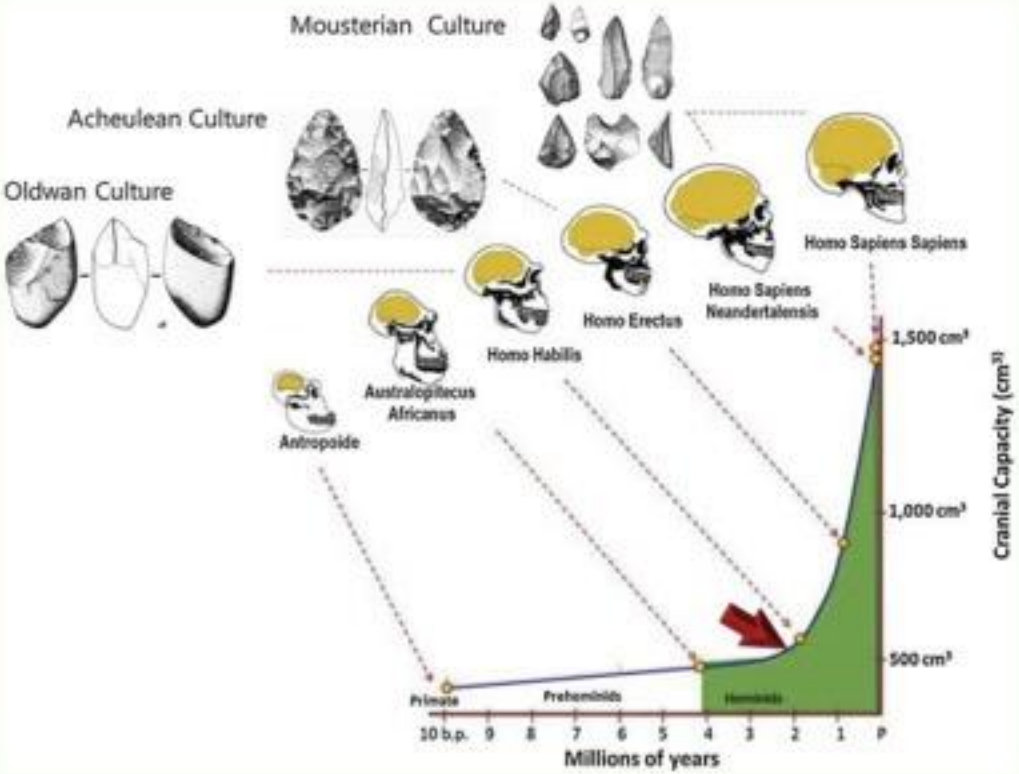


Why have a big brain?

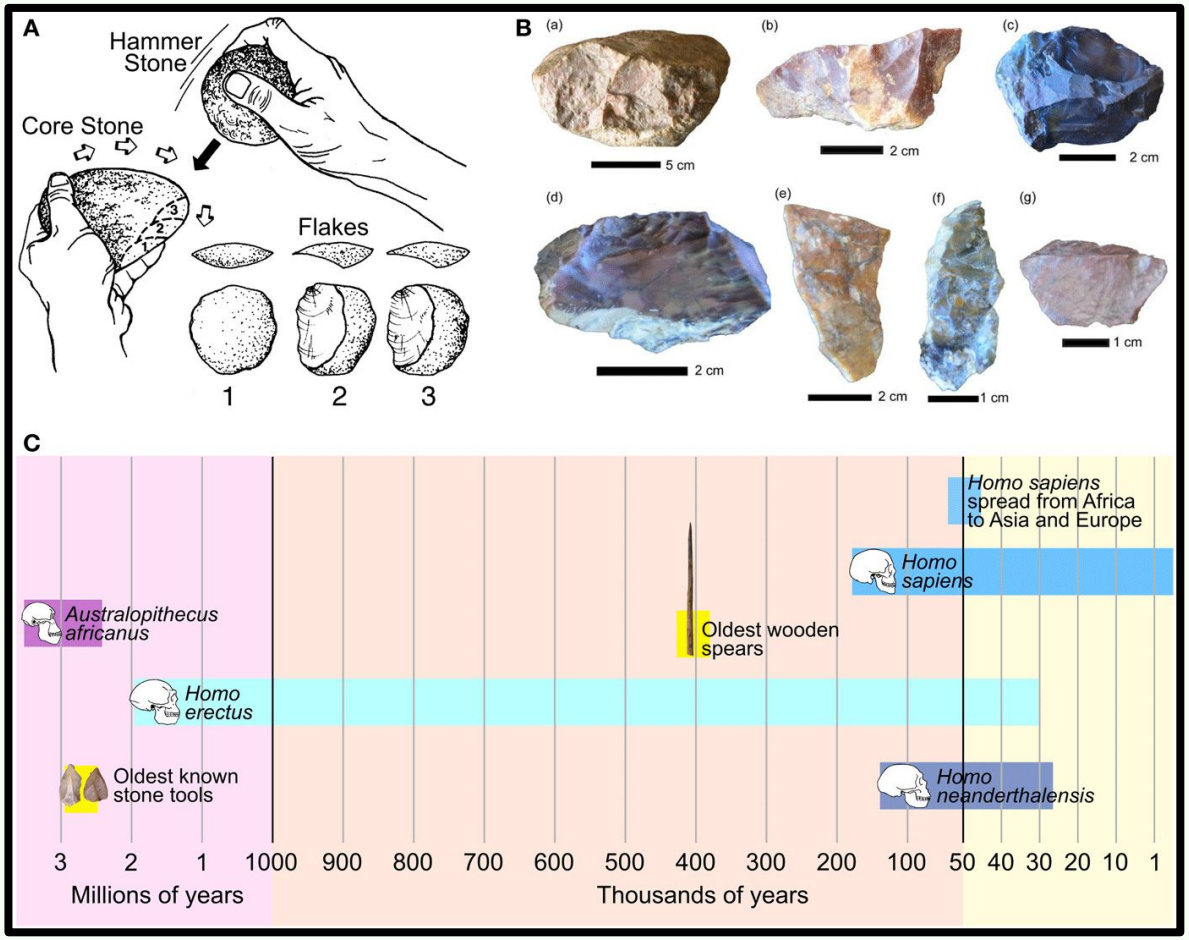
Possible explanations for the increasing brain size include **social networks, tools use, or an increasingly variable environment**



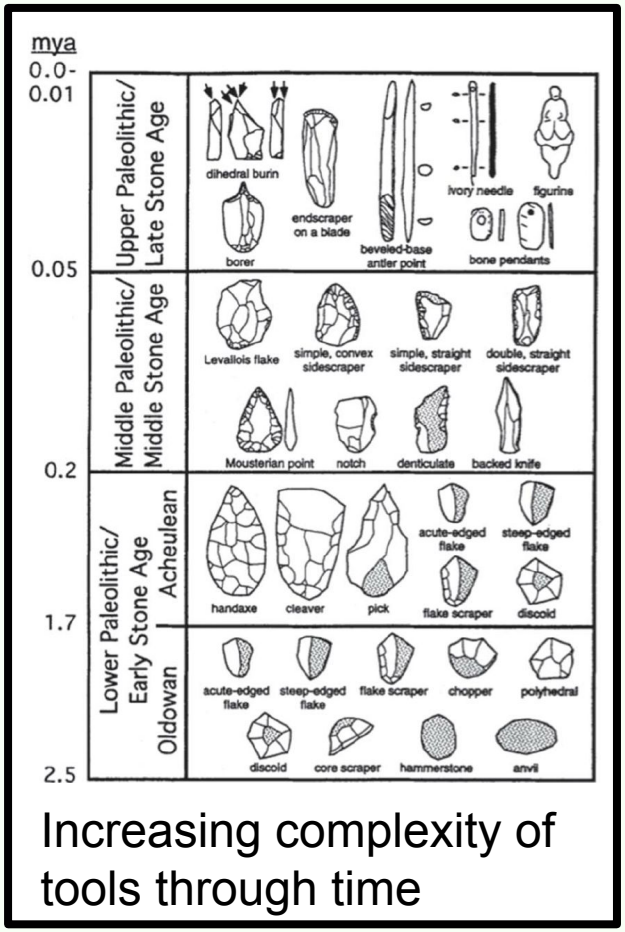




Tool Use



Kassuba et al 2014 Frontiers Young Minds

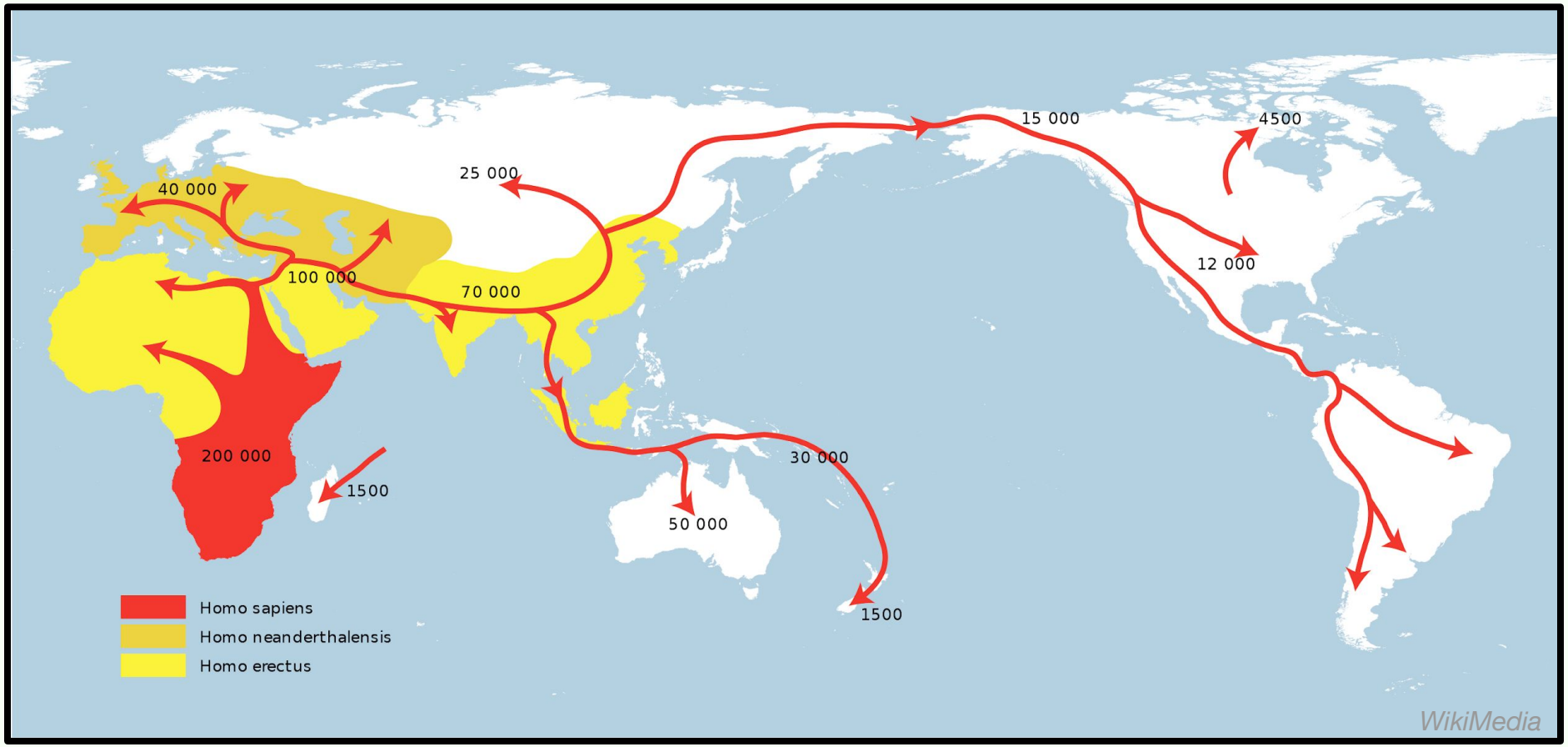


Increasing complexity of tools through time

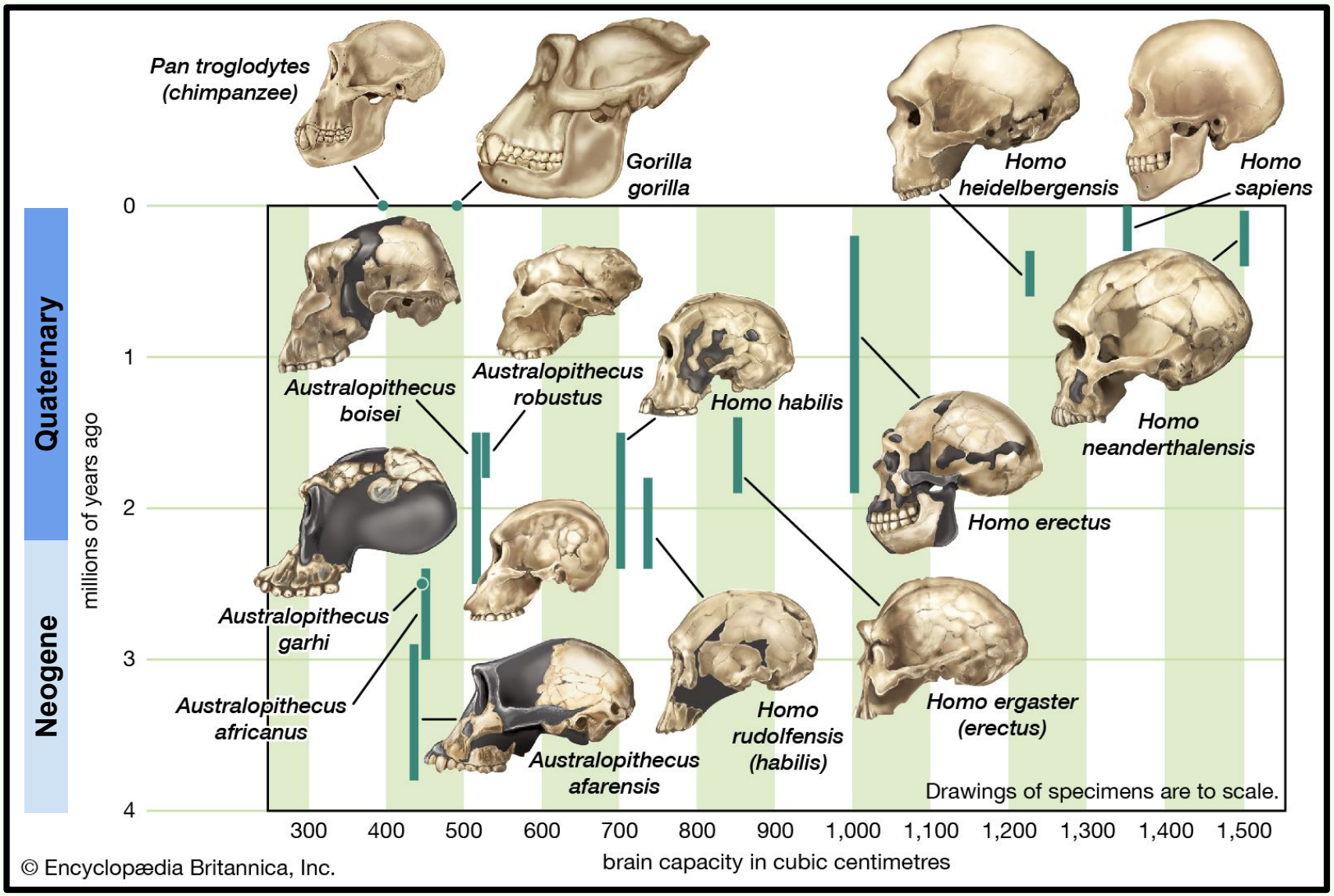
Neco et al 2015 Ciência & Ambiente

Homo

Homo erectus migrated to Eurasia first, then *H. neanderthalensis*, then *H. sapiens*



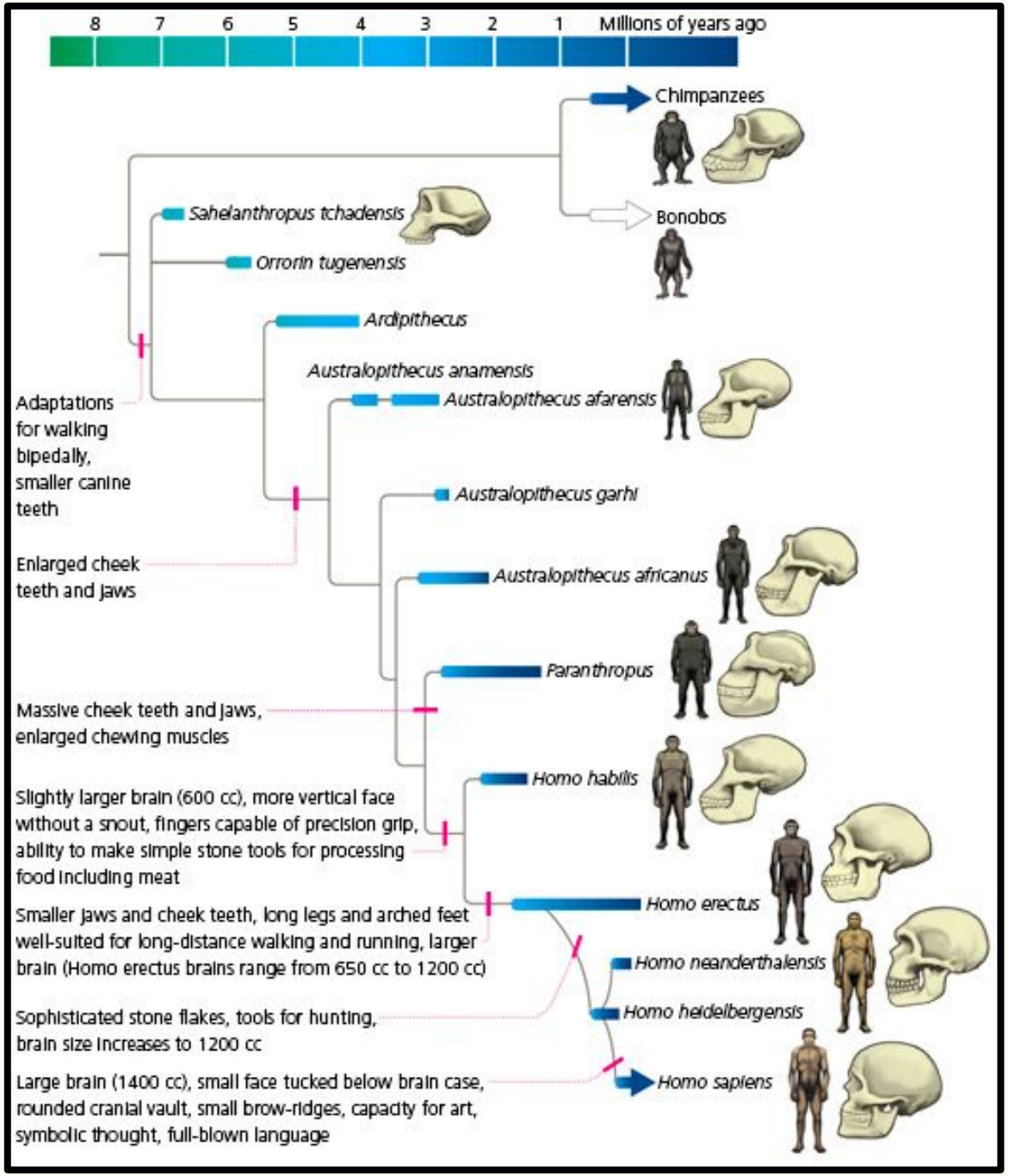
Homo


Pan troglodytes (chimpanzee)
Gorilla gorilla
Homo heidelbergensis
Homo sapiens
Australopithecus boisei
Australopithecus robustus
Homo habilis
Homo neanderthalensis
Homo erectus
Australopithecus garhi
Australopithecus africanus
Australopithecus afarensis
Homo rudolfensis (habilis)
Homo ergaster (erectus)
Quaternary
Neogene
millions of years ago
0
1
2
3
4
300
400
500
600
700
800
900
1,000
1,100
1,200
1,300
1,400
1,500
brain capacity in cubic centimetres
Drawings of specimens are to scale.
© Encyclopædia Britannica, Inc.

Synthesis

There is rich fossil evidence for the "Homo stem-lineage"

- Bipedalism
- Changes in teeth and jaws
- Increasing brain sizes
- Tool use and precision



The Age of Apes
Human Origins

Intended Learning Objective: Explain how fossil evidence reveals the origins and evolution of humans

https://docs.google.com/forms/d/16BeTLtCNvMwPIN8ybyZy44E4oZ0dgRtPHf-96nwlo5Q/edit#responses

Now take a look at the associated task/discussion

Robert Sansom, U. Manchester



Didier Descouens

The Age of Apes
Human Origins

Intended Learning Objective: Explain how fossil evidence reveals the origins and evolution of humans

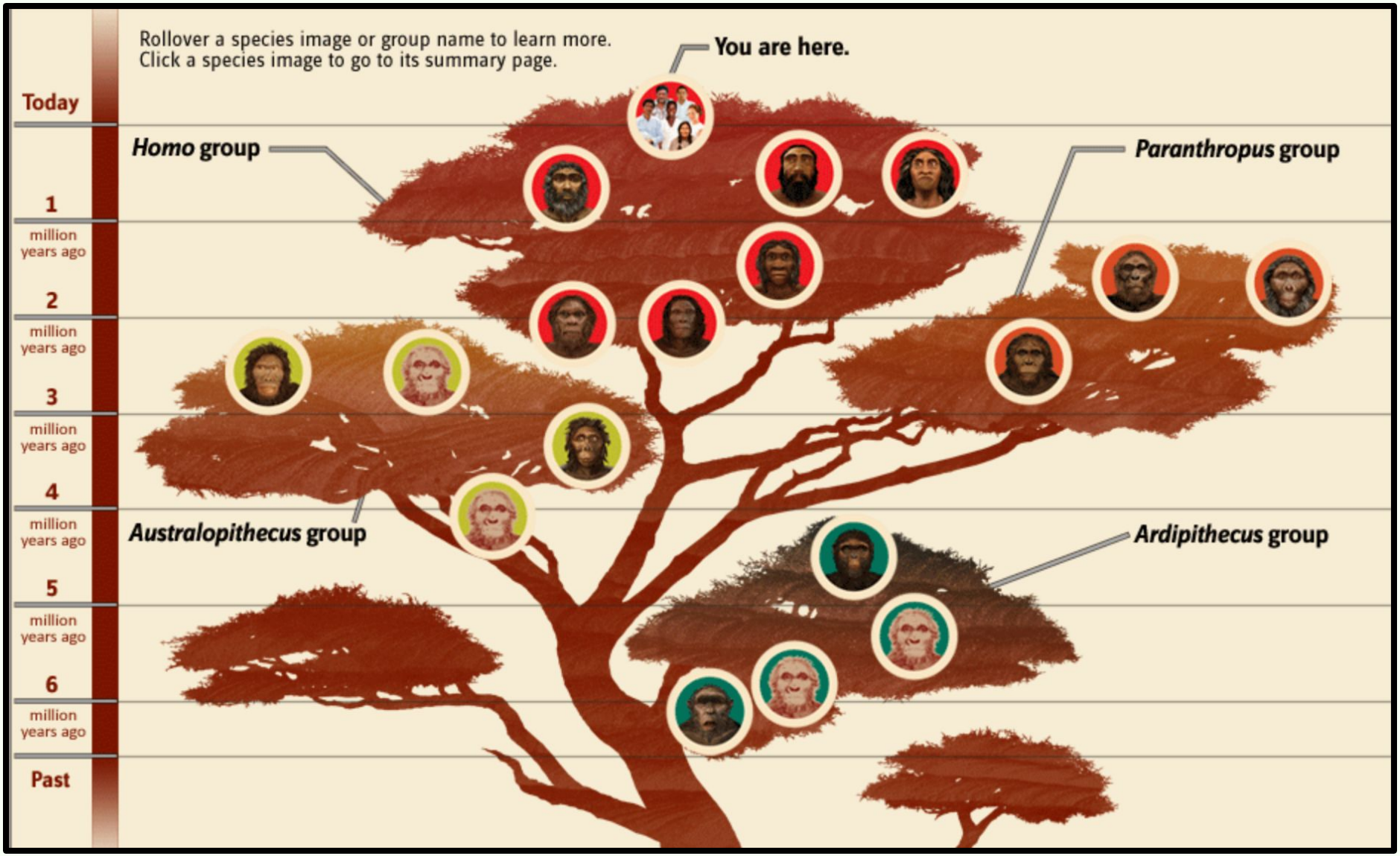
Associated Task: Explore the human family tree at this website from the Smithsonian Museum. Chose a taxon and an unanswered question - what data or findings would help us find the answers?

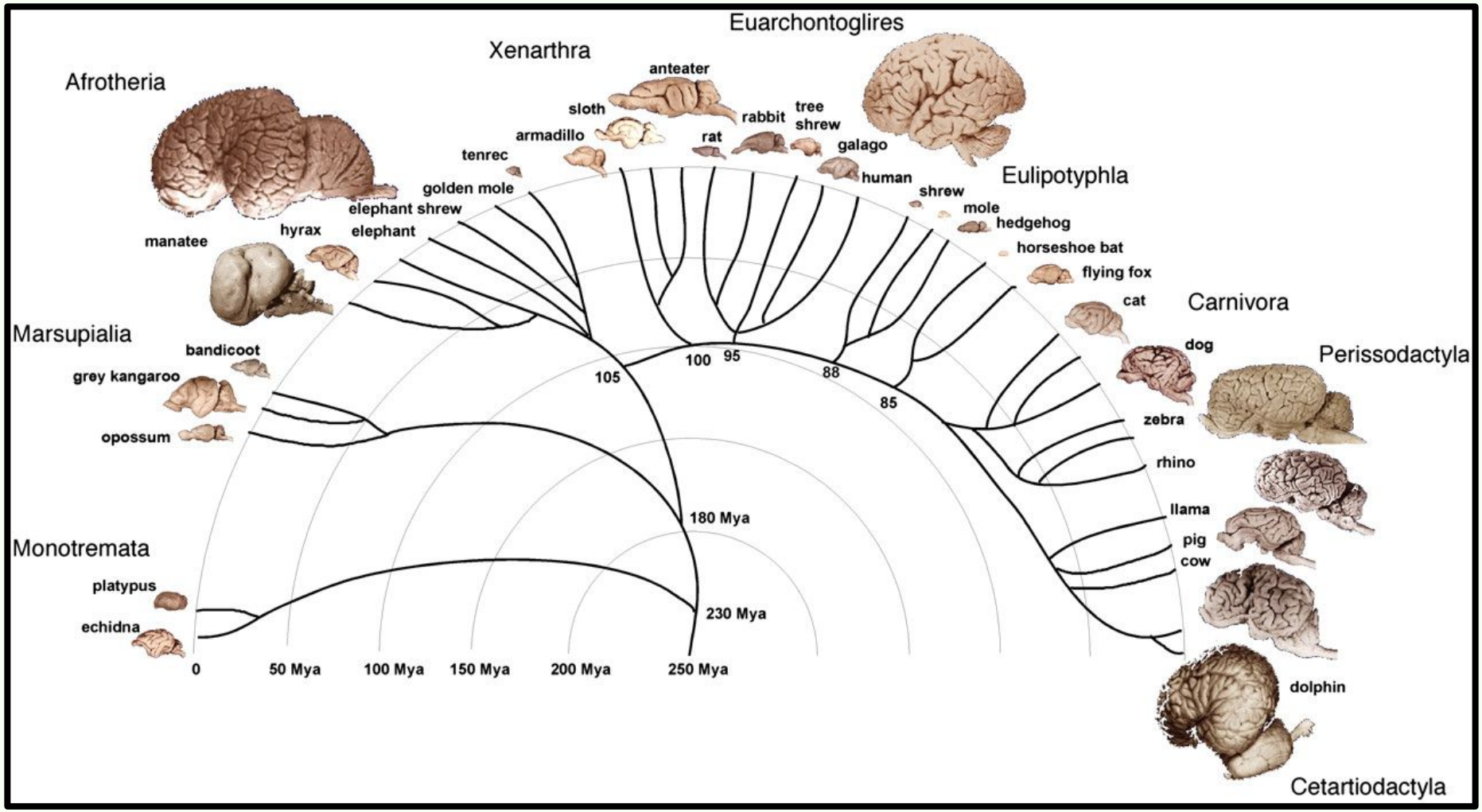
https://humanorigins.si.edu/evidence/human-family-tree

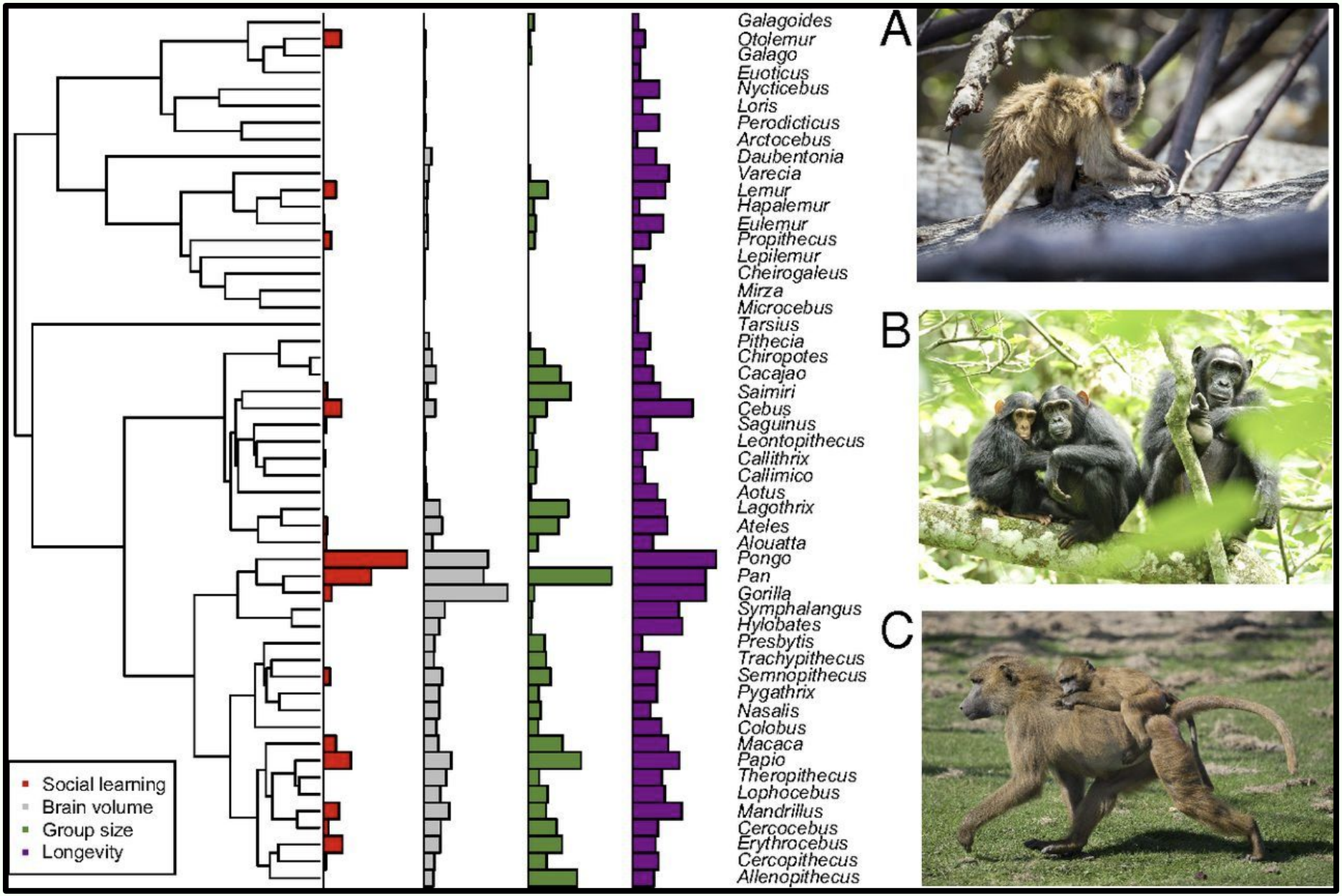
Robert Sansom, U. Manchester



Didier Descouens


Rollover a species image or group name to learn more.
Click a species image to go to its summary page.
You are here.
Today
Homo group
Paranthropus group
1 million years ago
2 million years ago
3 million years ago
4 million years ago
Australopithecus group
Ardipithecus group
5 million years ago
6 million years ago
Past


Euarchontoglires
Xenarthra
Afrotheria
anteater
sloth
armadillo
tenrec
rabbit
rat
tree shrew
galago
human
shrew
mole
hedgehog
horseshoe bat
flying fox
cat
golden mole
elephant shrew
elephant
hyrax
manatee
Eulipotyphla
Carnivora
Perissodactyla
Marsupialia
bandicoot
grey kangaroo
opossum
dog
zebra
rhino
llama
pig
cow
dolphin
105
100
95
88
85
180 Mya
230 Mya
Monotremata
platypus
echidna
0
50 Mya
100 Mya
150 Mya
200 Mya
250 Mya
Cetartiodactyla


Galagoides
Otolemur
Galago
Euoticus
Nycticebus
Loris
Perodicticus
Arctocebus
Daubentonia
Varecia
Lemur
Hapalemur
Eulemur
Propithecus
Lepilemur
Cheirogaleus
Mirza
Microcebus
Tarsius
Pithecia
Chiropotes
Cacajao
Saimiri
Cebus
Saguinus
Leontopithecus
Callithrix
Callimico
Aotus
Lagothrix
Ateles
Alouatta
Pongo
Pan
Gorilla
Symphalangus
Hylobates
Presbytis
Trachypithecus
Semnopithecus
Pygathrix
Nasalis
Colobus
Macaca
Papio
Theropithecus
Lophocebus
Mandrillus
Cercocebus
Erythrocebus
Cercopithecus
Allenopithecus
Social learning
Brain volume
Group size
Longevity
A
B
C

Part 10
The Age of Apes

Who are we?



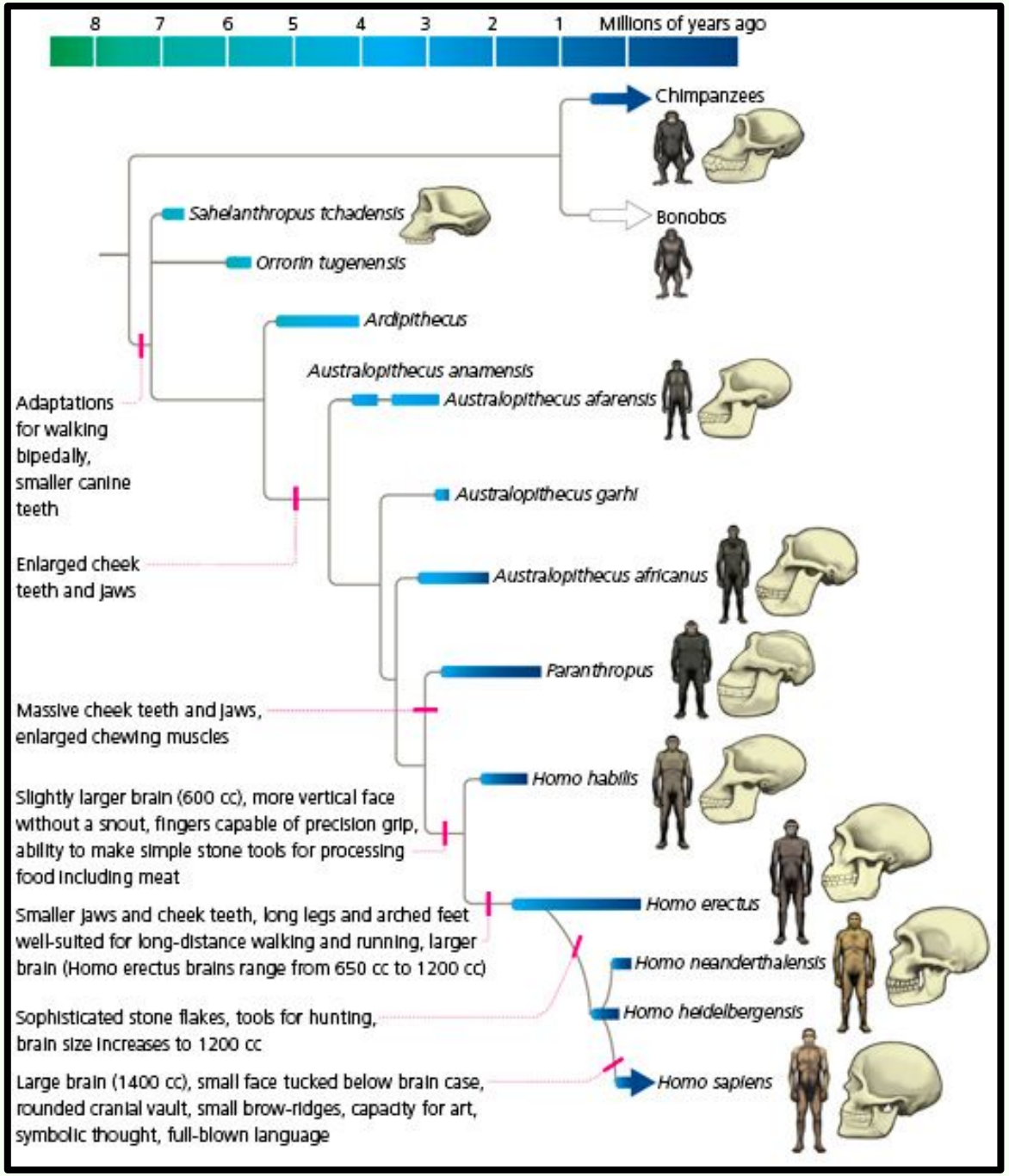
Didier Descouens

EART11100: How to Grow a Planet
Robert Sansom
University of Manchester

Synthesis

There is rich fossil evidence for the "Homo stem-lineage"

- Bipedalism
- Changes in teeth and jaws
- Increasing brain sizes
- Tool use and precision



Who are we?



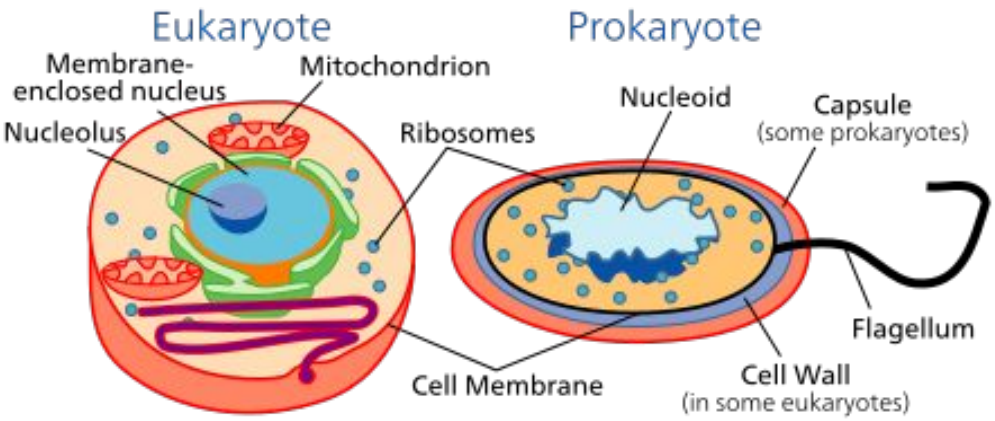
Didier Descouens

Classification of Life: Domain

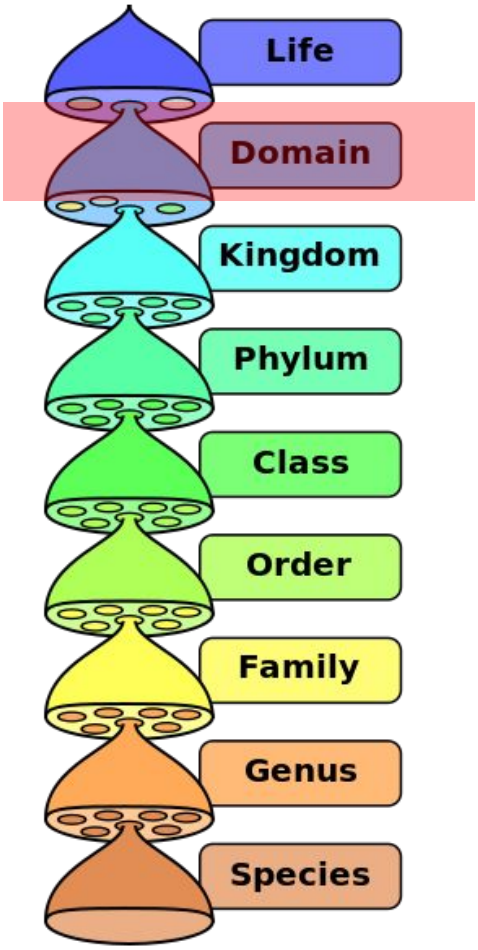
Bacteria:
unicellular prokaryotes with fatty ester lipids

Archaea:
unicellular prokaryotes with isoprene ether lipids

Eukaryotes:
have a cell nucleus and membrane bound organelles, can be multicellular



Taxonomy:
Hierarchical Classification



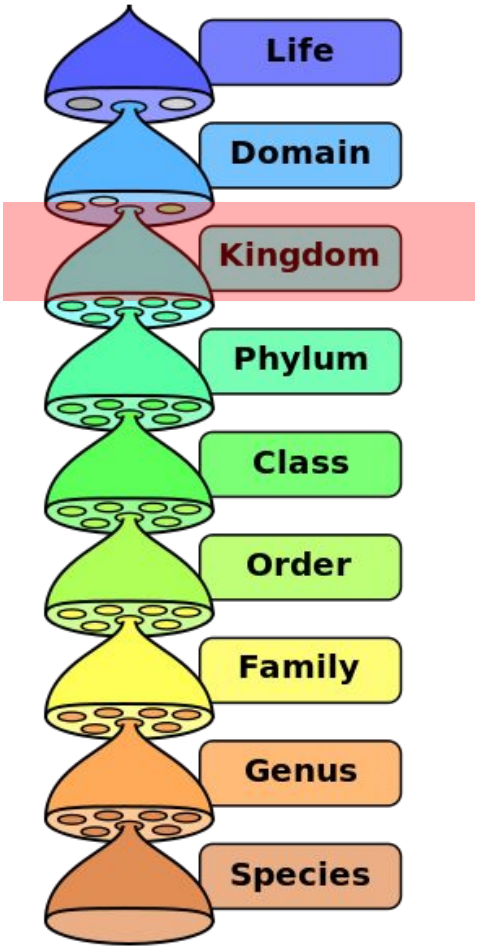
Classification of Life: Kingdom

Animalia (Metazoa): What is an animal?
Multicellular heterotrophs without a cell wall
Motility: they move, at least in part of the life cycle






Taxonomy:
Hierarchical Classification



Vertebrates

Jawless Vertebrates (agnathans)





Jawed Vertebrates (gnathostomes)





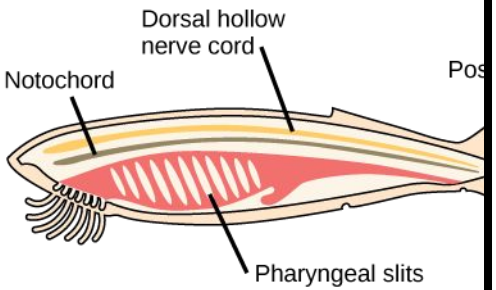
Agnathans

Gnathostomes

Vertebrates

Chordates

Vertebrates are in the phylum **Chordata**

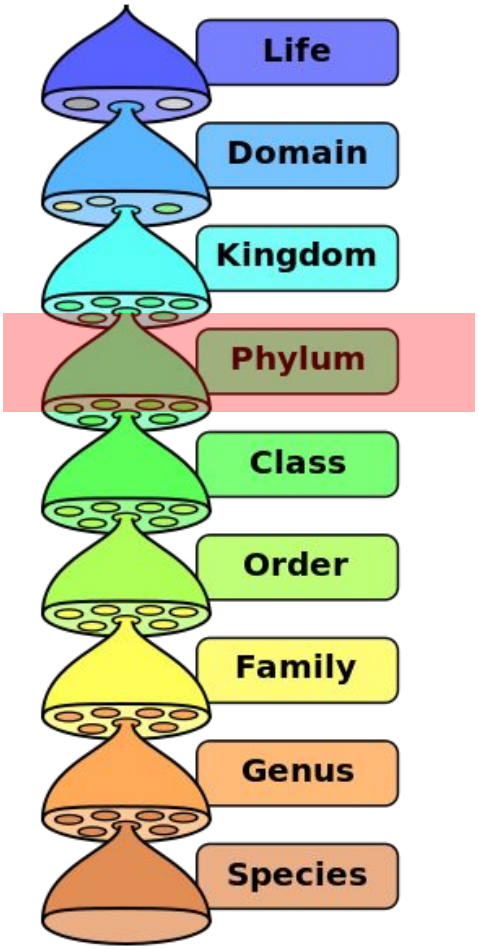


Chordates have:

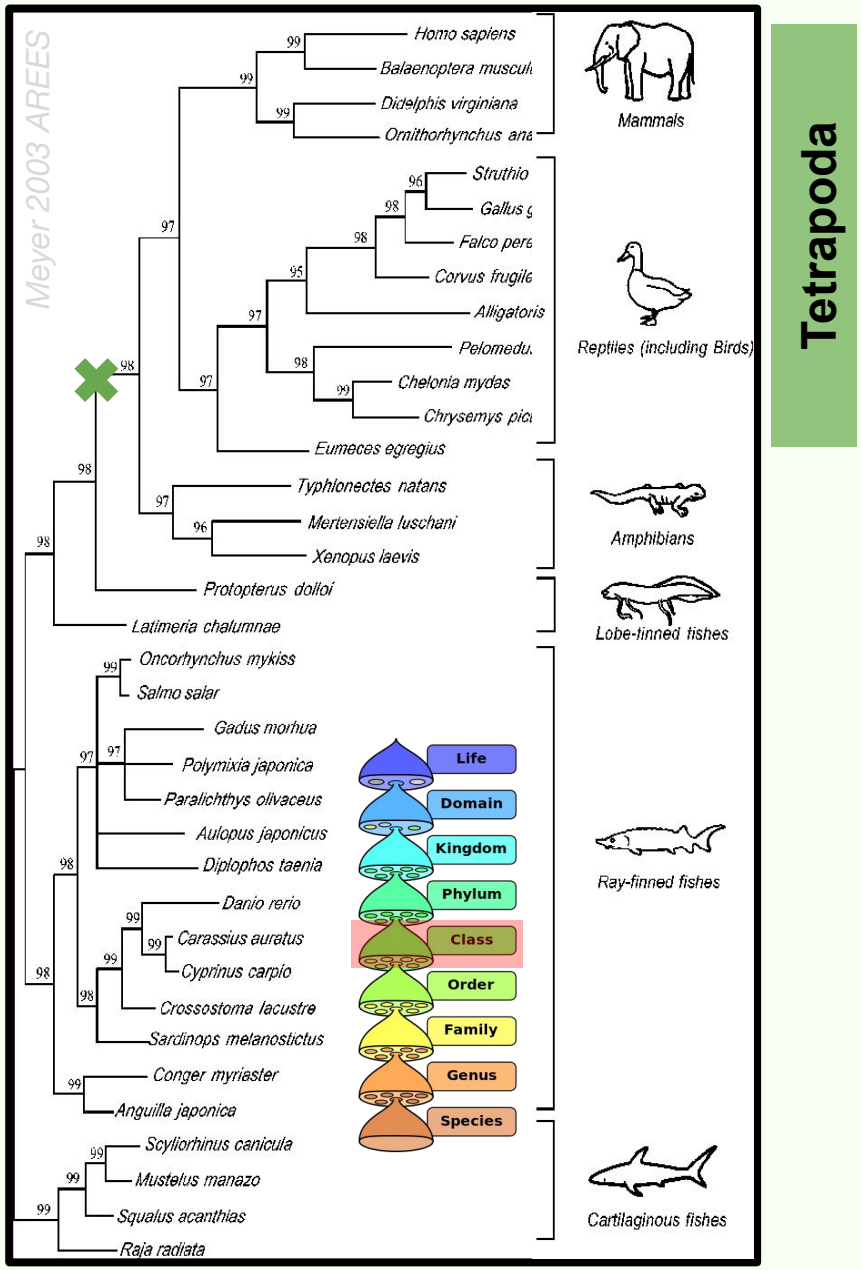
- **Notochord** (stiff ro along the back)
- **Dorsal nerve cord**
- **Pharyngeal slits** (f breathing)

These are **synapomorphies**: unic defining characteristics

Taxonomy:
Hierarchical Classification

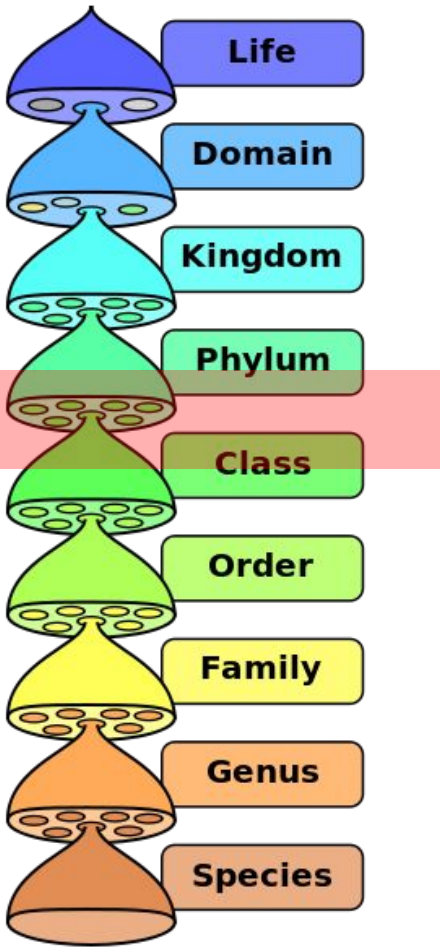


Terrestrial Arthropods and Vertebrates



Tetrapoda

Taxonomy:
Hierarchical Classification



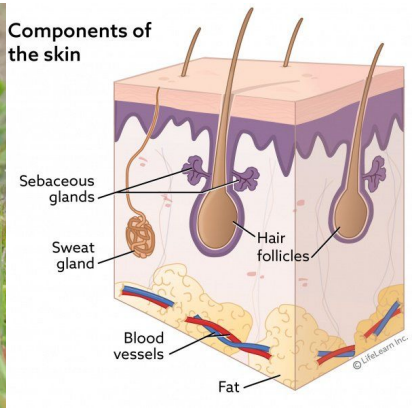
Mammals

Mammary Glands and Lactation

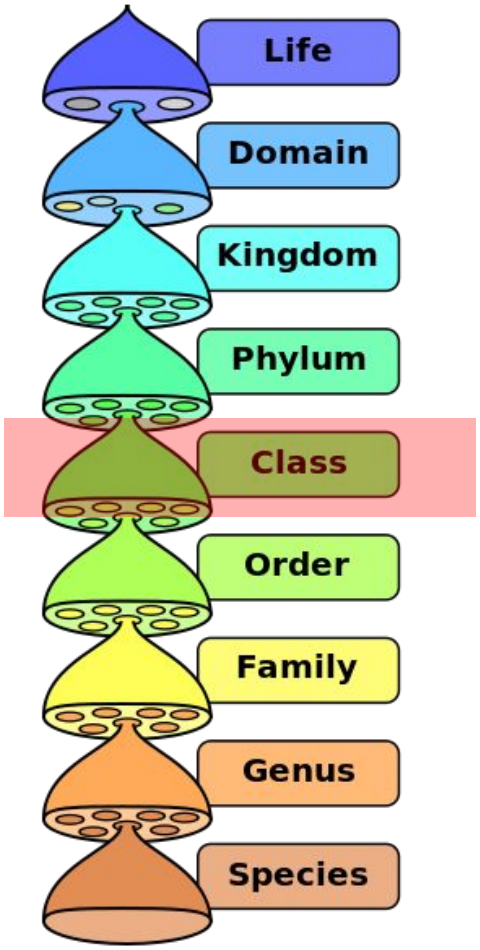


Hair and Sweat Glands





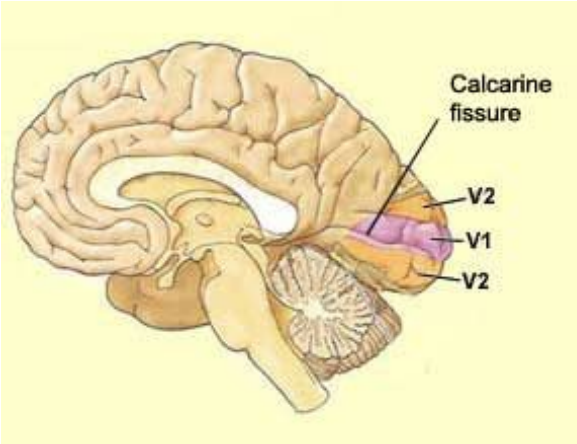
Taxonomy:
Hierarchical Classification



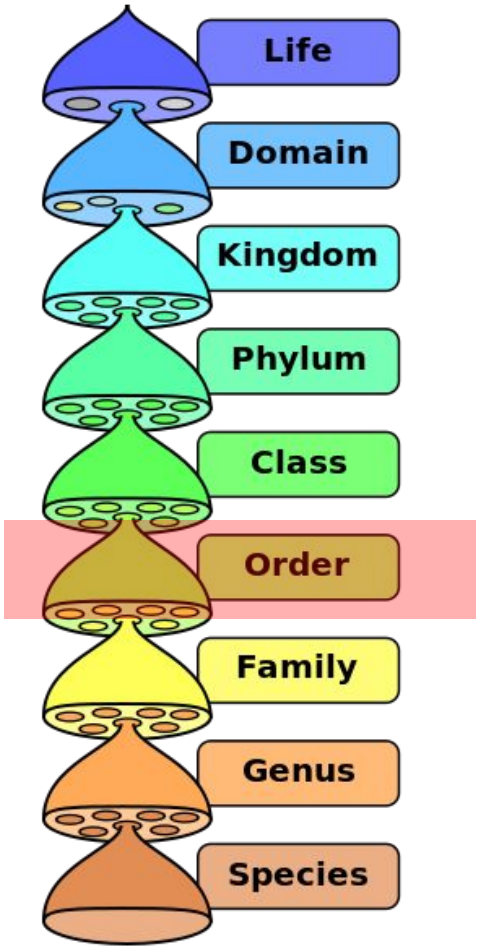
Primates



Primates are a diverse group of mammals with relatively large brains, specialized for vision, and tactile grasping feet and hands



Taxonomy:
Hierarchical Classification

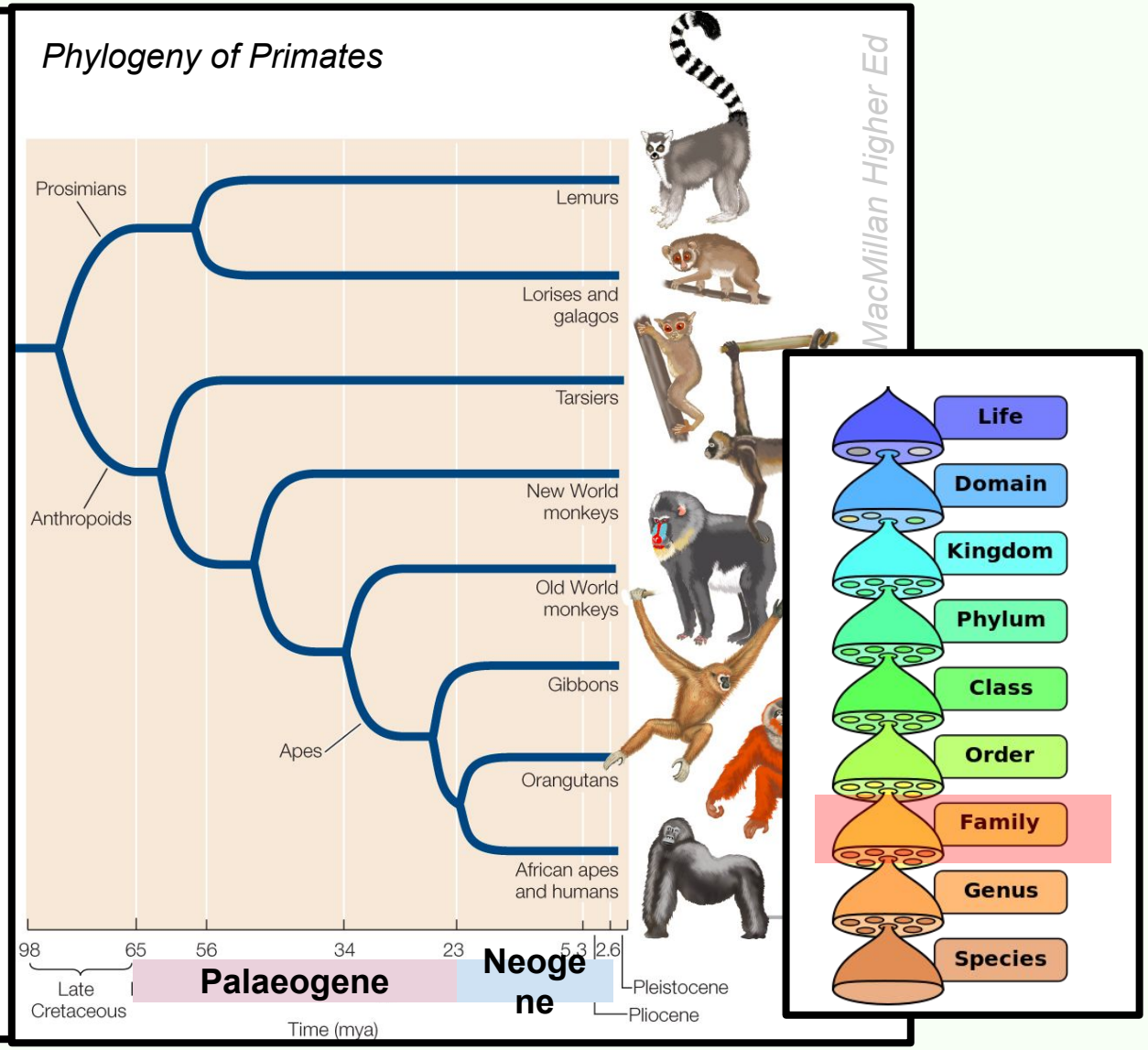


Hominidae

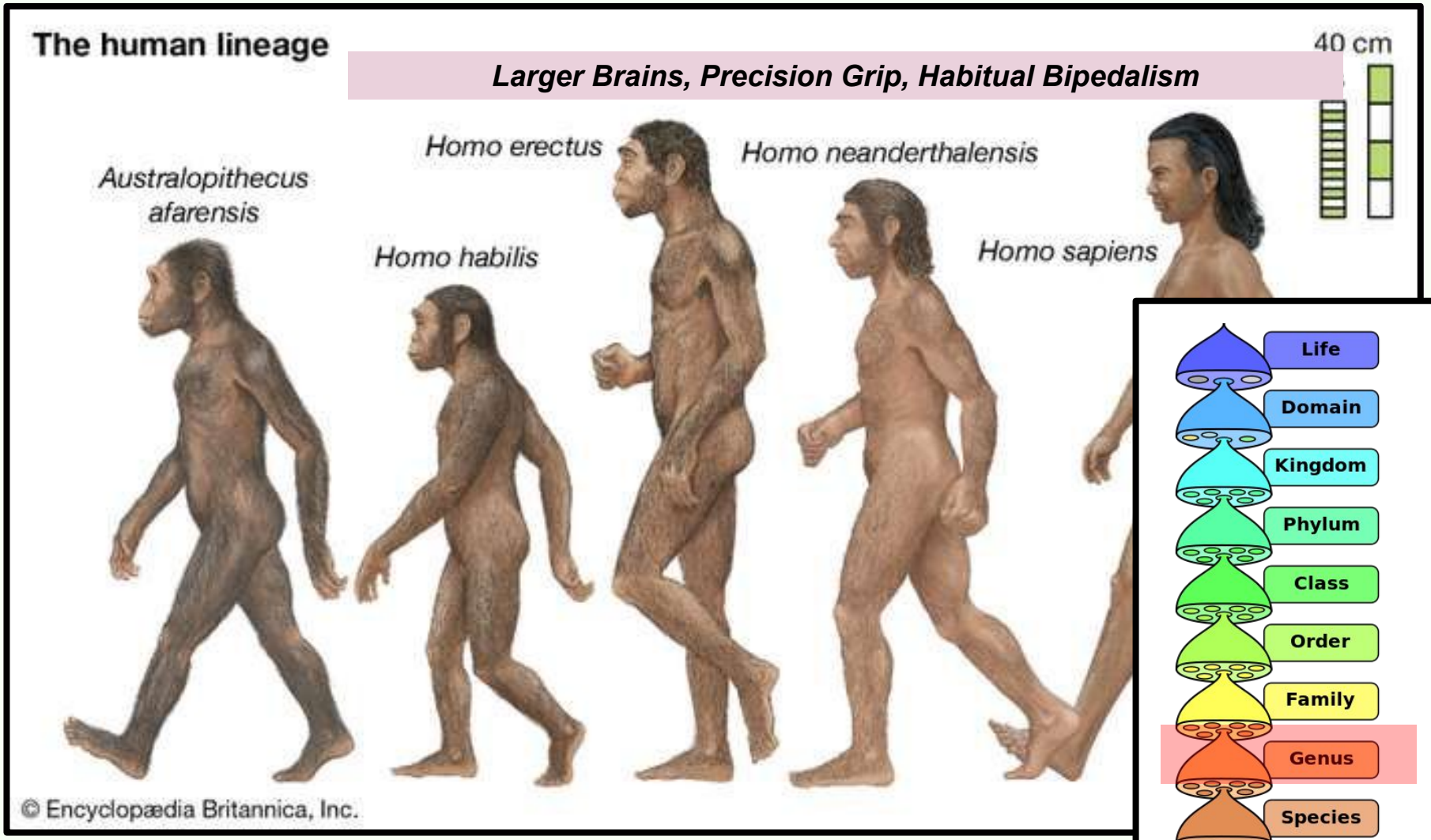
Within primates, we are part of the **Hominidae** (Great Apes), primates which have lost their tails, and exhibit complex behaviours

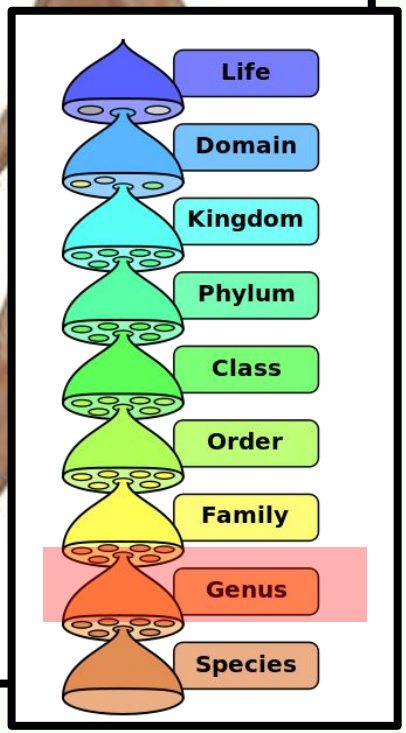


Living non-human species of Hominidae



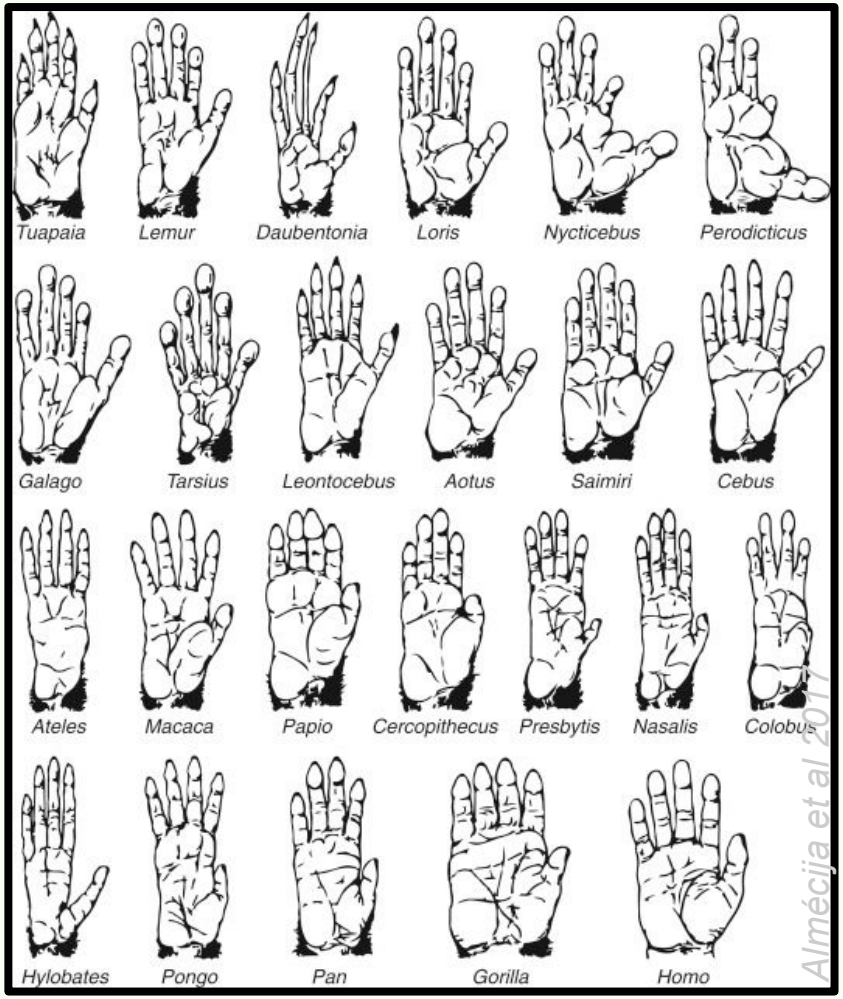
Genus *Homo*


The human lineage
40 cm
Larger Brains, Precision Grip, Habitual Bipedalism
Australopithecus afarensis
Homo habilis
Homo erectus
Homo neanderthalensis
Homo sapiens
© Encyclopædia Britannica, Inc.

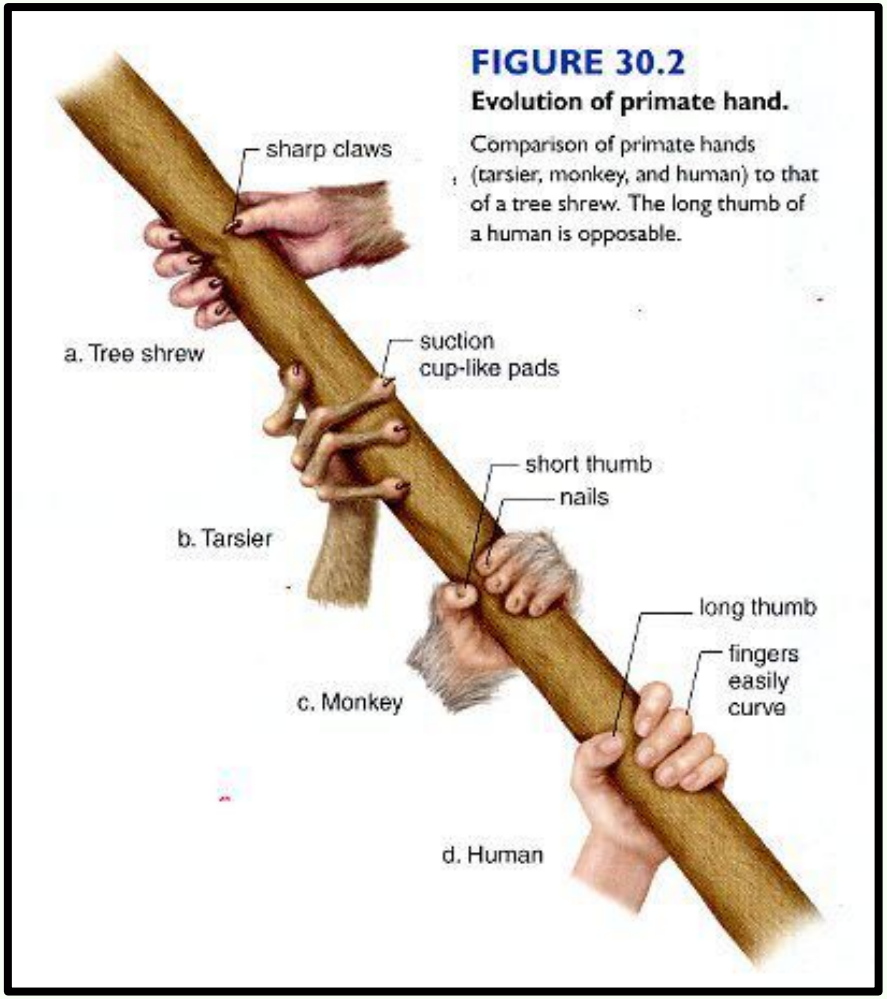

Life
Domain
Kingdom
Phylum
Class
Order
Family
Genus
Species

Grasping Hands

Primate hands show increasingly complexity of grasping due to flat keratin nails, opposable thumbs



Almécija et al 2017



Great Apes

Within primates, we are part of the **Hominidae** (Great Apes), primates which have lost their tails, and exhibit complex behaviours



Living non-human species of Hominidae



Chimpanzee



Monkey

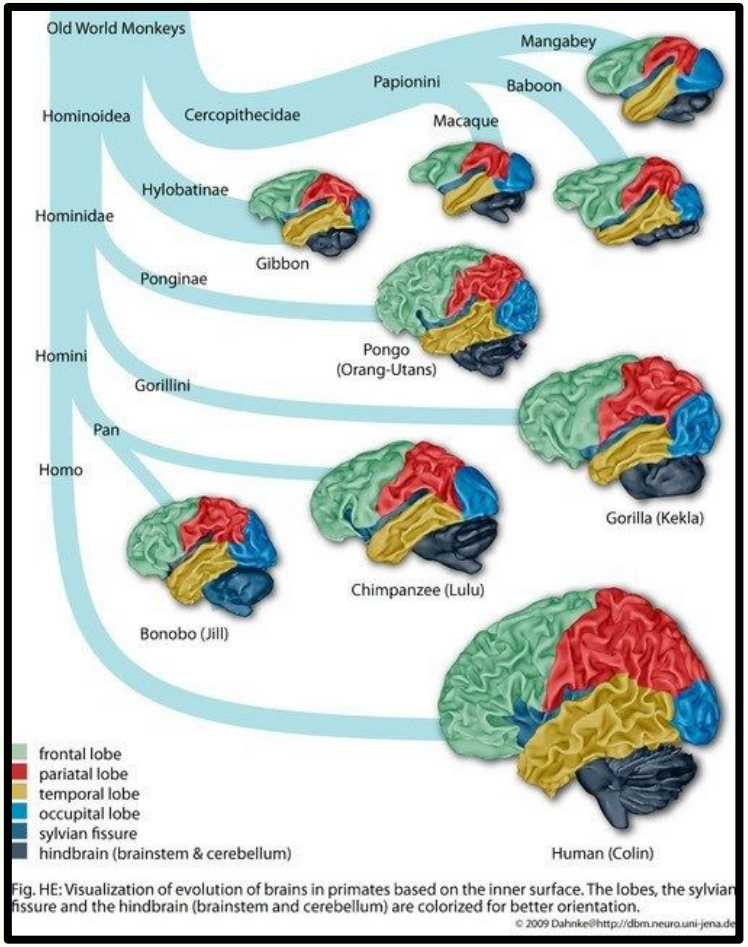


Fig. HE: Visualization of evolution of brains in primates based on the inner surface. The lobes, the sylvian fissure and the hindbrain (brainstem and cerebellum) are colorized for better orientation.

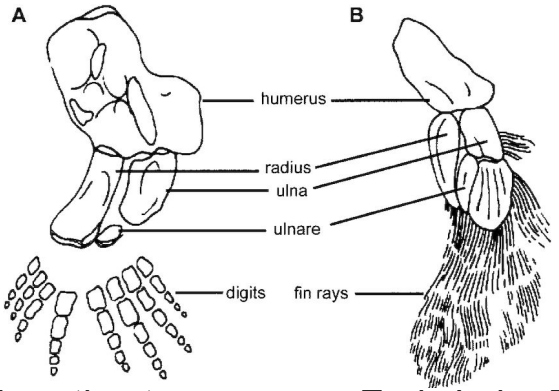
© 2009 Dahnke@http://dbm.neuro.uni-jena.de

C Gaser/U. Jena

Tetrapod Transition

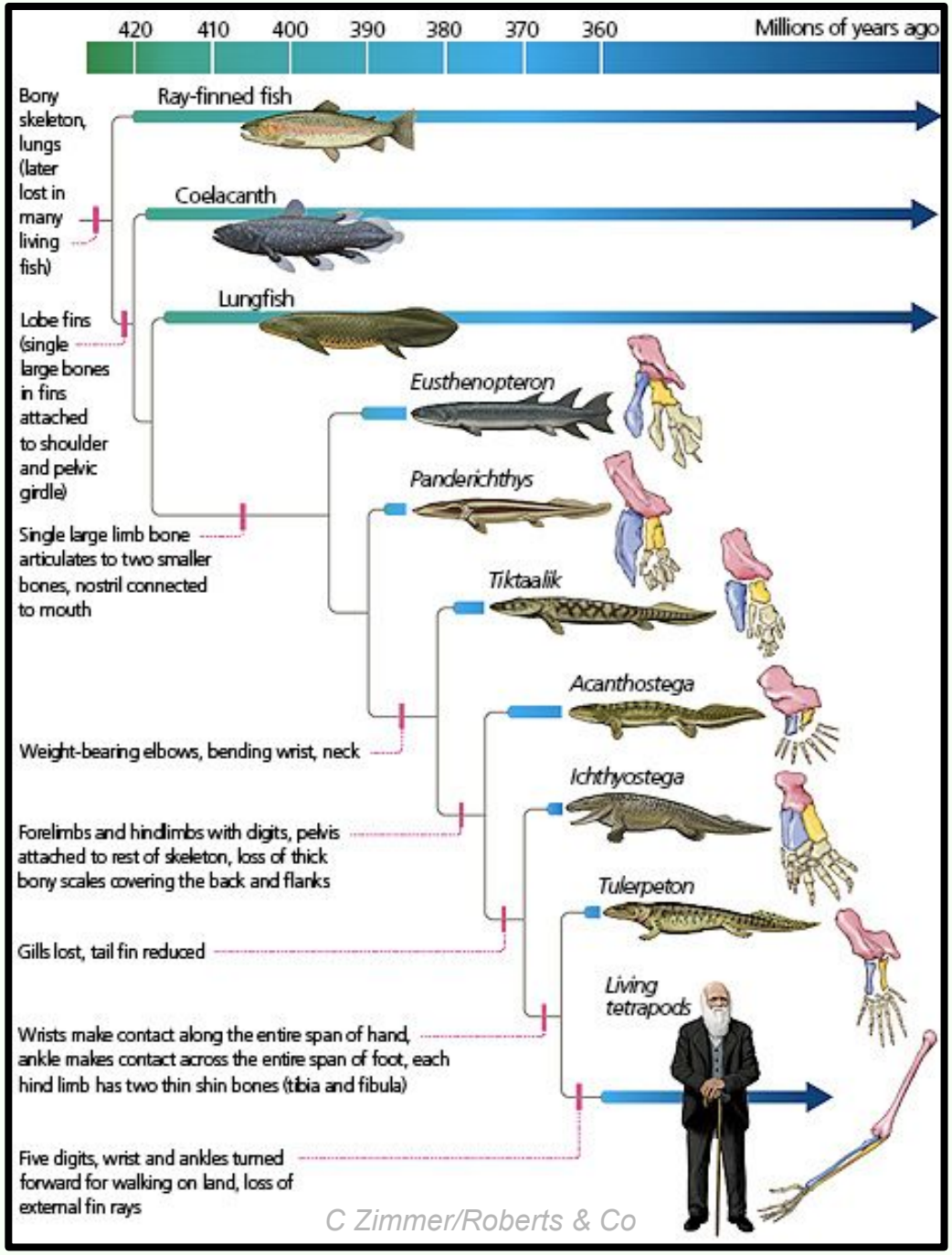
There are a whole suite of fossil lobe-finned fish revealing the stepwise acquisition of tetrapod characters and transition to terrestrial habitat

Changes in limbs, shoulders, ribs, head, fins and appearance of first **digits**

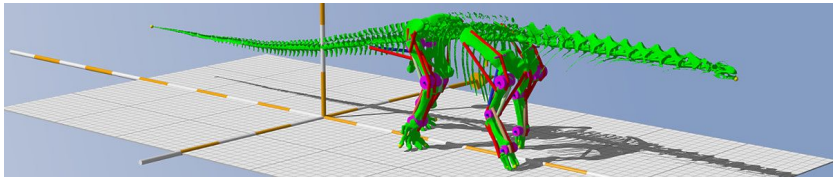


Acanthostega Early Lobe Fin

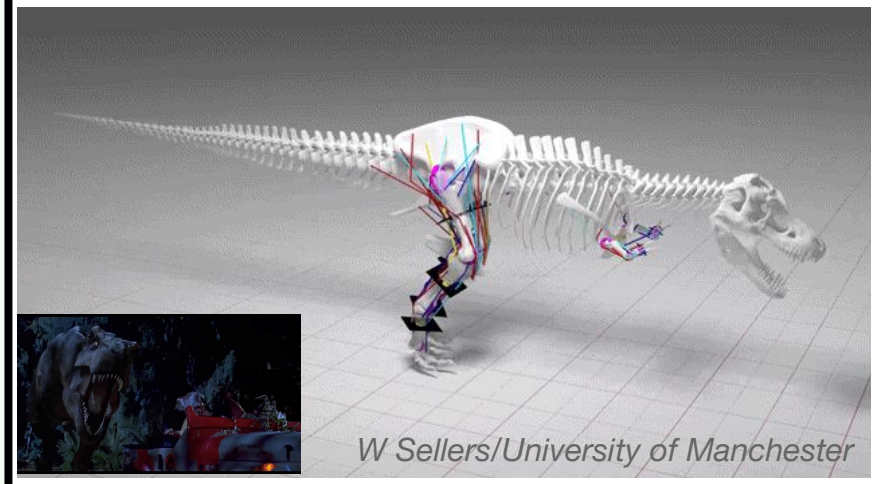
Long et al 2004 Physio Zoology



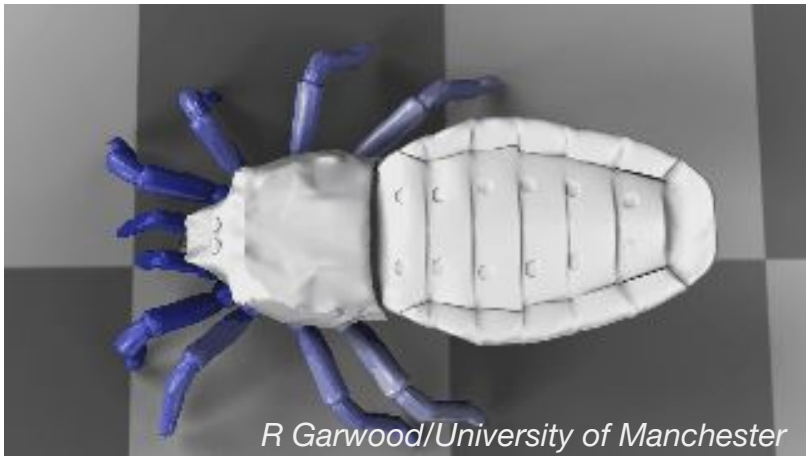
Biomechanics of Walking



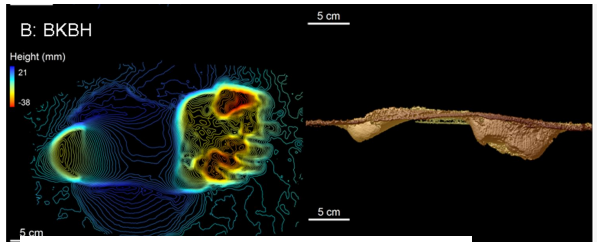
Digital models can be used to model walking e.g. applying virtual muscles and tendons to the skeleton



Gait reconstruction shows *T. rex* was unlikely to be able to run, and therefore not a hunter



Gait reconstruction of trigonotarbids show they walked a bit like spiders



Laetoli foot prints 4.5mya

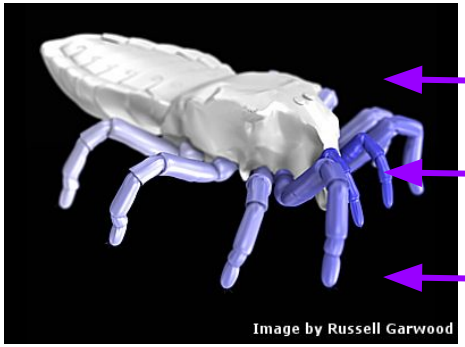




Raichlan et al 2010 PlosOne *Leakey Foundation*

Foot prints and models demonstrate **bipedalism** in *Australopithecus*

Physiology of Living on Land

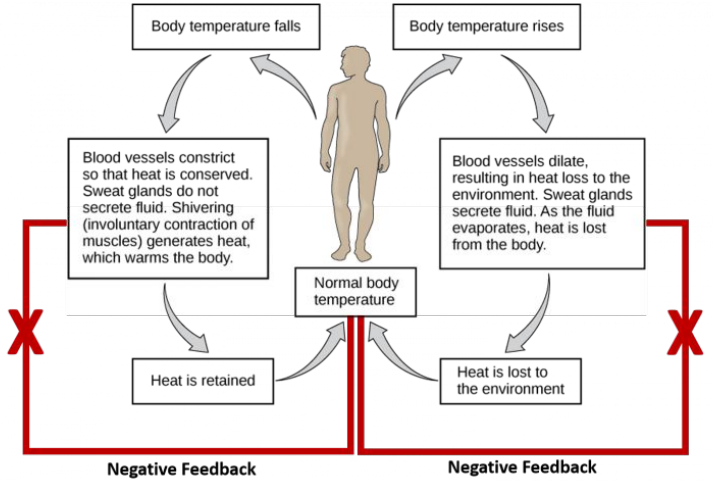


Oxygen

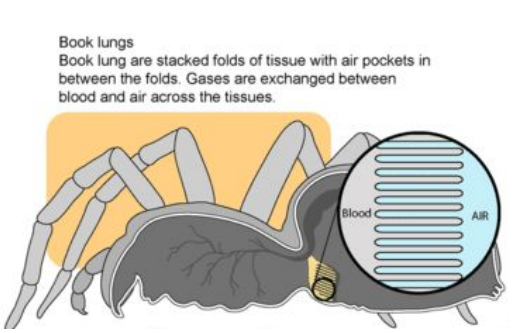
Water

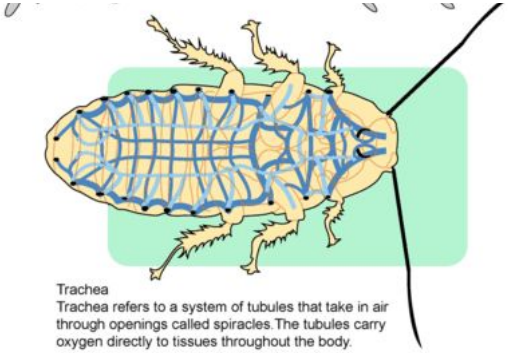
Temperature

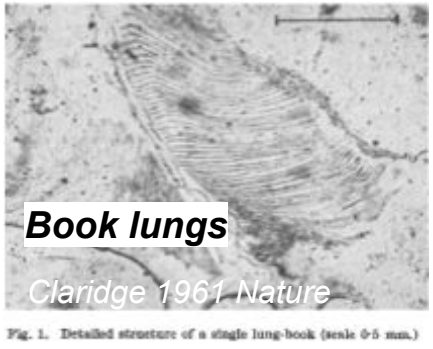
Transition from marine to terrestrial brings challenges in regulation and coping with environmental changes

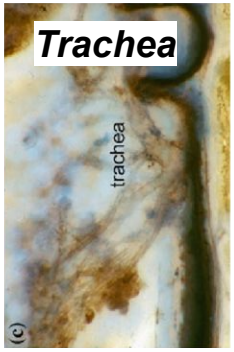


Homeostasis: maintaining a steady internal state







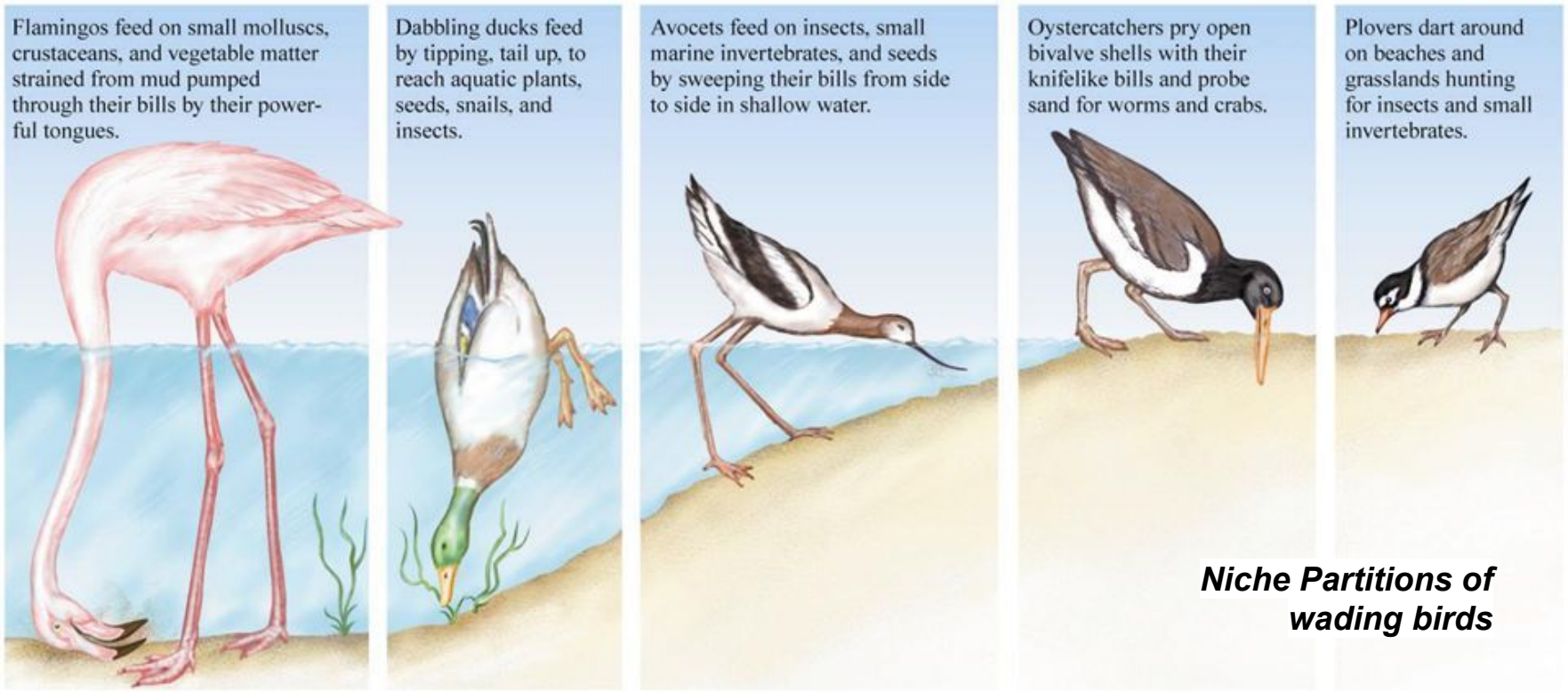


Dunlop et al 2004

Breathing air rather than water is a challenge with different solutions

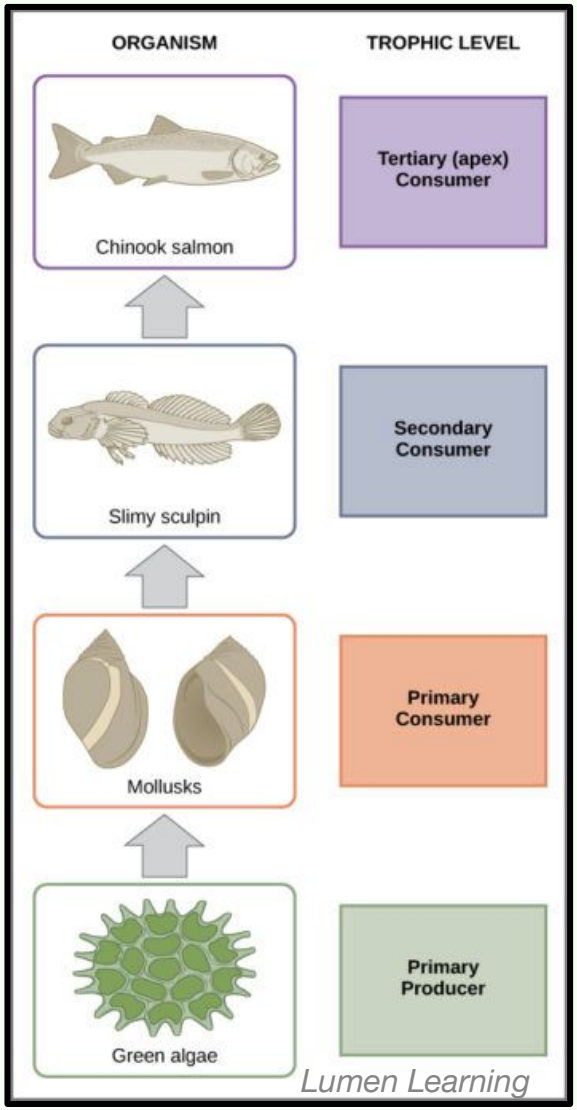
Ecological Niche

Each organism has an **ecological niche**: the *position* of a species within an ecosystem in terms of the range of conditions it can persist in
This includes the **biotic environment** (interactions with other species such as competitors, predators, etc) and **abiotic environment** (e.g. temperature, O_2 etc).



Niche Partitions of wading birds

Ecology and Food webs



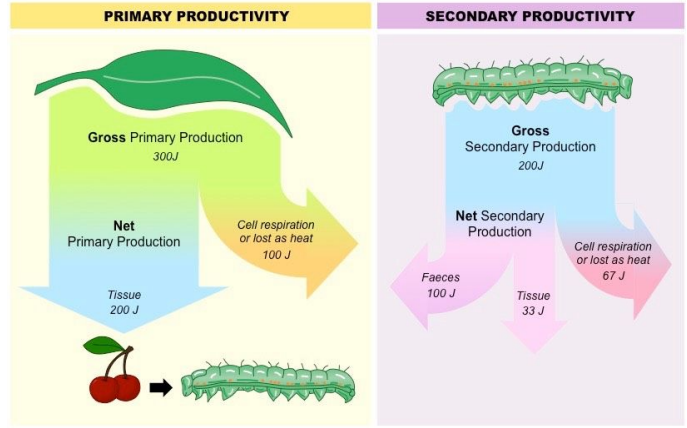
Apex predators only consume carnivores, at the top of the chain

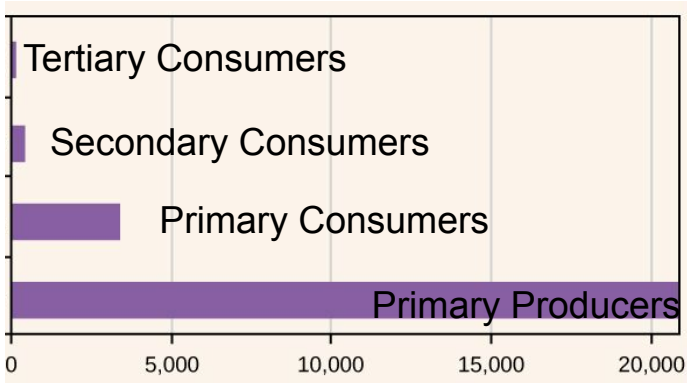
Carnivores consume the herbivores, Omnivores consume primary produces as well as herbivores

Herbivores consume the primary producers (e.g. grazing, filter feeding, etc.)

Autotrophs generate energy and organic matter from sunlight (photosynthesis)

Energy loss at each stage. Most matter is primary producers





Relative Energy Content in Trophic Levels (kcal/m2/year)

How to define a species?

Can you define a species with reference to characteristics or specimens?



A **Holotype** specimen


Edward Drinker Cope

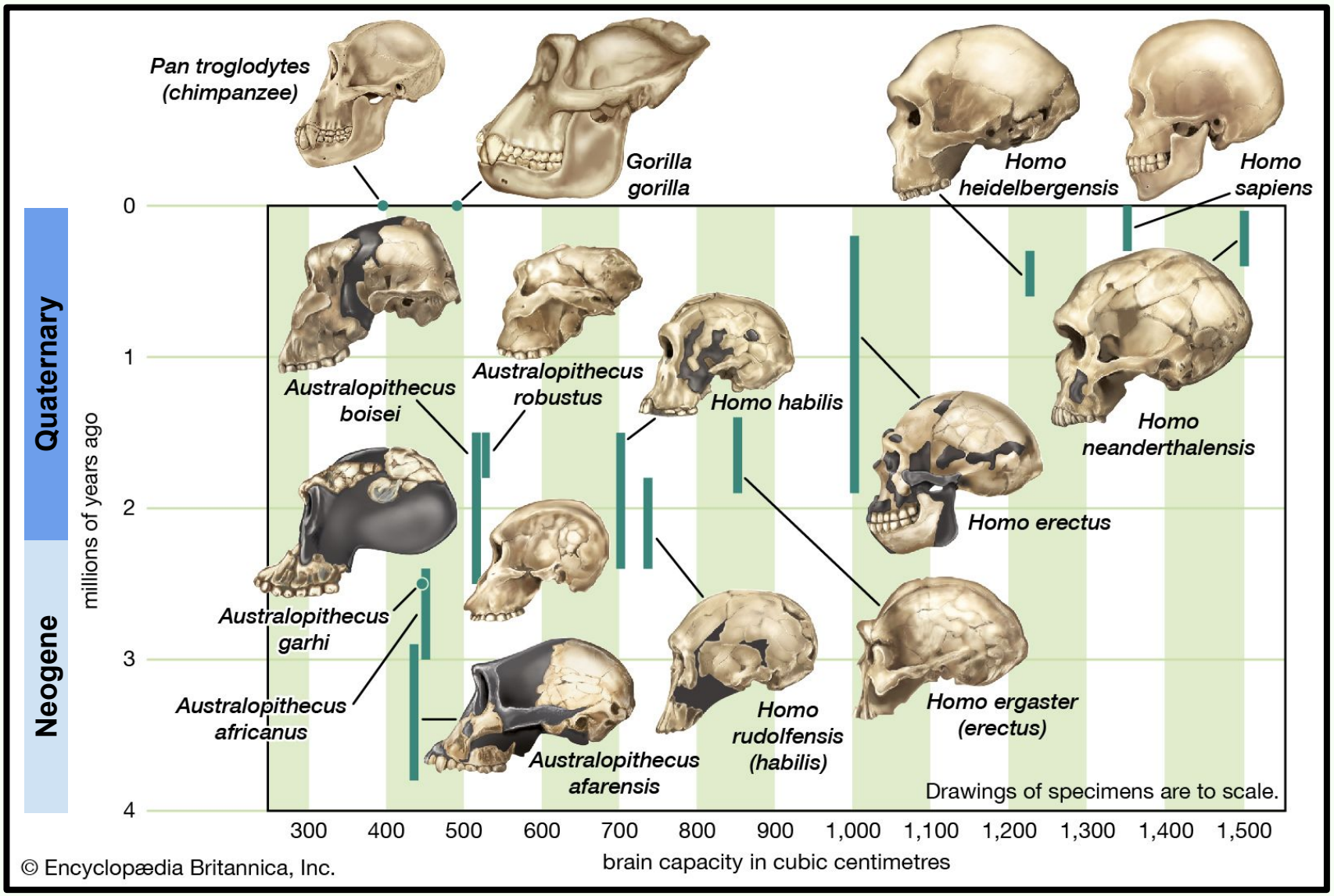
There is one living species of Homo, ***Homo sapiens***



Compared to other apes:

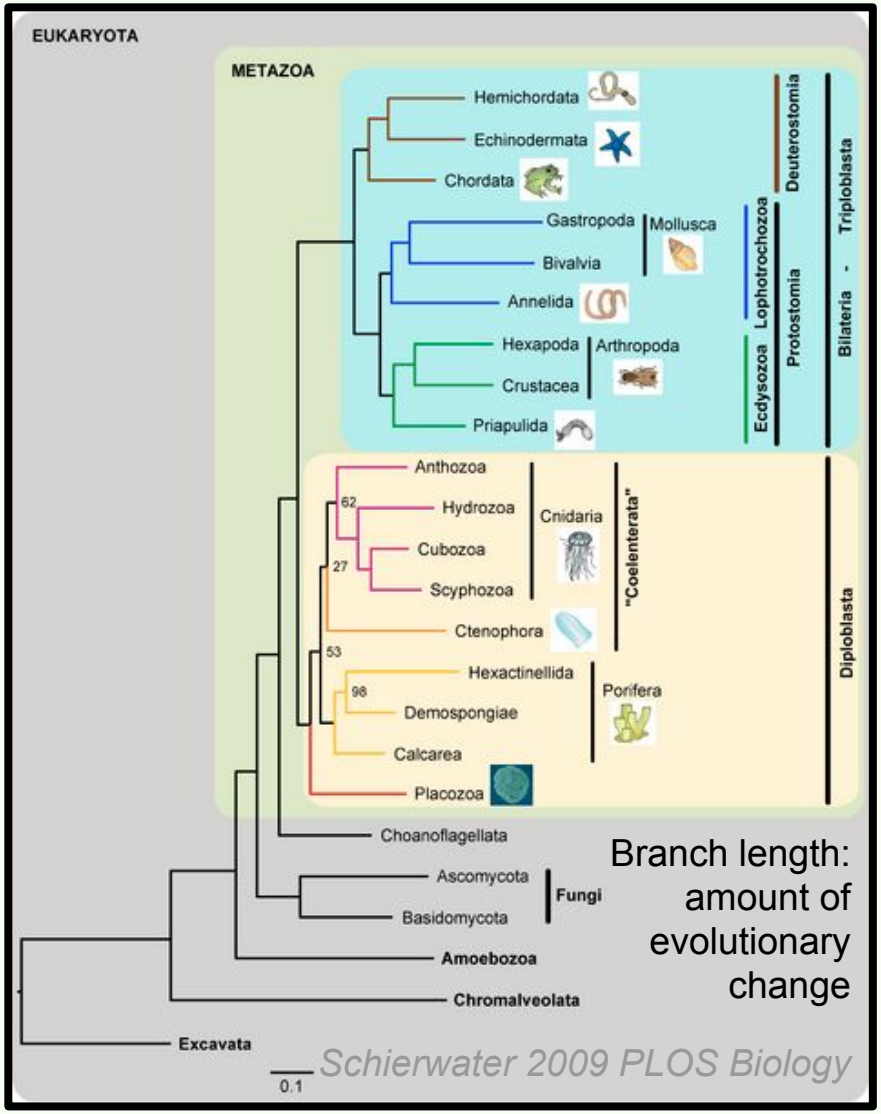
- Larger brains and skulls
- Bipedal walking gait
- Protruding nose and chin
- Smaller jaws and teeth

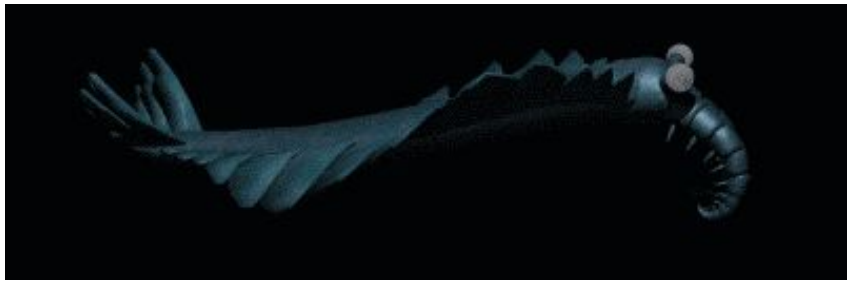
Homo


Pan troglodytes (chimpanzee)
Gorilla gorilla
Homo heidelbergensis
Homo sapiens
Australopithecus boisei
Australopithecus robustus
Homo habilis
Homo neanderthalensis
Homo erectus
Australopithecus garhi
Australopithecus africanus
Australopithecus afarensis
Homo rudolfensis (habilis)
Homo ergaster (erectus)
Quaternary
Neogene
millions of years ago
0
1
2
3
4
300
400
500
600
700
800
900
1,000
1,100
1,200
1,300
1,400
1,500
brain capacity in cubic centimetres
Drawings of specimens are to scale.
© Encyclopædia Britannica, Inc.

Phylogeny

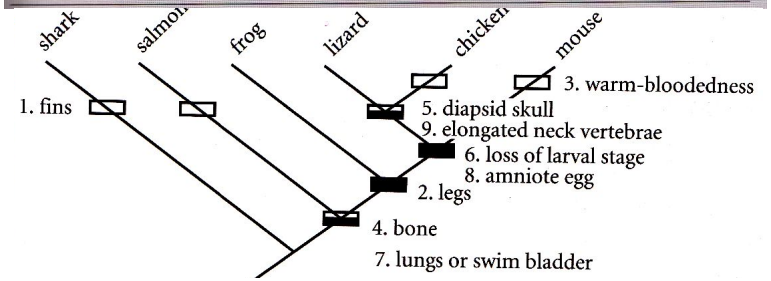
A **phylogeny** is a evolutionary tree: a hypothesis of evolutionary history and relationships between species





Ancient fossils don't have DNA, so we need to use **morphological** data

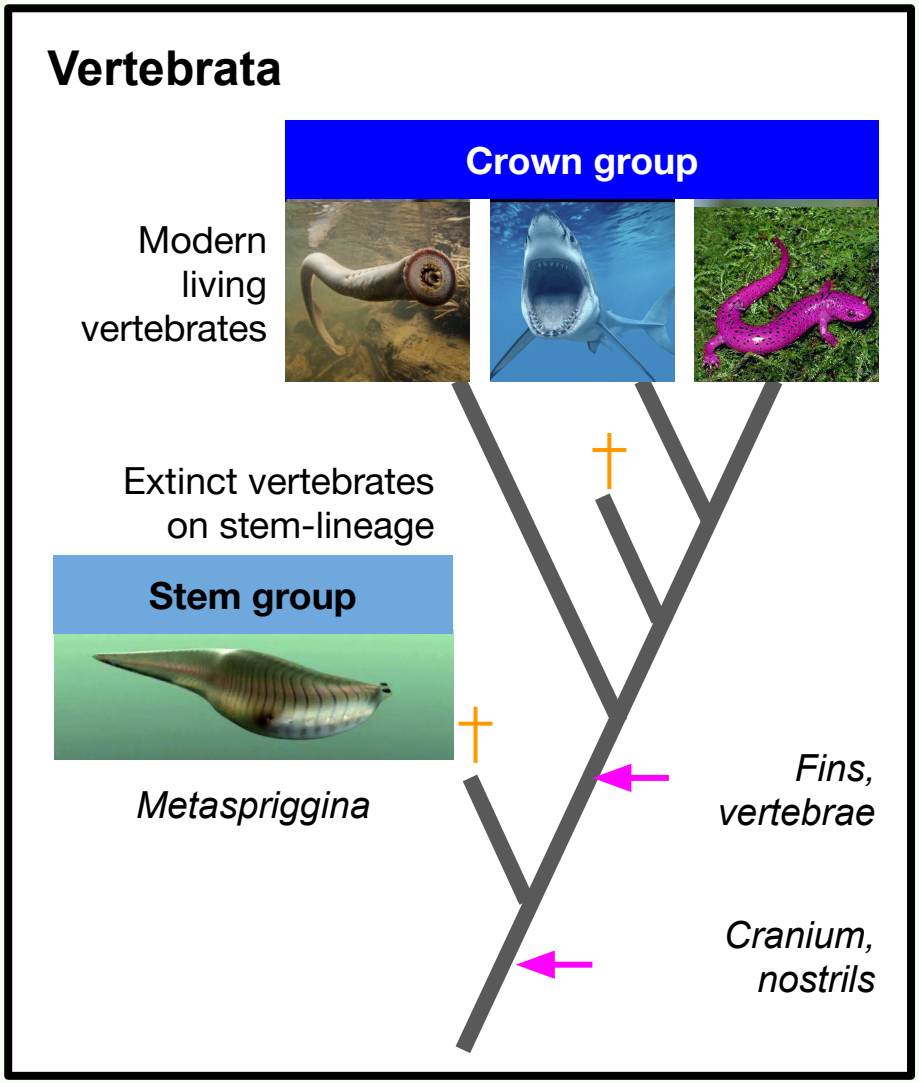
Character	Shark	Salmon	Frog	Lizard	Chicken	Mouse
1 Fins	1	1	0	0	0	0
2 Legs	0	0	1	1	1	1
3 Warm-bloodedness	0	0	0	0	1	1
4 Bone	0	1	1	1	1	1
5 Diapsid skull	0	0	0	1	1	0
6 Loss of larval stage	0	0	0	1	1	1
7 Lung or swim bladder	0	1	1	1	1	1
8 Amniote egg	0	0	0	1	1	1
9 Elongated neck vertebrae	0	0	0	1	1	0
10 Marginal teeth	0	1	1	1	1	1

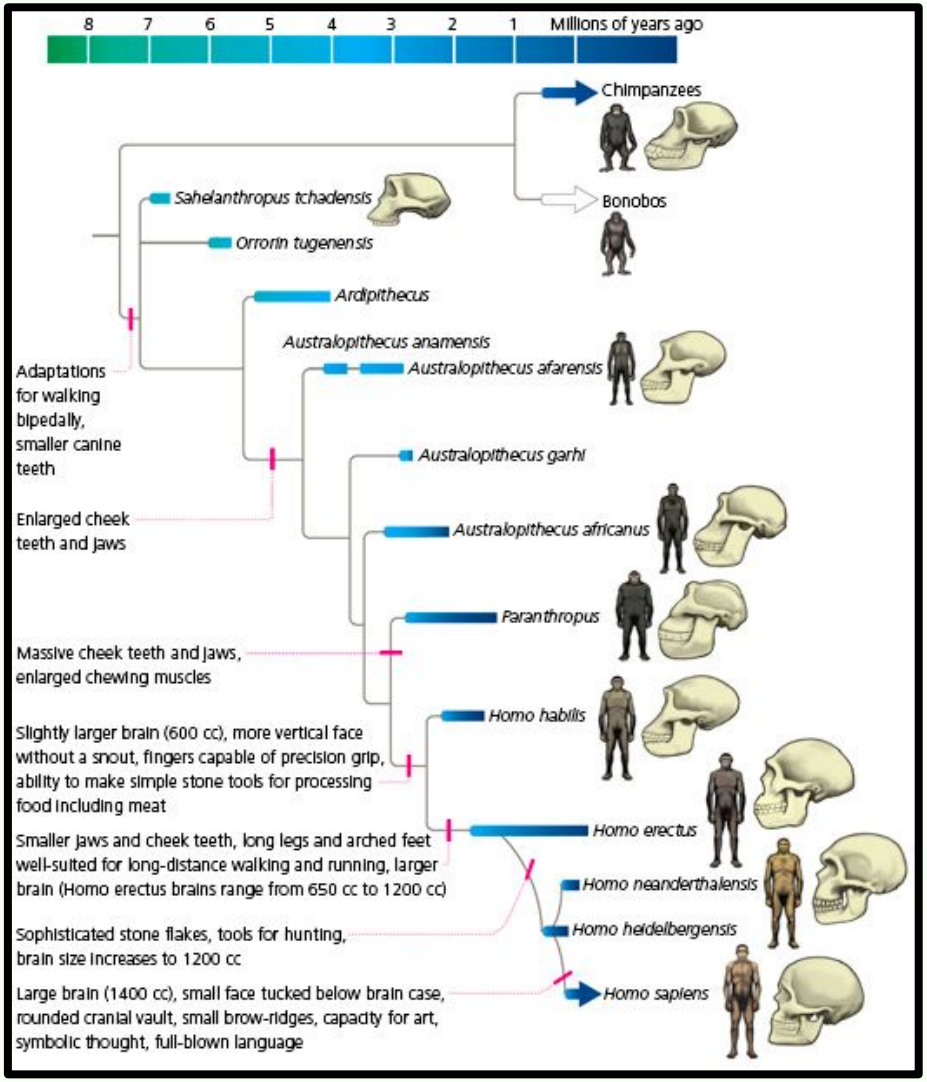


Anatomy broken down into binary characters (e.g. presence/absence), used to build relationships

Stem-groups

A **phylogeny** is a evolutionary tree: a hypothesis of evolutionary history and relationships between species



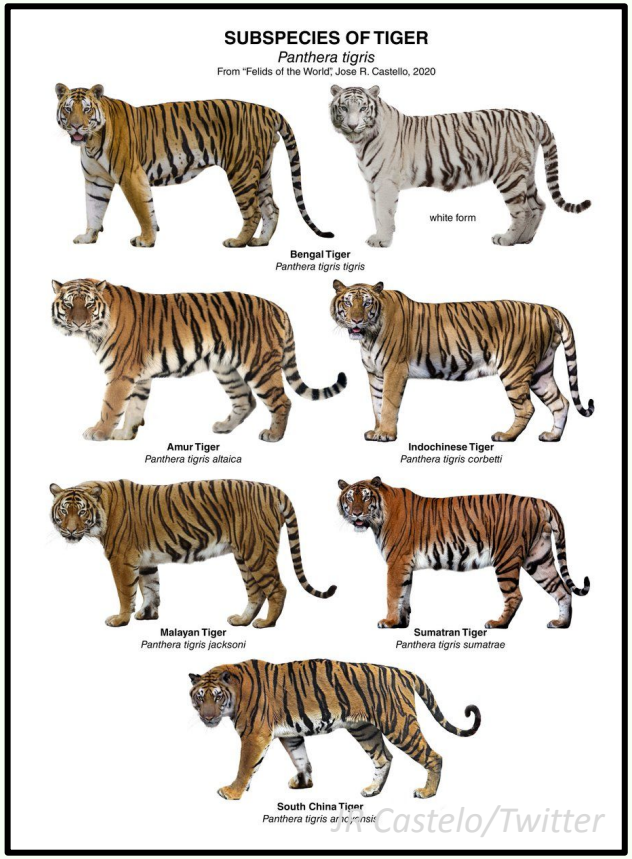


Biological Species Concept

Group of actually or potentially breeding natural populations that are **reproductively isolated** from other groups (Mayr 1940)

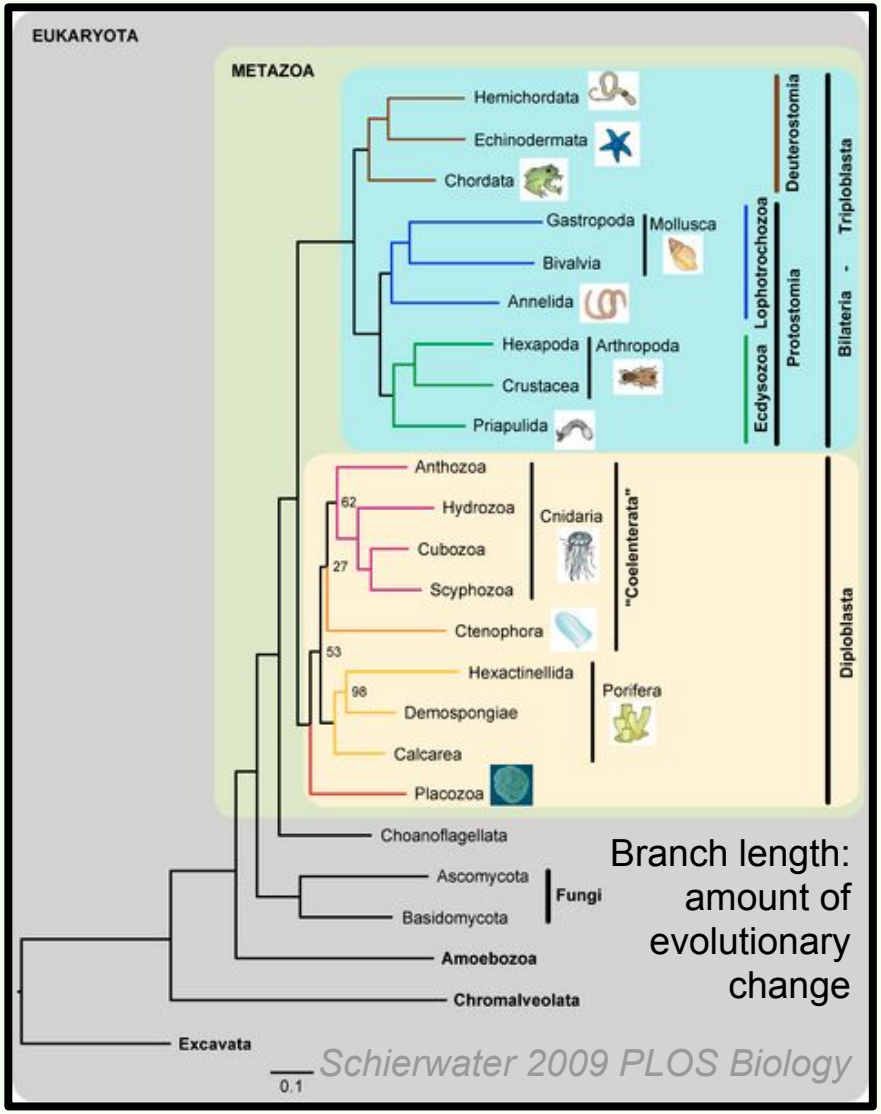


Tiger: *Panthera tigris*



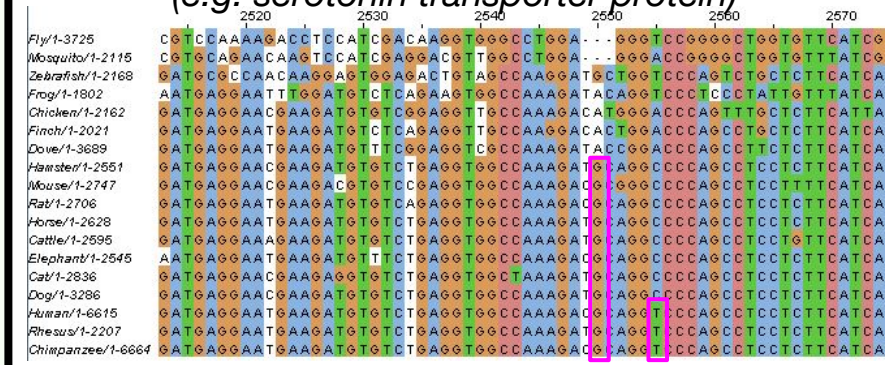
Phylogeny

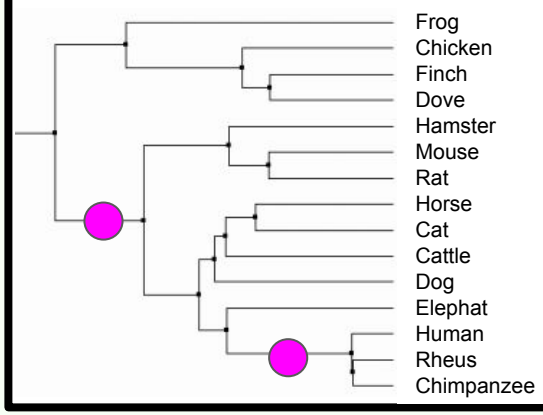
A **phylogeny** is a evolutionary tree: a hypothesis of evolutionary history and relationships between species



This is a **molecular phylogeny** - it is based on the similarities and differences in the genetic sequences

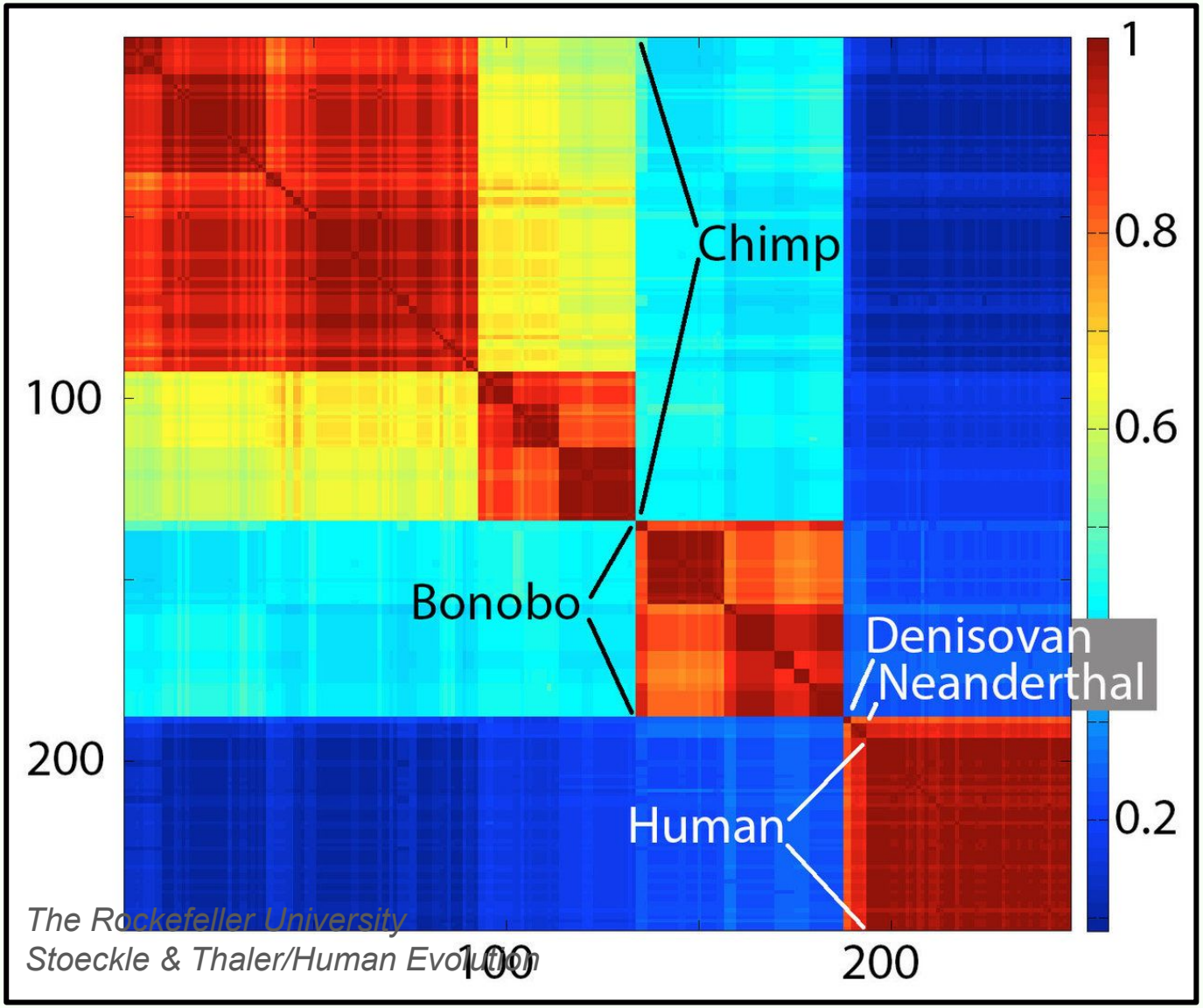
Genetic sequences of genes or whole genome (e.g. serotonin transporter protein)





Computer algorithms use the similarity of genetic sequences to reconstruct the most likely evolutionary tree

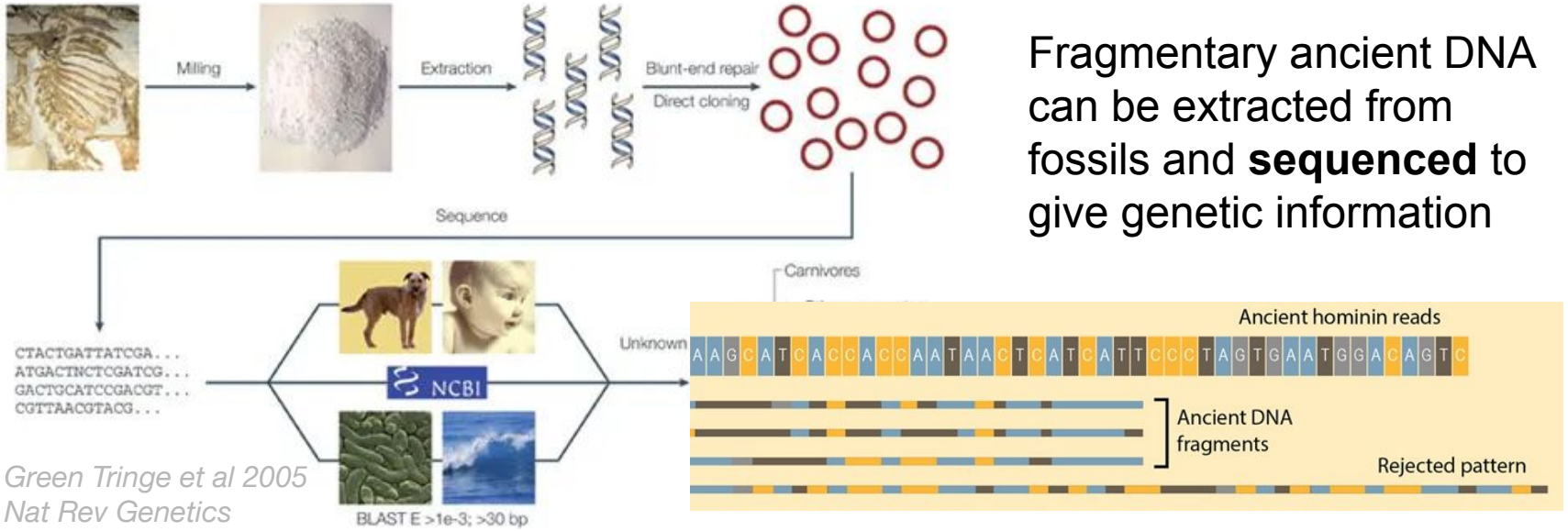
Genetic Similarity with Human Population



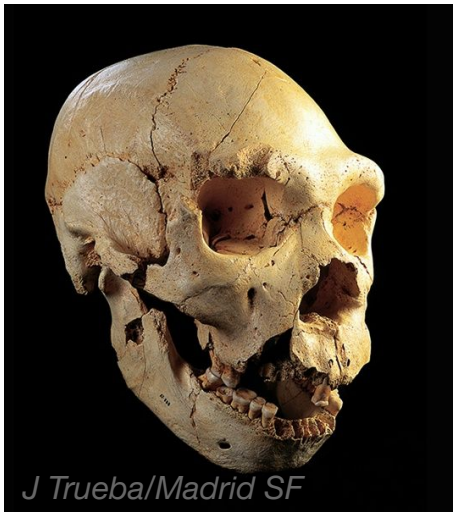
Humans show a very high degree of genetic similarity

Here is example of mitochondrial sequence similarity

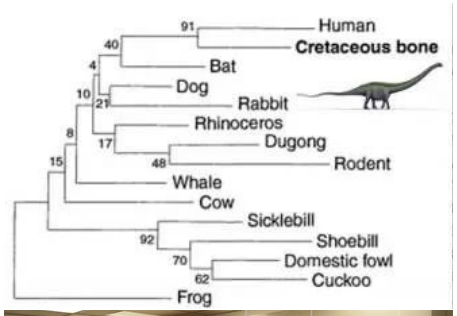
Ancient DNA



Fragmentary ancient DNA can be extracted from fossils and **sequenced** to give genetic information



DNA sequences extracted from Neanderthals reveal extensive interbreeding between neanderthals & *Homo sapiens*

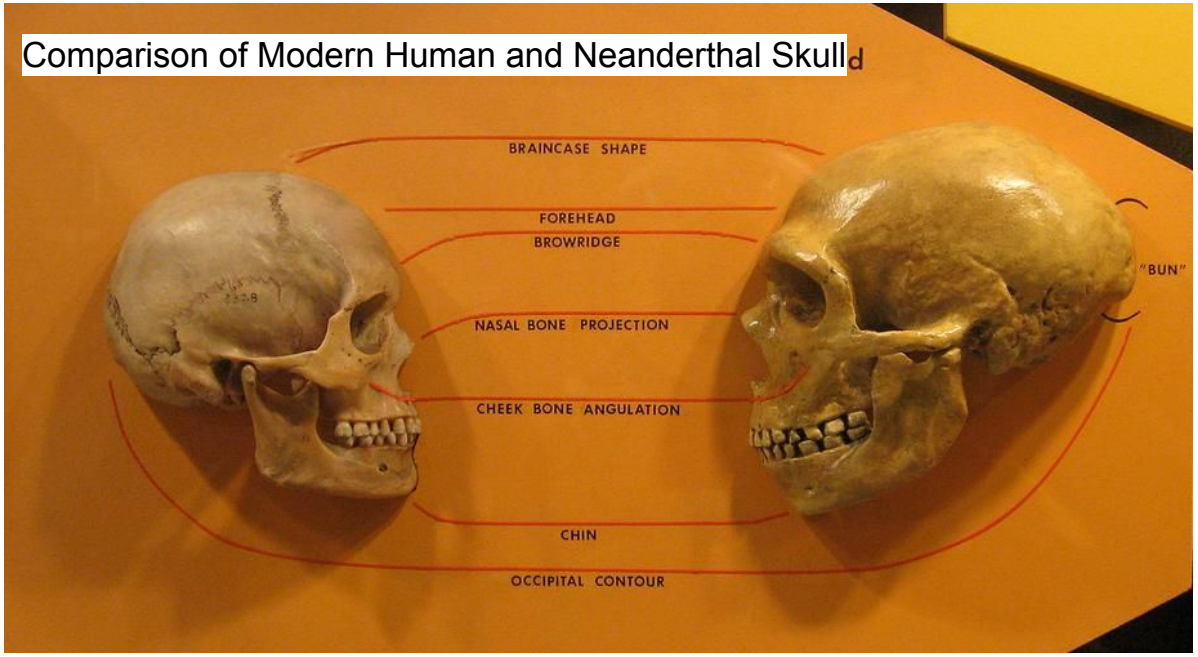




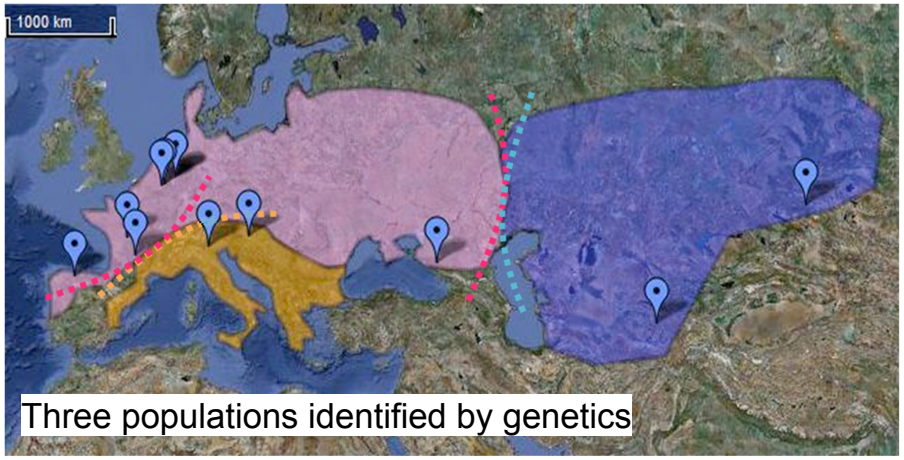
Highly sensitive to contamination and highly labile

Oldest known reliable ancient DNA is 800,000 years old

Neanderthals

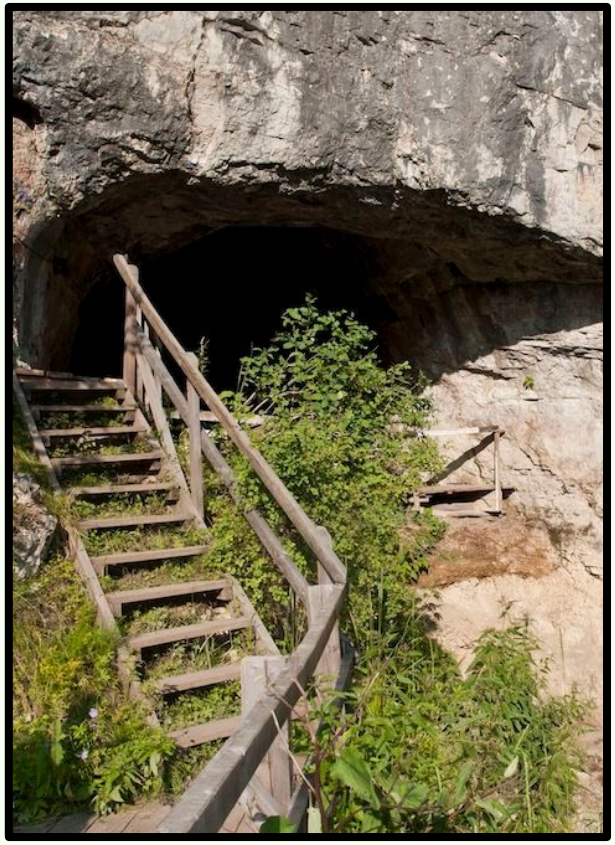








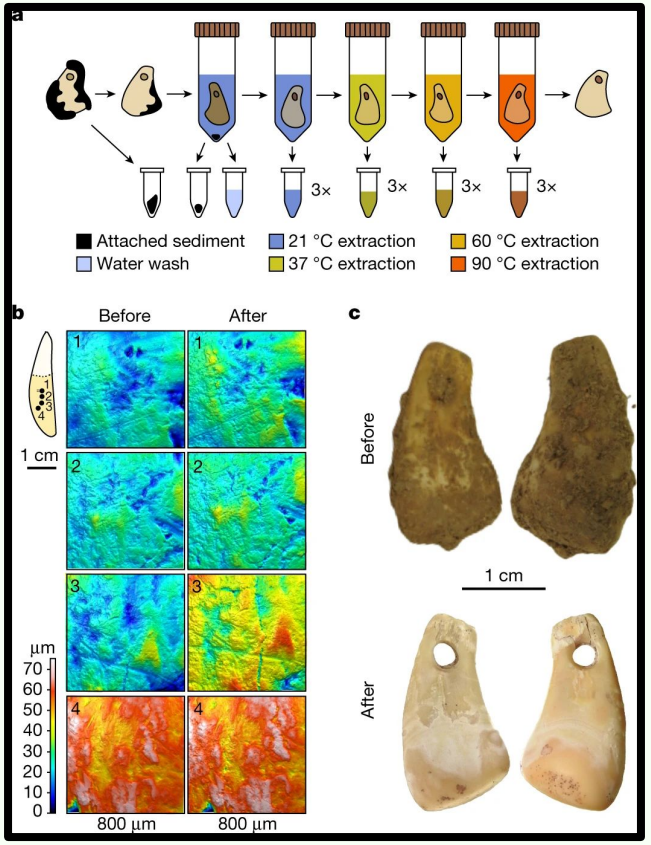
Denisovans



Denisova cave in Siberia



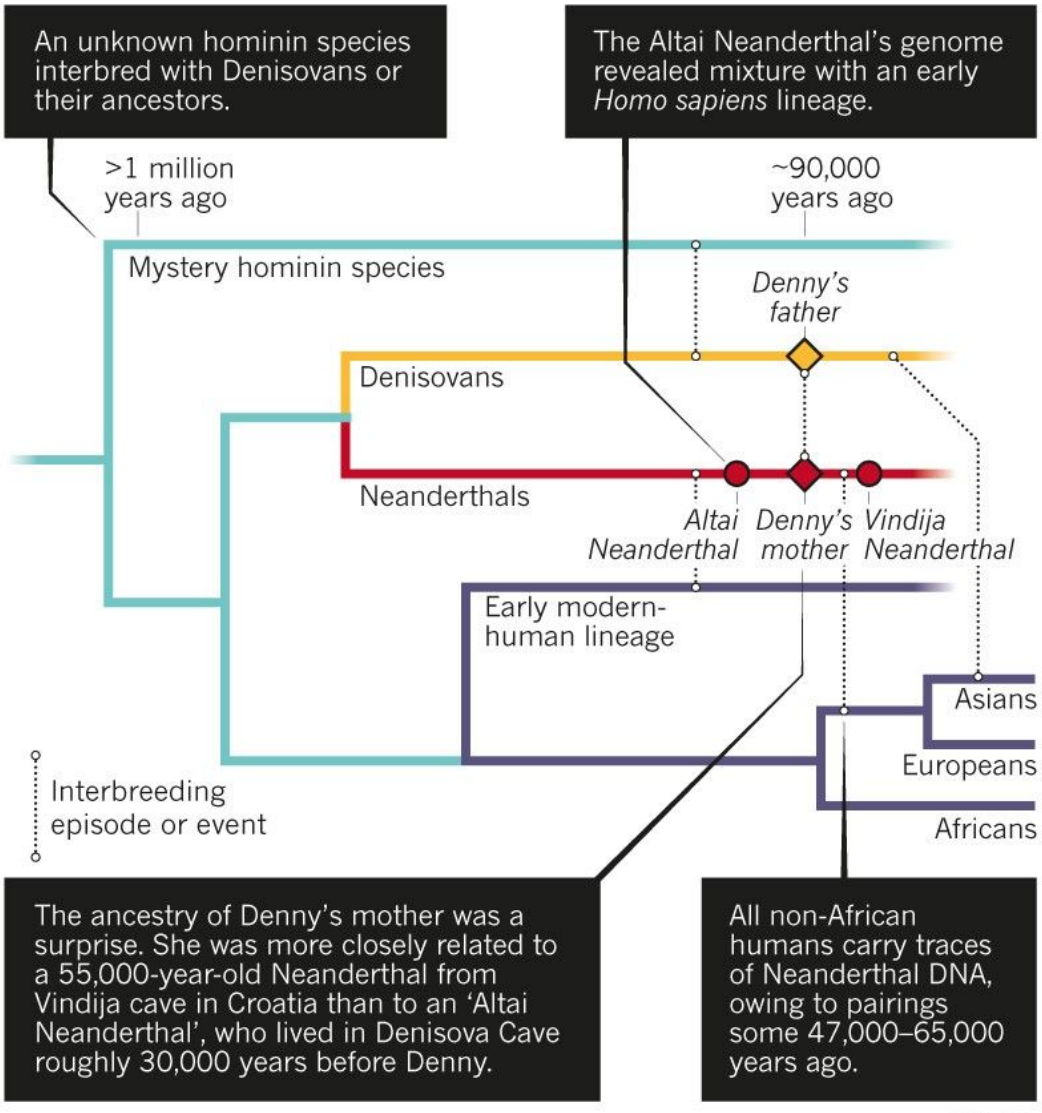
Isolated fragmentary remains ~40k years old



Extraction of ancient DNA sequences

TANGLED TREE

A female born to a Neanderthal mother and Denisovan father roughly 90,000 years ago — nicknamed Denny — is one of many examples of interbreeding between ancient human groups.





Hominin interbreeding

Research probing the link between us and our extinct "cousins"

1 Early human and eastern Neanderthal

Around 100,000 years ago

Genome of a female Neanderthal showed up sections of human DNA

2 Eastern Neanderthal and Denisovan

50,000 years ago

New DNA research finds remains of a 13-year-old girl from the Denisova cave in Siberia who was half Neanderthal and half Denisovan

3 Human and Denisovan

45,000 years ago

Humans reached Papua New Guinea bringing Denisovan genes with them

4 Human and western Neanderthal

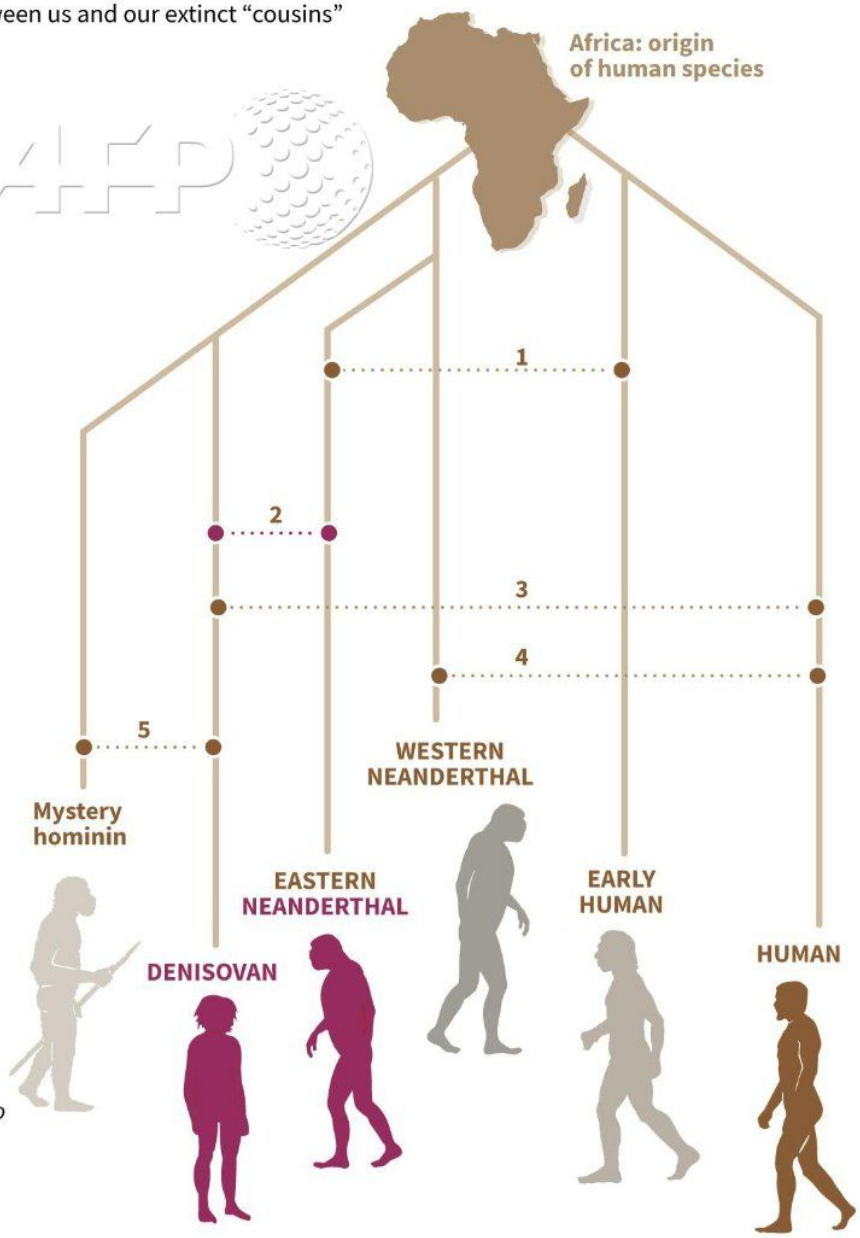
40,000 years ago

DNA from a human skeleton from Romania suggested the subject had a Neanderthal great-great grandparent

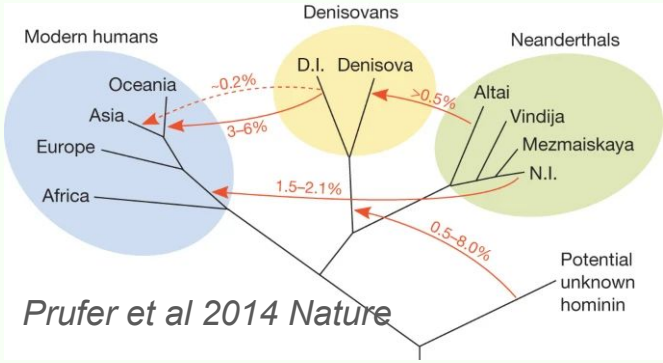
5 Denisovan and mystery hominin

More than 30,000 years ago

Ancient genomes suggest interbreeding between human-like groups in Europe and Asia

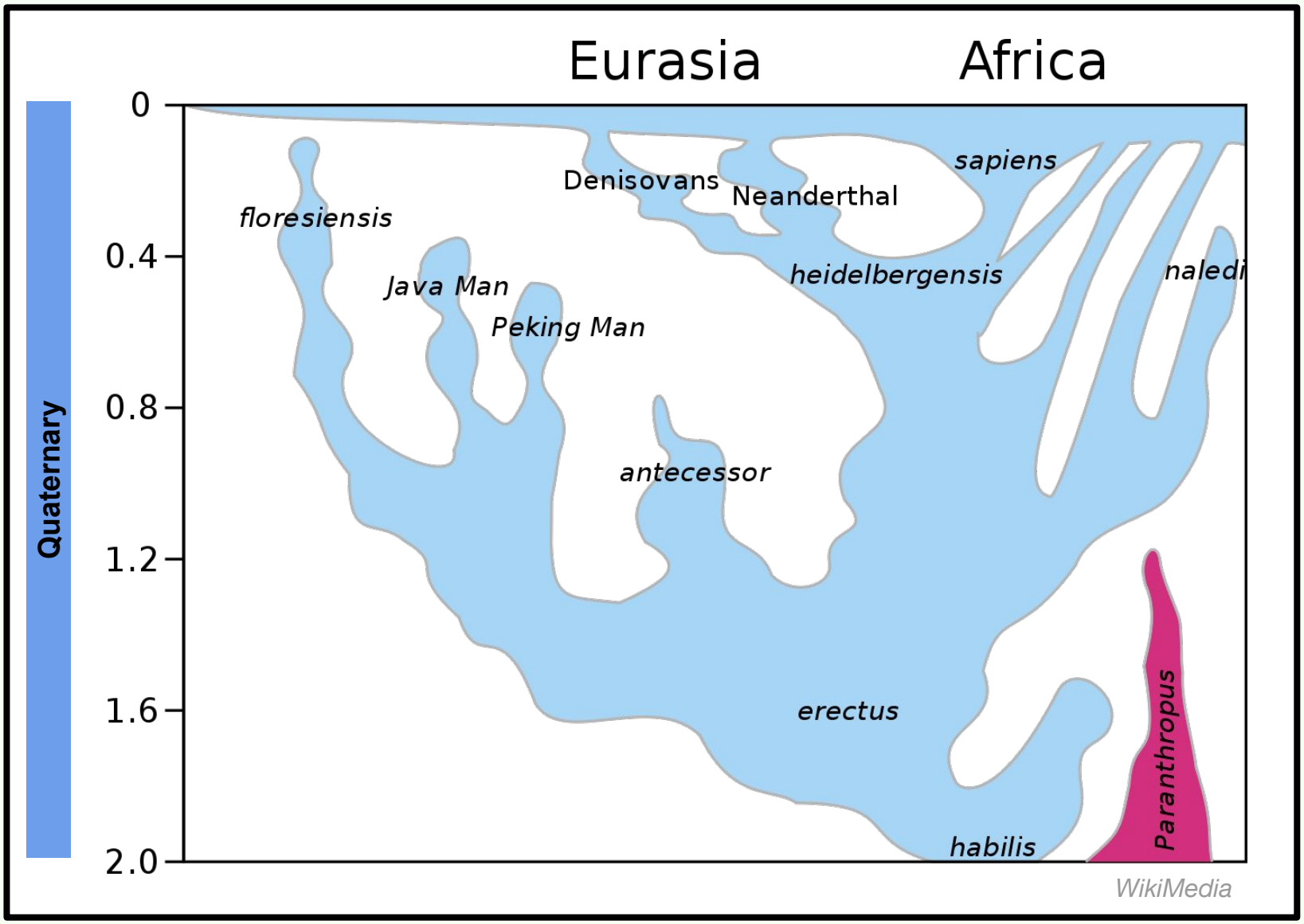


© AFP Source: Nature/NationalGeographic



Prufer et al 2014 Nature

Interbreeding




© Timothy Allen/BBC

Part 11
The Age of Apes

Our Species: Who are we?



Didier Descouens

EART11100: How to Grow a Planet
Robert Sansom
University of Manchester

The Age of Apes
Our Species: Who are we?

Intended Learning Objective: Discuss species definitions in relation to the history of *Homo sapiens* and interbreeding populations

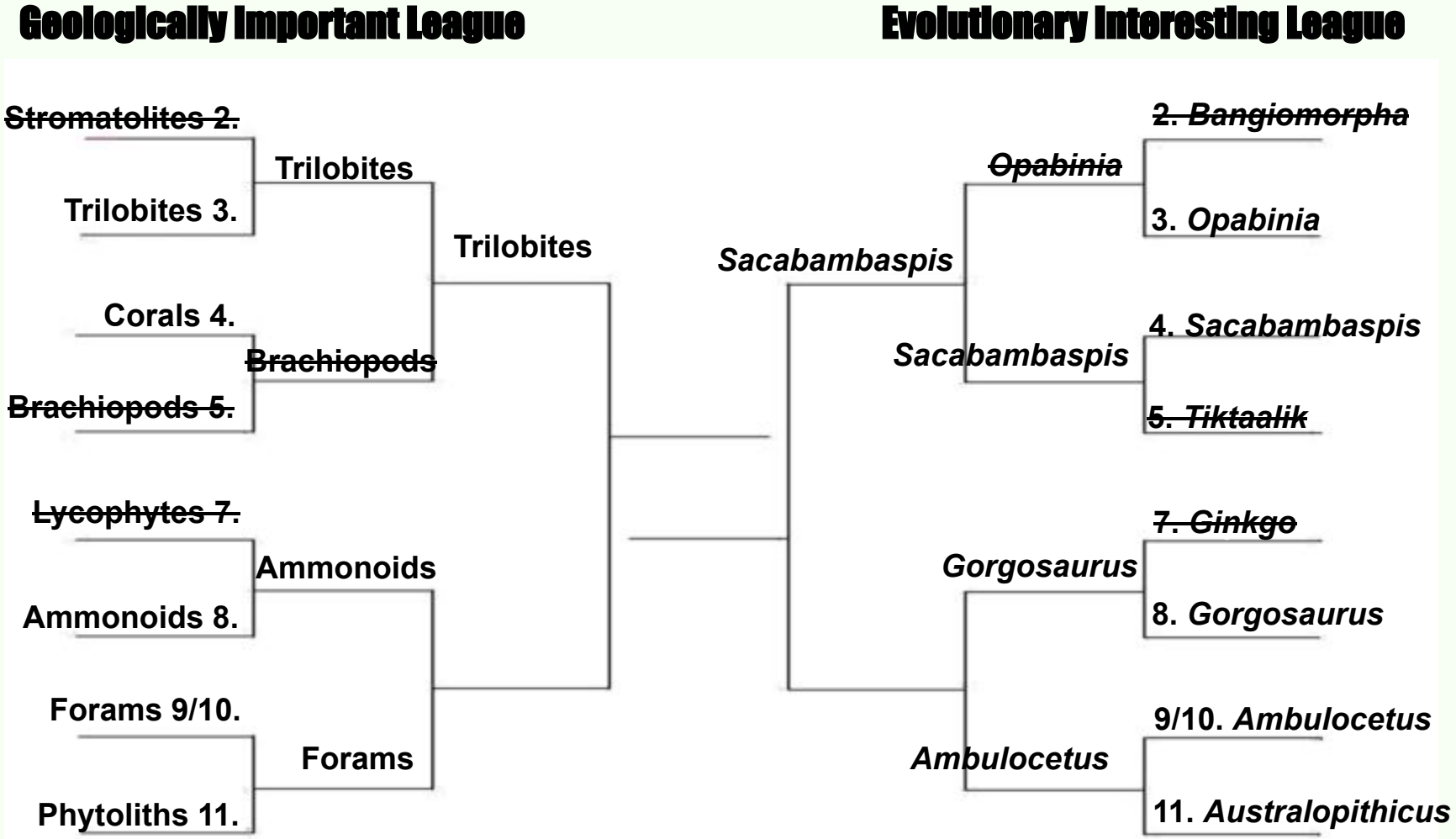
No Associated Task

Robert Sansom, U. Manchester

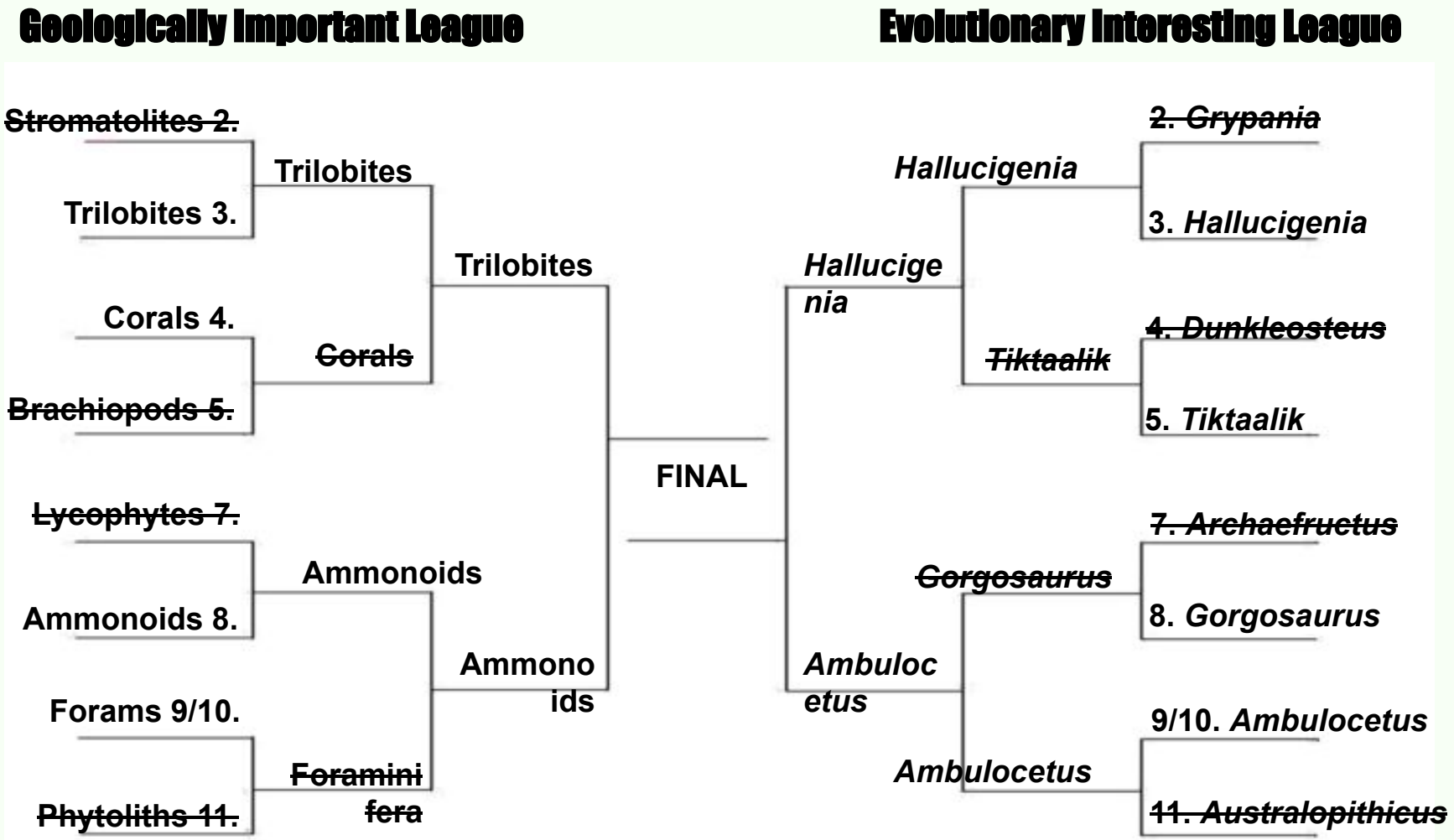


Didier Descouens

Fossil of the Week: PLAYOFFS 2024

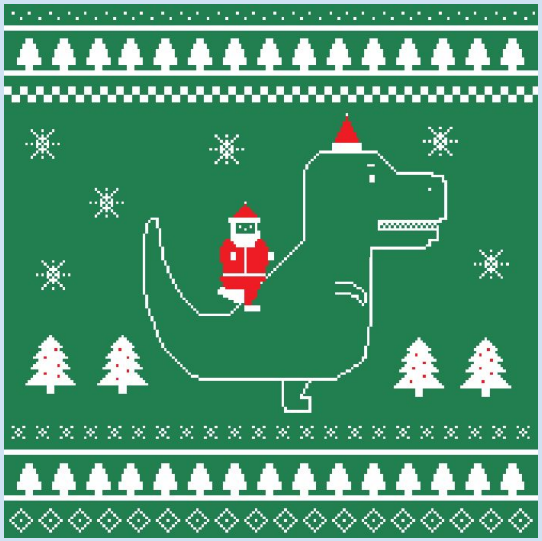


Fossil of the Week: PLAYOFFS 2021



Part 10
The Age of Apes

Live Session

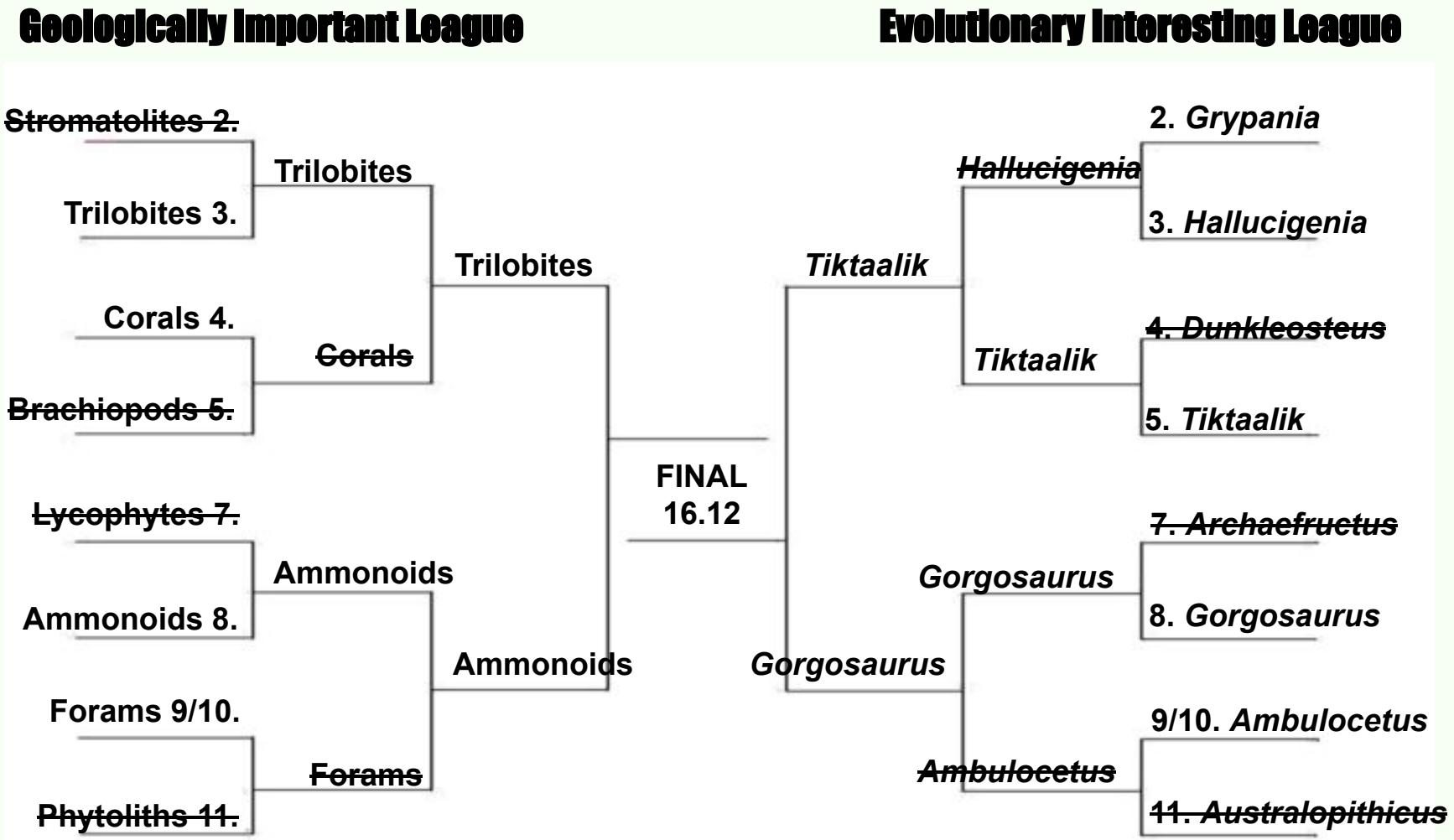


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University of Manchester



Didier Descouens

Fossil of the Week: PLAYOFFS 2020



Part 10
The Age of Apes

Primates



Didier Descouens

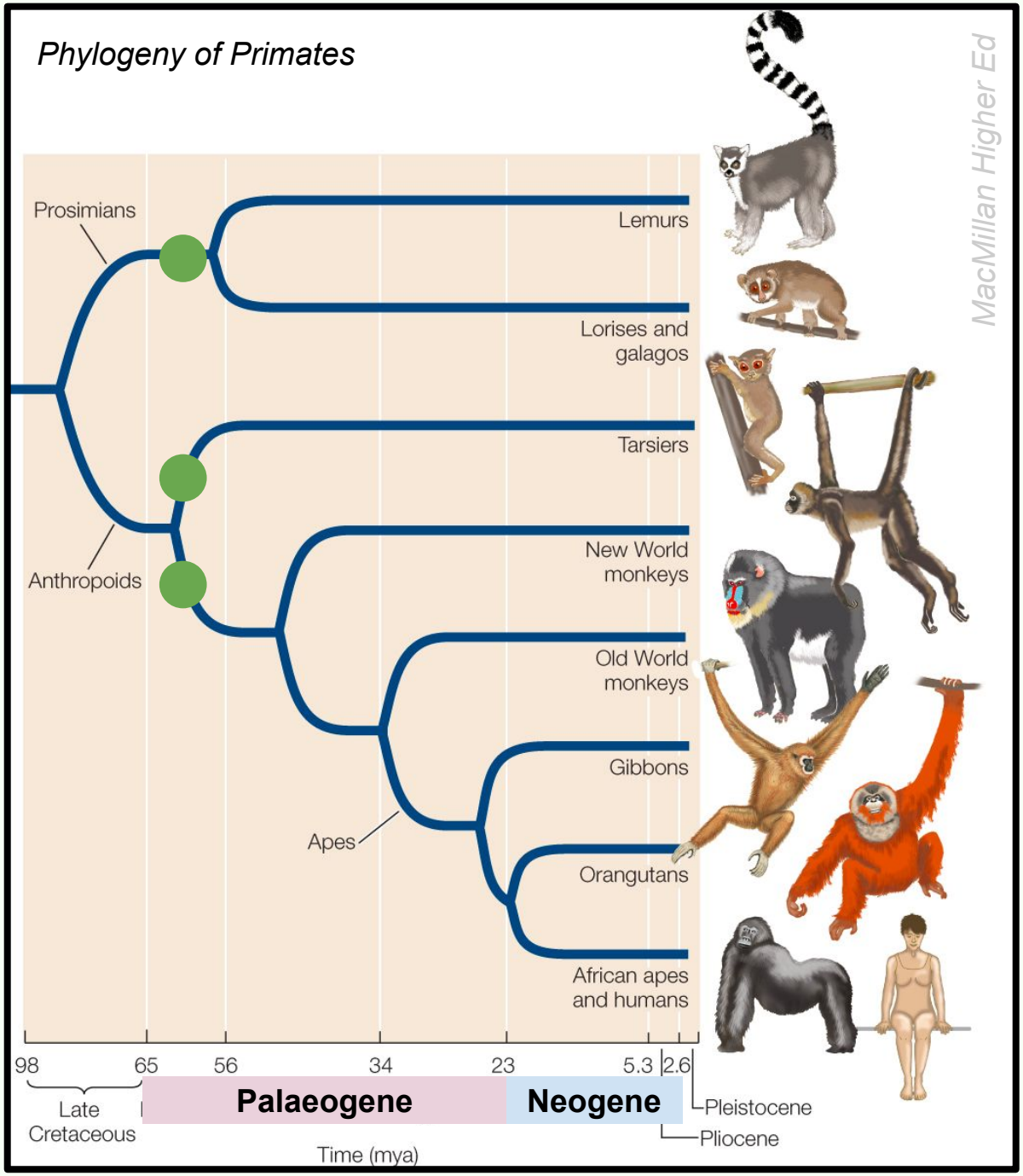
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Primates

Primates have a long evolutionary history, divergences taking place in Palaeogene

Group comprises of Lemuriformes, Tarsiiformes, and Simiiformes

Primary specializations of primates relate to vision and thumbs



Great Apes

Within primates, we are part of the **Hominidae** (Great Apes), primates which have lost their tails, and exhibit complex behaviours



Living non-human species of Hominidae



Chimpanzee



Monkey

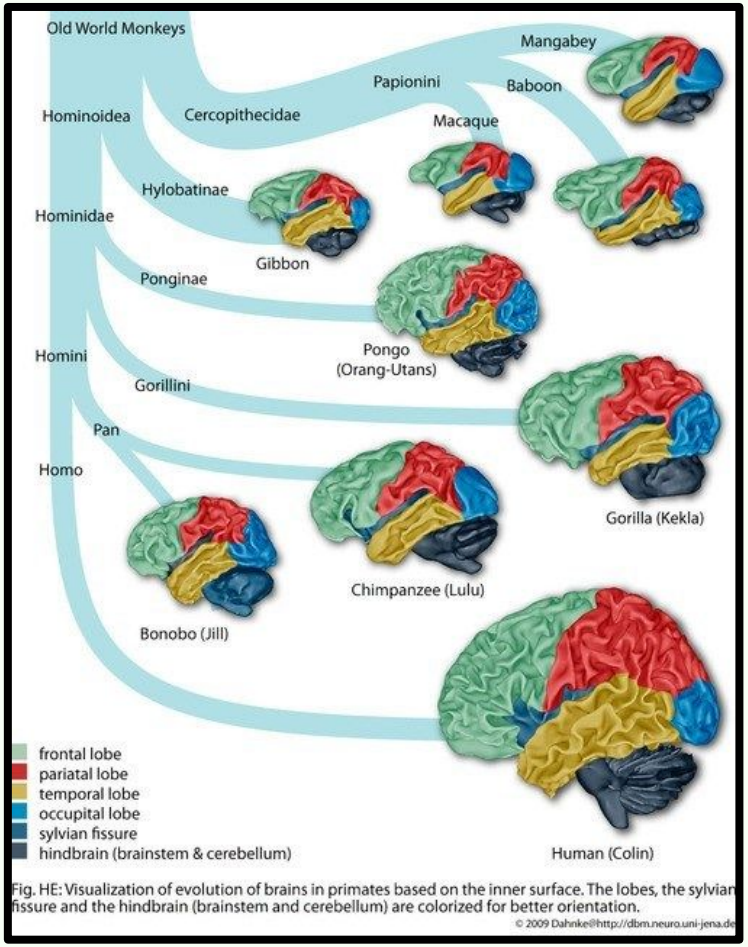


Fig. HE: Visualization of evolution of brains in primates based on the inner surface. The lobes, the sylvian fissure and the hindbrain (brainstem and cerebellum) are colorized for better orientation.

© 2009 Dahnke@http://dbm.neuro.uni-jena.de

C Gaser/U. Jena

The Age of Apes

Primates

Intended Learning Objective: Outline the biology of primates in the context of human evolution

Associated Task: Take a look at the phylogeny of mammal brains and their sizes below the video. What factors might be related to the evolution of big brains?

Robert Sansom, U. Manchester



Didier Descouens

Part 10
The Age of Apes

Human Origins



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The Age of Apes
Human Origins

Intended Learning Objective: Explain how fossil evidence reveals the origins and evolution of humans

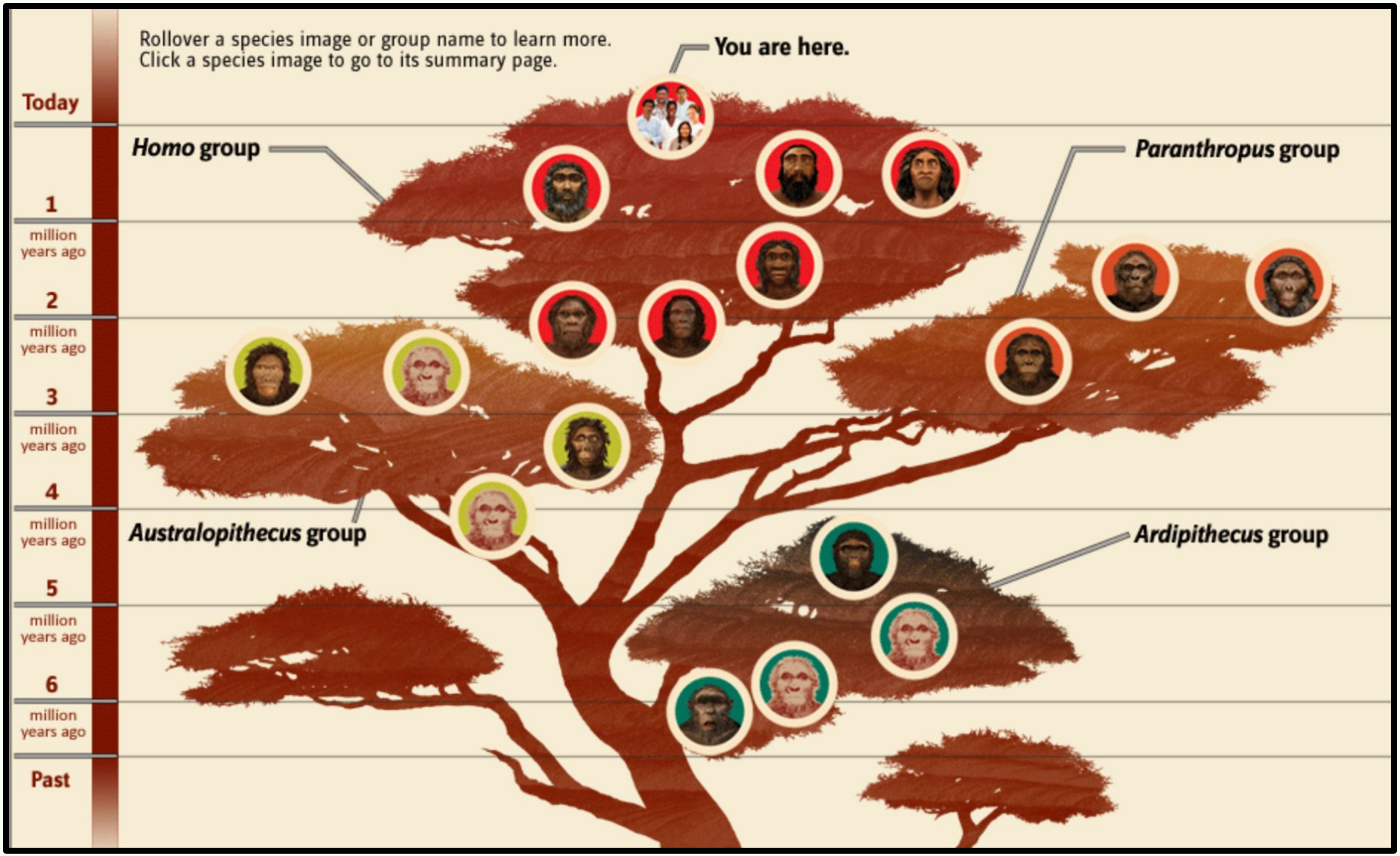
Associated Task: Explore the human family tree at this website from the Smithsonian Museum. Chose a taxon and an unanswered question - what data or findings would help us find the answers?

https://humanorigins.si.edu/evidence/human-family-tree

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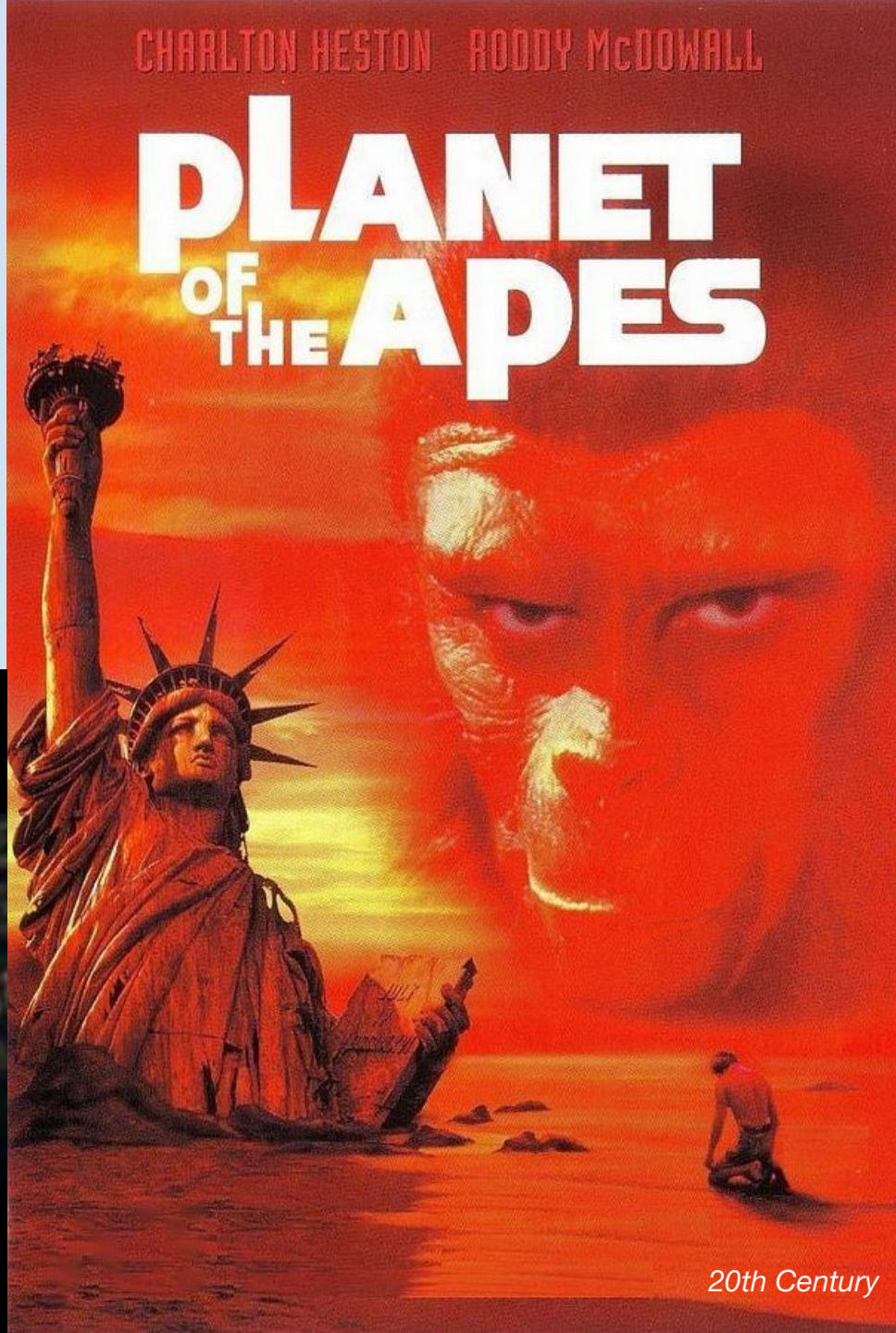

Rollover a species image or group name to learn more.
Click a species image to go to its summary page.
You are here.
Today
Homo group
Paranthropus group
1 million years ago
2 million years ago
3 million years ago
4 million years ago
Australopithecus group
Ardipithecus group
5 million years ago
6 million years ago
Past

The Age of Apes

Planet of the Apes (1968)

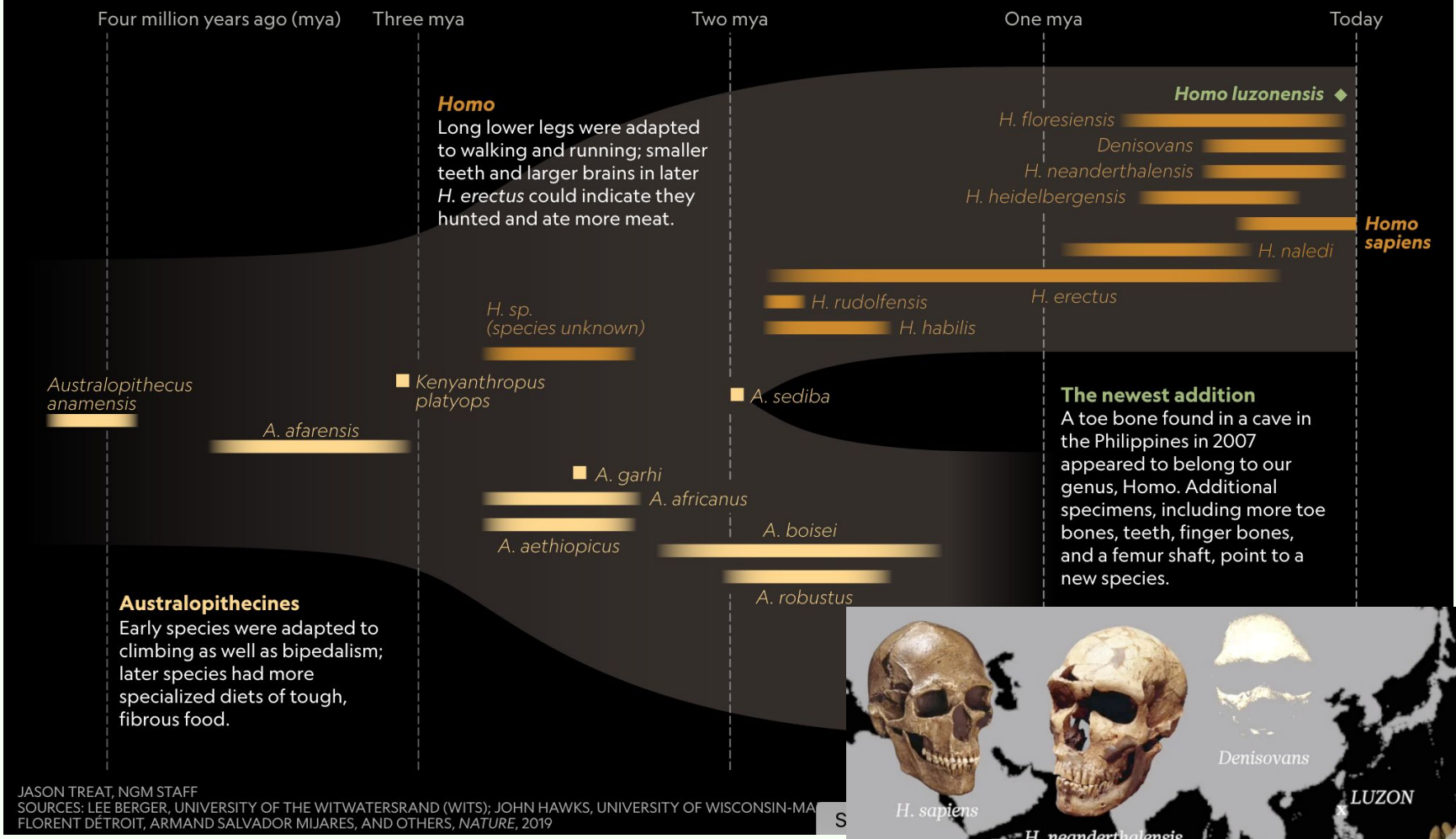
“Somewhere in the universe, there must be something better than man!”

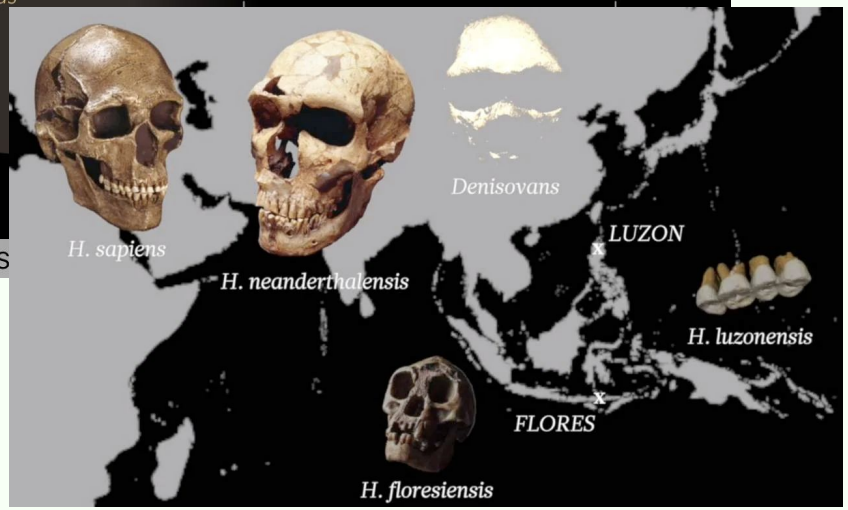




Human origins

Over millions of years, Africa, Asia, and Europe incubated a dazzling array of ancient human relatives. Today only one branch of the family tree remains: us.

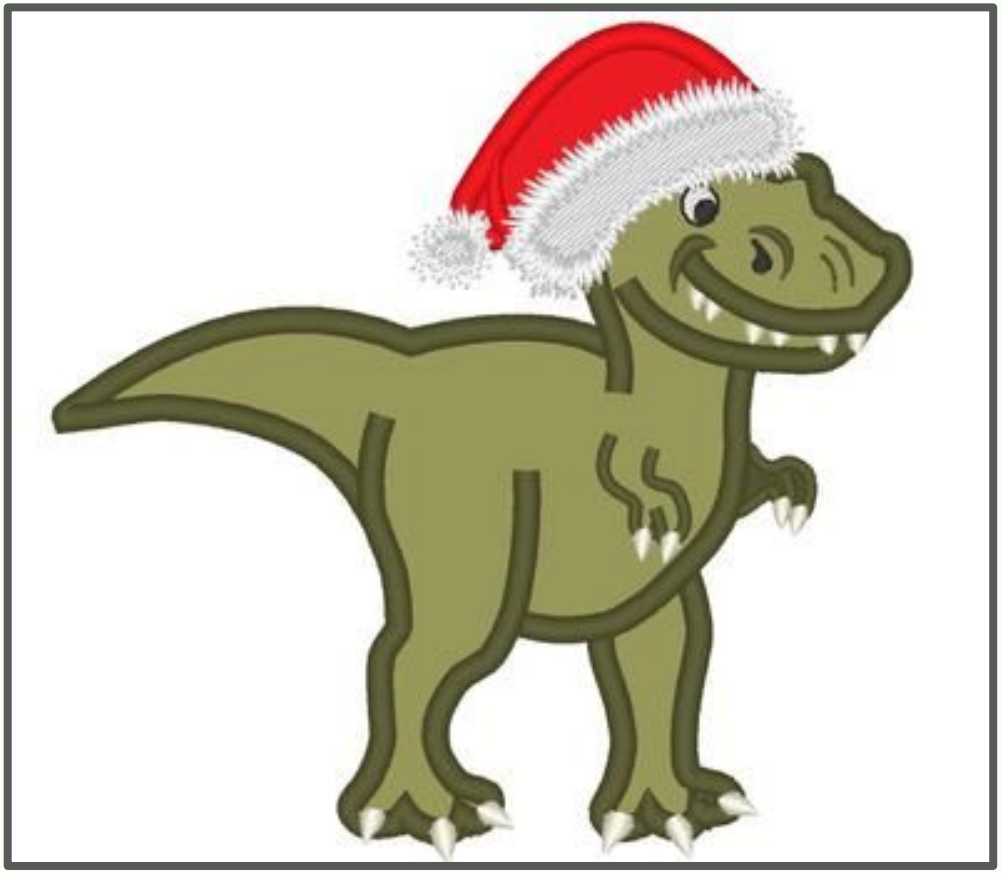




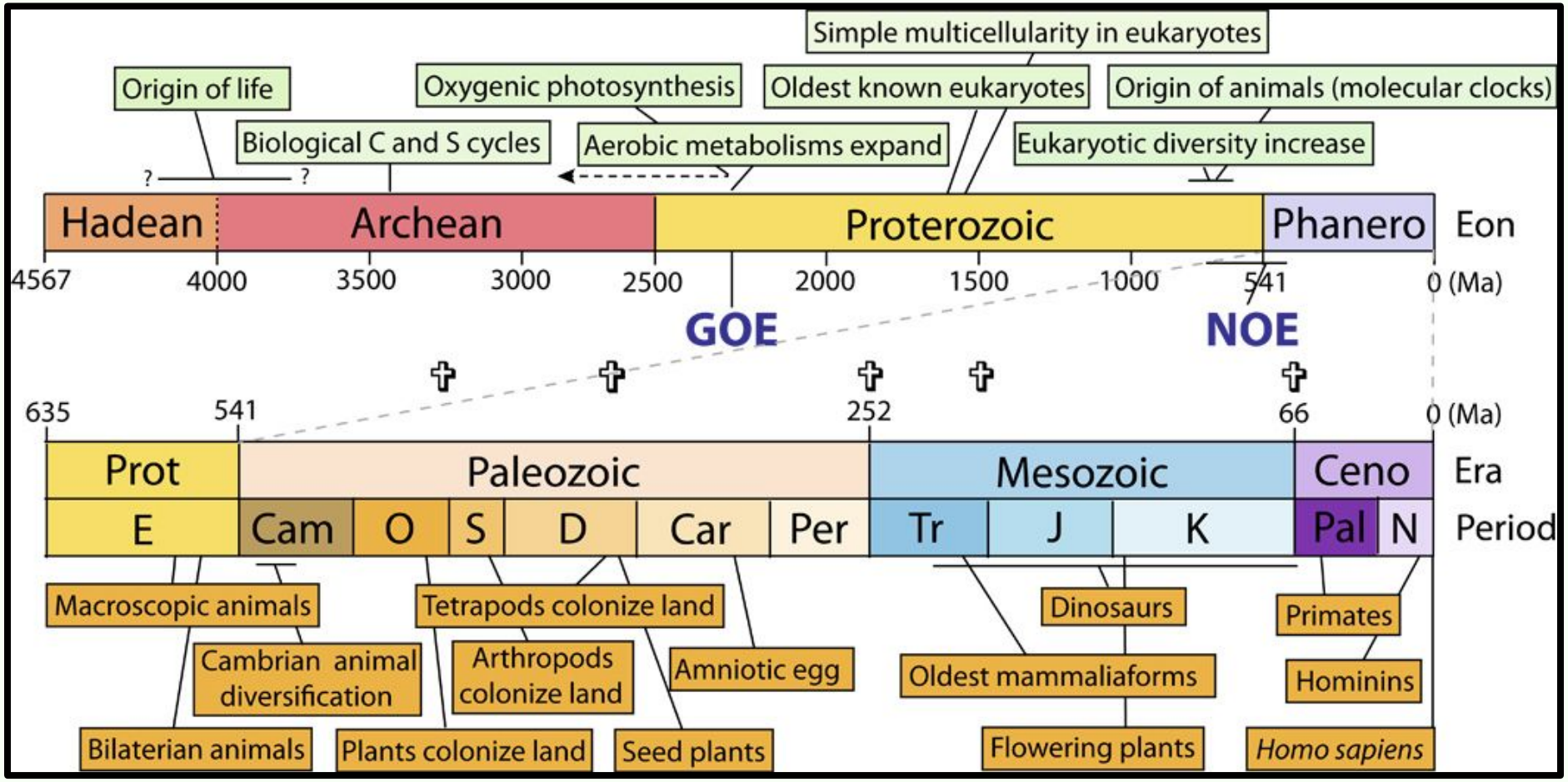
JASON TREAT, NGM STAFF
SOURCES: LEE BERGER, UNIVERSITY OF THE WITWATERSRAND (WITS); JOHN HAWKS, UNIVERSITY OF WISCONSIN-MA
FLORENT DÉTROIT, ARMAND SALVADOR MIJARES, AND OTHERS, *NATURE*, 2019

Exam Time

- 100 multiple choice questions in January Exam
- Only slight changes from your mid-semester test
- Consult the ILOs for an idea of what you will be examined on



Happy to talk Palaeobiology Pathway



Knoll & Nowak 2017 Science Advances